BeneVision

Central Monitoring System

Operator's Manual

CE 0123

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- The product is used in accordance with the instructions for use.

WARNING

- **This equipment must be operated by skilled/trained clinical professionals.**
- **It is important for the hospital or organization that employs this equipment to carry out a reasonable service/maintenance plan. Neglect of this may result in machine breakdown or personal injury.**

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Preface

Manual Purpose

This manual contains the instructions necessary to operate the product safely and in accordance with its function and intended use. Observance of this manual is a prerequisite for proper product performance and correct operation and ensures patient and operator safety.

This manual is based on the maximum configuration and therefore some contents may not apply to your product. If you have any question, please contact Mindray.

This manual is an integral part of the product. It should always be kept close to the equipment so that it can be obtained conveniently when needed.

Intended Audience

This manual is geared for clinical professionals who are expected to have a working knowledge of medical procedures, practices and terminology as required for monitoring of critically ill patients.

Illustrations

All illustrations in this manual serve as examples only. They may not necessarily reflect the setup or data displayed on your central monitoring system.

The "XX" symbol in an illustration may refer to parameter numeric values, waveform data, or descriptive text for a functionality.

Conventions

- *Italic text* is used in this manual to quote the referenced manuals, chapters, sections and formulas.
- **Bold text** is used to indicate the screen texts and names of hard keys.
- → is used to indicate operational procedures.

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1 Safety

1.1 Safety Information

The safety statements presented in this chapter refer to the basic safety information that the operator of the BeneVision Central Monitoring System (hereinafter referred to as the CMS) shall pay attention to and abide by. There are additional safety statements in other chapters or sections, which may be the same as or similar to the following, or specific to the operations

WARNING

- **Indicates a potential hazard or unsafe practice that, if not avoided, could result in death or serious injury.**

CAUTION

- **Indicates a potential hazard or unsafe practice that, if not avoided, could result in minor personal injury or product/property damage.**

NOTE

- **Provides application tips or other useful information to ensure that you get the most from your product.**

1.1.1 Warnings

WARNING

- **The CMS is intended for use only by clinical professionals or under their guidance. It must only be used by persons who have received adequate training in its use. Anyone unauthorized or untrained must not perform any operation on it.**
- **The physiological waveforms, parameters and alarms displayed on the CMS screen are for clinician's reference only and cannot be directly used as the basis for clinical treatment. Before giving invasive treatment to a patient, you must go to the corresponding monitor to confirm the results you have obtained from the CMS.**
- **If any value displayed on the CMS screen is abnormal or questionable, first determine the patient's vital signs by alternative means and then verify that the CMS and the monitoring devices connected to the CMS are working properly.**
- **The CMS is a clinical information device. Except for using such components as the mouse, touchscreen and keyboard to perform normal operations, do not touch or disassemble any other component, especially the power component; otherwise, it may result in personnel injury.**
- **The computer running the CMS software must comply with local relevant regulations. The CMS is intended to connect with Mindray monitoring devices only. Connecting devices made by other manufacturers may cause the values displayed on the CMS to be inaccurate.**
- **The CMS can be constructed using a wireless local area network (WLAN) for connecting patient monitors. When data is transmitted via radio frequency (RF) signals, it may impact the performance of other WLANs in the environment or the use of other RF equipment. Therefore, the wireless RF equipment must comply with CE, FCC and other local relevant standards and regulations.**

WARNING

- **When the CMS is transmitting data via a WLAN, loss of patient data may occur due to interference from other RF sources.**
- **The CMS software copyright is solely owned by Mindray. No organization or individual shall resort to altering, copying, or exchanging it or to any other infringement on it in any form or by any means without due permission.**
- **Data communication must be performed within a closed network or within a virtually isolated network provided by a hospital for all network functions. The hospital is responsible for ensuring the security of the virtually isolated network.**
- **Do not connect any external audio device to the CMS.**
- **Do not block the speaker of the CMS.**
- **Do not rely exclusively on audible alarm system. Setting alarm volume to a low level may result in a hazard to the patient.**
- **PACEMAKER PATIENTS – On ventricular paced patients, episodes of Ventricular Tachycardia may not always be detected. Do not rely entirely upon the system's automated arrhythmia detection algorithm. Keep pacemaker patients under close surveillance.**
- **A hazard can exist if different alarm presets are used for the same or similar equipment in any single area.**

1.1.2 Cautions

CAUTION

- **The service life of the CMS depends on its hardware. Rough treatment, dropping or collision should be avoided in the operation on the keyboard, mouse and computer; otherwise, the service life of the CMS may be shortened.**
- **Components of the CMS, such as the keyboard and mouse, may be contaminated by microorganisms during transport, storage and use.**
- **Before removing components of the CMS from their packaging, the packaging should be inspected for damage. In case of any damage, contact the carrier or Mindray immediately.**
- **All CMS equipment must utilize the hospital emergency power system. Failure to do so will result in loss of monitoring during extended periods of power failure. Hospitals without an emergency power system should use an Uninterruptible Power Supply (UPS) to power the CMS. When there is a power failure, the system should be shut down by following the specified shutdown procedure before the UPS is turned off. If the system has a sudden power failure, system failure may occur and consequently the system will not work correctly next time or may even have a serious result.**
- **Do not transport a CMS host or displays while they are running. Doing so may cause damage.**
- **System time should be set before the CMS is put into use. If the system time is changed when the CMS is in use, it may result in patient data loss.**
- **The CMS is capable of connecting up to 64 monitoring devices. The CMS should be connected to the network utilizing a network designed according to the installation guide.**
- **Restart the CMS every three months. Long time operation of the system may lead to a failure of the operating system.**
- **The CMS should be installed only using Mindray provided Microsoft Windows operating system, service packs, and patches. Use of unauthorized software may lead to abnormal system operation or failure.**
- **Do not install any third party applications not approved by Mindray or utilize the CMS host for any other purpose.**

CAUTION

- **Using the "Show Desktop" feature in Windows is prohibited. Installing or using any other software not provided by Mindray is prohibited. Software that has not been tested or verified by Mindray may cause the instability of the system. Mindray assumes no responsibility for this.**
- **When printing data through an external printer, be sure to follow the printer's instructions. In case any problem occurs during printing, consult the printer's instructions.**
- **At the end of its service life, the CMS must be disposed of in compliance with the guidelines regulating the disposal of such products. If you have any questions concerning disposal of the equipment, please contact Mindray.**

1.1.3 Notes

NOTE

- **This manual describes all features and options. Your equipment may not have all of them.**
- **Keep this manual in the vicinity of the CMS so that it can be obtained conveniently when needed.**
- **During normal use, the operator is expected to face the front of the equipment.**
- **All the CMS host, parts, or accessories should be from Mindray or acknowledged by Mindray.**

1.2 Equipment Symbols

NOTE

- **Some symbols may not appear on your equipment.**
- **This manual describes a series of typical computer symbols, which may be slightly different from what are on your computer. The computer symbols usually have their specific meanings. Refer to your computer's operator's manual or contact our service personnel if you have any question.**

Symbol	Description	Symbol	Description
	Refer to instruction manual/booklet		CAUTION: To reduce the risk of electric shock, do NOT remove cover. Refer servicing to qualified service personnel.
	Alternating current		Power switch
	Keyboard port		Mouse port
	Serial communication (COM) port		Display port
	Printer port		USB port or device
	Network port		Microphone port

Symbol	Description	Symbol	Description
	Sound input port		Sound output port
	Date of manufacture		Manufacturer
SN	Serial number		Equipotentiality
CE 0123	This product bears CE mark indicating its conformity with the provisions of the Council Directive 93/42/EEC concerning medical devices and fulfils the essential requirements of Annex I of this directive. Note: The product complies with the Council Directive 2011/65/EU.		
	The following definition of the WEEE label applies to EU member states only. This symbol indicates that this product should not be treated as household waste. By ensuring that this product is disposed of correctly, you will help prevent bringing potential negative consequences to the environment and human health. For more detailed information with regard to returning and recycling this product, please consult the distributor from whom you purchased it. * For system products, this label may be attached to the main unit only.		

2 System Overview

2.1 Intended Use

The BeneVision Central Monitoring System (hereinafter referred to as the CMS) is intended to conduct centralized monitoring of vital sign information from patient monitors, defibrillator/monitors, and telemetry transmitters through wired or wireless LAN in hospitals or medical institutions.

It is intended for use in a hospital or other clinical environment under the direct supervision of a licensed healthcare practitioner.

WARNING

- **The CMS is intended for use by qualified physicians or well-trained clinicians. Anyone unauthorized or untrained must not perform any operation on it.**
- **The physiological waveforms, parameters and alarms displayed on the screen of the CMS are for reference only and cannot be directly used as the basis for clinical treatment.**
- **If any value displayed on the CMS screen is abnormal or questionable, first determine the patient's vital signs by alternative means and then verify that the CMS and monitoring devices are working properly.**

2.2 Contraindications

None

2.3 System Components

The CMS mainly consists of the following components:

- CentralStation
- ViewStation (optional)
- WorkStation (optional)
- Central Monitoring System Viewer (hereinafter referred to as the CMS Viewer) (optional)
- Computer platform
- USB dongle
- Network devices (optional)
- Remote display and data transmission device (optional)
- Recorder (optional)
- Printer (optional)

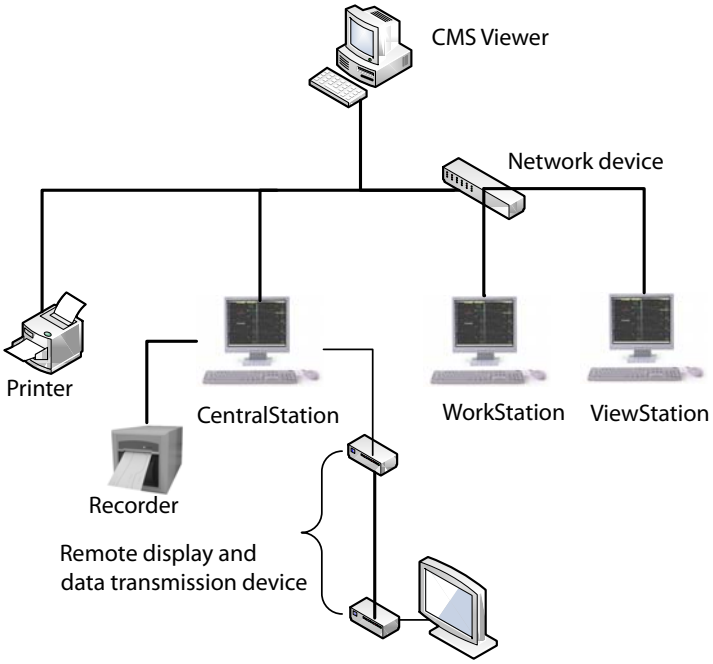

Figure 2-1 CMS Components

2.3.1 CentralStation

The CentralStation can admit Mindray physiological monitoring devices via wired or wireless connection. It can display, store, print, and review the patient clinical data derived from compatible monitoring devices.

The CentralStation can be running on virtual machines.

NOTE

- **When the CentralStation is installed on a virtual machine, do not use the CentralStation as the primary device where alarms are annunciated. Because when installed on a virtual machine, the CentralStation cannot detect the state of sound card and therefore may not annunciate an alarm sound.**

2.3.2 WorkStation

The WorkStation can display and manage patient data with monitoring devices assigned from the host CentralStation (the host CentralStation refers to the CentralStation which the WorkStation,the ViewStation, or the CMS Viewer is connected to). All the patient data comes from the CentralStation.

The WorkStation can be running on virtual machines.

You can enter or modify patient demographic information, configure alarm limits, alarm priorities, and discharge patients at the WorkStation based on the granted permissions.

2.3.3 ViewStation

The ViewStation can display patient data with monitoring devices assigned from the host CentralStation. All the patient data comes from the CentralStation. You can view patient data at the ViewStation only.

The ViewStation can be running on virtual machines.

2.3.4 CMS Viewer

The CMS Viewer is used to view a single patient at the host CentralStation. It is independent of the CMS and shall be installed separately.

The CMS Viewer is for remote browsing only and cannot be used to set or control the host CentralStation or the monitoring devices at the host CentralStation.

For more information on the CMS Viewer, see *BeneVision Central Monitoring System Viewer Operator's Manual (P/N: 046-010296-00)*.

2.3.5 Computer Platform

The CentralStation, the WorkStation, and the ViewStation consist of system software and computers. They have the same computer components: display, host computer, mouse, and keyboard.

2.3.5.1 Display

Each of the CentralStation, the WorkStation, and the ViewStation can be equipped with up to four external displays. For display specifications, see Appendix *A Technical Specifications* in this manual.

If you are using a touchscreen display, you can select any selectable screen object by tapping the object.

NOTE

- **Displaying right click menus is not supported by the touchscreen.**
- **Do not use any sharp objects such as pencils to tap the screen.**

2.3.5.2 Host Computer

A host computer is used to run the CMS software program. The CentralStation, the WorkStation, and the ViewStation need to be installed on different host computers. For computer specifications, see Appendix *A Technical Specifications* in this manual.

2.3.5.3 Mouse

The mouse can be used to to select a screen element by moving the cursor on the element and then clicking on it.

NOTE

- **Clicking refers to positioning the mouse pointer on a selection and pressing the left mouse button once.**

2.3.5.4 Keyboard

A keyboard is used to type text into a data entry field.

2.3.6 USB Dongle

A USB dongle is a prerequiste for using the CMS. It offers the following capabilities:

- Provide copyright protection for the CMS software.
- Configure functionalities.

NOTE

- **Do not exchange the USB dongles among different CMS systems.**
- **If your USB dongle is lost, please contact Mindray to get a new one.**

2.3.7 Network Device

A network device such as a switch and access point is used to connect a monitoring device to the CMS.

2.3.8 Remote Display and Data Transmission Device

The CMS supports connecting a video extender to display near-end videos at a remote site. The recommended remote display and data transmission device is ATEN KVM (Keyboard Video Mouse) products.

2.3.9 Recorder

The CMS can be equipped with a thermal recorder. The thermal recorder records patient information, measurement numerics and waveforms. The thermal recorder has a separate power supply. It can be connected to the CMS via a RS232 interface. For more information, see *10 Recording*.

2.3.10 Printer

The CMS can be equipped with a printer to output various reports. The printer has a separate power supply. For more information, see *9 Printing*.

2.4 Networking Mode

The CMS network consists of the bedside device network and the CMS network. The CMS supports networking among multiple central monitoring systems so that the patient data displayed on one CMS can be viewed by other networked CMS.

The typical networking diagram of the CMS is as shown below.

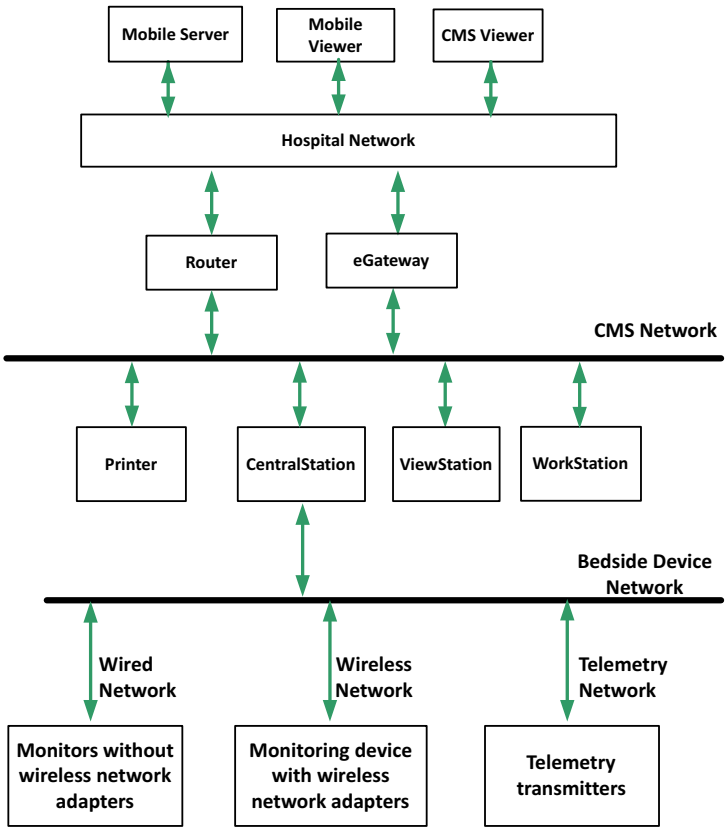

Figure 2-2 Networking Mode

2.4.1 Bedside Device Network

The bedside device network connects:

- Up to 96 monitoring devices that support the CMS+ protocol and broadcast communication
- Up to 400 monitoring devices that support the CMS+ protocol and multicast communication
- Up to 400 monitoring devices that support the ELAN protocol
- Up to 1200 monitoring devices that support the MD2 protocol

2.4.1.1 Communication Modes

Monitoring devices communicate with the CentralStation in one of the following ways:

- Wired network

The CentralStation and patient monitors interconnect through an Ethernet network switch.

- Wireless network

The CentralStation and patient monitors or TM80 telemetry monitors can interconnect through a wireless network consisting of Wireless Access Points and Ethernet switches.

- Telemetry network

The CentralStation is connected to the telemetry receiver through an Ethernet network switch. The telemetry receiver connects multiple telemetry transmitters via wireless RF.

2.4.1.2 Supported Monitoring Devices

The bedside device network supports the following devices:

- Bedside monitors
  - BeneVision N series patient monitors
  - BeneView T series patient monitors
  - iPM series patient monitors
  - iMEC series patient monitors
  - uMEC series patient monitors
  - VS series vital signs monitors
  - PM series patient monitors
  - MEC series patient monitors
- BeneHeart series defibrillator/monitors
- Telemetry devices
  - TMS-6016
  - TM80

2.4.2 CMS Network

The CMS network mainly consists of the CentralStation, the WorkStation, the ViewStation, and the CMS Viewer. The CentralStation admits monitoring devices, collects and stores patient data, enunciates alarms, and transfers the patient data to the WorkStation, ViewStation, CMS Viewer, and Mobile Server.

The CMS viewer is used to view real-time data and historical data of an online patient at the host CentralStation. An online patient refers to a patient who is being monitored by the CMS.

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3 Getting Started

3.1 Overview

This chapter describes preparation before using the CMS and gives an overview of the CMS.

3.2 Equipment Preparation Safety Information

WARNING

- **The CMS software copyright is solely owned by Mindray. No organization or individual shall resort to altering, copying, or exchanging it or to any other infringement on it in any form or by any means without due permission.**
- **Connect only approved devices to the CMS. Devices connected to the equipment must meet the requirements of the applicable IEC standards (e.g. IEC 60950 safety standards for information technology equipment and IEC 60601-1 safety standards for medical electrical equipment). The system configuration must meet the requirements of the IEC 60601-1 medical electrical systems standard. Any personnel who connect devices to the equipment's signal input/output port are responsible for providing evidence that the safety certification of the devices has been performed in accordance to the IEC 60601-1. If you have any questions, please contact Mindray.**
- **If it is not evident from the CMS specifications whether a particular combination with other devices is hazardous, for example, due to summation of leakage currents, please consult the manufacturer or an expert in the field. A determination must be made that the proposed combination will not negatively affect the devices themselves or the patient's safety.**

CAUTION

- **The CMS should be installed by authorized Mindray personnel.**
- **Before unpacking, examine the packing case carefully for signs of damage. If any damage is detected, contact the carrier or Mindray.**
- **Make sure that the CMS operating environment meets the specific requirements. Otherwise unexpected consequences, e.g. damage to the equipment, could result.**
- **Components of the CMS, such as the keyboard and mouse, may be contaminated by microorganisms during transport, storage and use. Before removing them from their packaging, the packaging should be inspected for damage. In case of any damage, contact the carrier or Mindray immediately.**
- **When disposing of the packaging material, be sure to observe the applicable waste control regulations and keep it out of children's reach.**
- **Observance of this manual is a prerequisite for proper product performance and correct operation and ensures patient and operator safety.**

NOTE

- **Never place the CMS within a patient environment.**
- **Do not block the displays of the CMS.**
- **Put the CMS in a location where you can easily view and operate the equipment.**
- **Keep this manual in the vicinity of the equipment so that it can be conveniently referenced when needed.**
- **Save the packing case and packaging material as they can be used if the CMS must be reshipped.**

3.3 Environmental and Power Requirements

This section describes the environmental requirements and power requirements for installing and using the CMS.

3.3.1 Environmental Requirements

The environment where the CMS is installed should be reasonably free from noises, vibration, dust, and corrosive, flammable, explosive substances. If the CMS is installed in a cabinet, sufficient space in front and behind should be left for convenient operation, maintenance and repair. Moreover, to maintain good ventilation, the CMS should be at least 2 inches (5 cm) away from the cabinet.

When the CMS is moved from one place to another, condensation may occur as a result of temperature or humidity difference. In this case, never start the system before the condensation disappears.

Each component of the CMS must work under the specified environment.

3.3.2 Power Requirements

To protect hospital personnel from electric shock, the CMS (including the host and displays) and its recorder must have their casings properly grounded. The CMS host is provided with a 3-wire power cord. Plug it into a properly grounded 3-wire receptacle.

Each component of the CMS must be powered by the specified power source.

WARNING

- **Select appropriate power supply according to the setup of the system power voltage. Otherwise serious damage may be caused to the CMS.**
- **Never use a 3-wire to 2-wire adapter with any unit of the CMS.**

3.4 Turning On the System

Perform safety checks before starting your system. For more information, see Section*13.3 General Inspection*.

To start the CMS system, follow this procedure:

1. Connect the power cord to the AC power source. When an uninterruptible power supply (UPS) is available, connect the UPS to the power supply and then connect the system to the UPS.
2. Turn on the host to start the operating system and displays. The system will perform a series of self-tests.
   - If the self-tests pass, the system will beep one time and enter the multibed screen immediately.
   - If the self-tests fail, an error message is displayed. In this case, consult the manual provided with the computer for assistance or contact Mindray.

CAUTION

- **To prevent damaging the system from a sudden power failure, it is recommended that your CMS is equipped with a UPS.**

3.5 Multibed Screen

Once started, the CMS displays the multibed screen where multiple patients are monitored collectively.

3.5.1 Example Multibed Screen

The regular multibed screen is as shown in Figure 3-1.

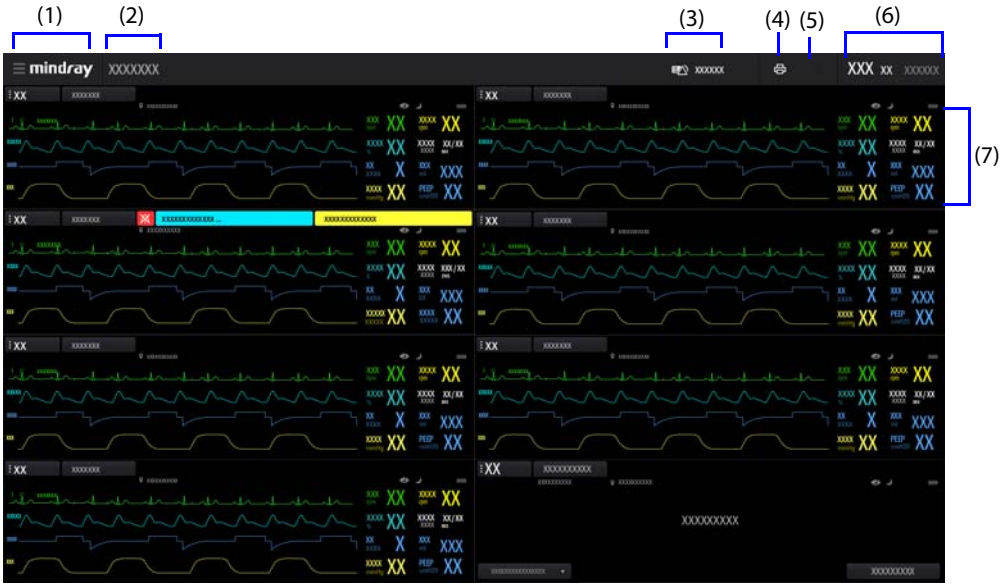


Figure 3-1 Example Multibed Screen

(1)	System menu area: selecting this area opens the drop-down list where you can select different functions.
(2)	Hospital information area/system alarm area: when system level alarms are not triggered, this area displays the hospital name and department where the CMS is located. If you need to change the hospital name or department, see Section *12.12.9 Setting A Device Location*. When system level alarms are triggered, this area displays the highest priority alarm message. Selecting this area opens the list of currently triggered system alarm messages. You can also review the system alarm messages in the alarm list area. For information on system level alarms, see Section *B.2 Alarm Messages in the System Alarm Area*.
(3)	Alarm list area: selecting this area opens the alarm list that includes physiological and technical alarms for all the devices being monitored by the CMS.
(4)	Print button: selecting this buttondisplays **Print Summary** and **Print Queue**. For more information on the two options, see *9 Printing*.
(5)	System silence symbol: selecting this symbol silences the system. When the system is silenced, the symbol is displayed at the top of the screen.
(6)	System time area
(7)	Patient sector: displays real-time monitoring data. One patient sector displays data from one patient only. For more information, see *3.5.3 Patient Sector Status*.

3.5.2 Symbols on the Multibed Screen

The following table lists the symbols displayed on the multibed screen.

Symbol	Description	Symbol	Description
	The color of the symbol changes to indicate different patient gender status: ■ Blue: indicates this patient is a male adult. ■ Grey: indicates that this patient is an adult without specified gender.		Adult, female
	The color of the symbol changes to indicate different patient gender status: ■ Blue: indicates this patient is a male pediatric. ■ Grey: indicates that this patient is a pediatric without specified gender.		Pediatric, female
	The color of the symbol changes to indicate different patient gender status: ■ Blue: indicates this patient is a male neonate. ■ Grey: indicates that this patient is a neonate without specified gender.		Neonate, female
	Indicate the Wi-Fi signal strength after a monitoring device is connected to an AP. More white arcs indicate a stronger Wi-Fi signal strength.		Indicate the received signal strength of telemetry devices. The color of the symbol changes to indicate received signal strength changes. ■ White: indicates the received signal strength is normal. ■ Yellow: indicates the received signal strength is weak. ■ Red: indicates no signal is received.
	Current patient sector is idle and can admit a monitoring device.		Contain more than one alarm messages.
	Selecting this symbol enters the ViewBed screen.		Indicate that the monitoring device is from a remote CentralStation.
	All the alarms are paused.		Alarms for some parameter are turned off or the monitoring device is in the alarm off state.
	Audible alarms are paused.		Audible alarms are turned off
	Alarms are acknowledged and the alarm system is reset.		The system is silenced.
	Selecting this symbol releases the system from silenced state.		Selecting this symbol silences the system.

Symbol	Description	Symbol	Description
	A patient's pacing status is not set at the monitoring device. This symbol will flash in the waveform area when a pacer pulse is detected. You can set the pacing status on the patient management screen.		A patient's pacing status is set to **Yes**. The pace pulse markers "|" are shown on the ECG wave when a pacer pulse is detected.
	A patient's pacing status is set to **No**.		This is the nurse call symbol. ■ After the nurse call button is pressed on a monitoring device, this symbol will continuously flash in the waveform area and a corresponding prompt tone will sound. This prompt tone will automatically terminate after two beeps. ■ Selecting this symbol clears this symbol and stops the prompt tone. ■ The nurse call will be stored on the event review page.
	The battery has sufficient power. The white grid indicates the remaining battery power.		The battery has low charge and needs to be charged or changed.
	The battery has critically low charge and needs to be charged or changed immediately. Otherwise, the monitoring device will shut down.		No battery is installed into the monitoring device.
	This monitor is in privacy mode.		This monitor is in night mode.

NOTE

- **When a battery symbol listed in the table above is displayed, see appropriate monitoring device's operator's manuals for details about the battery time and alarms.**

3.5.3 Patient Sector Status

The patient sector may stay in one of the following operating states:

- Offline: indicates that this patient sector has a patient admitted but its corresponding monitoring device may be turned off or disconnected from the CentralStation.
- Monitoring: indicates that the patient is being monitored and the monitoring device is communicating with the CMS.
- Standby: indicates that the monitoring device is in standby mode.
- Discharged: indicates that the patient has been discharged from the monitoring device or from the CMS.
- Idle: indicates that no monitoring device has been associated with this patient sector. The patient sector is available for admitting a new monitoring device. For details on how to admit monitoring devices, see *4 Patient Management*.

3.5.4 Patient Sector in the Monitoring State

When a patient sector is in the monitoring state, it displays real-time patient data transmitted from the monitoring device, as shown in Figure 3-2.

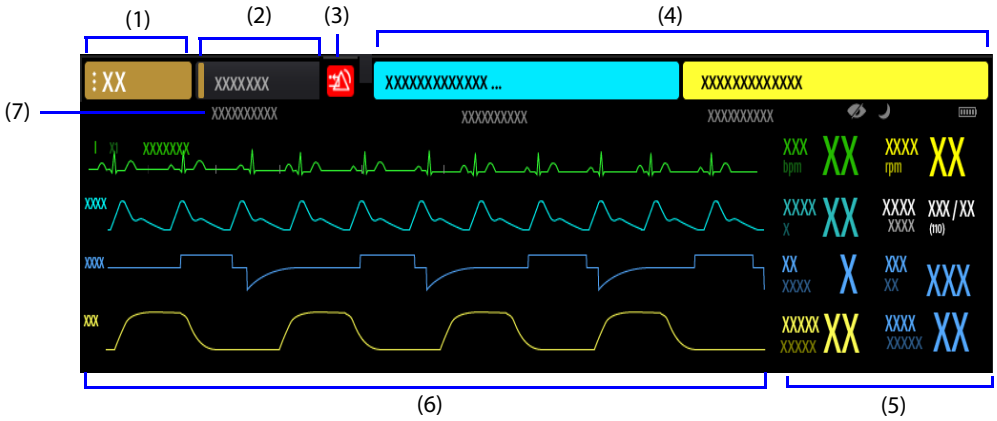


Figure 3-2 Example Patient Sector in Monitoring Status

(1)	Bed number and room number area: displays the bed number and room number. You can also assign a patient to a care group. After selecting a care group, the bed number and room number area is filled with the background color selected for **Care Group** on the patient management screen. For more information, see Section *3.5.4.1 Bed Number and Room Number Area*.
(2)	Patient information area: it can be set to display patient name or patient ID. For details on how to perform this setting, see Section *12.5.6 Setting Other Patient Information Items*. You can also assign a patient to a patient group. After selecting a patient group, the starting portion of the patient information area is filled with the color selected for **Patient Group** on the patient management screen. Selecting this area enters the patient management screen.
(3)	Alarm status symbol: indicates current alarm status. For more information, see Section *3.5.2 Symbols on the Multibed Screen*.
(4)	Alarm information area: displays the highest priority alarms. This area can be divided into the technical alarm information area and physiological alarm information area when there is sufficient space to display the patient sector. The technical alarm information area is displayed on the left side of the alarm information area. The physiological alarm information area is displayed on the right side of alarm information area. The ellipsis symbol "..." indicates that there are more than one alarm message.
(5)	Parameter area: displays parameter numerics transmitted from a monitoring device. Selecting this area enters the ViewBed screen.
(6)	Waveform area: displays waveforms transmitted from a monitoring device. When one or all lethal arrhythmia alarms are set to off, a corresponding message is displayed under the first ECG waveform. When the nurse call and/or event function of a telemetry device is switched off, a corresponding message is displayed under the first ECG waveform. Selecting this area enters the ViewBed screen.
(7)	Status information area: displays information such as notes,device signal, device name, device location, operating mode, and battery power. For description of symbols, see Section *3.5.2 Symbols on the Multibed Screen*.

NOTE

- **Contents displayed in a patient sector are subject to the number of patient sectors. The less the number of patient sectors, the more contents displayed in a patient sector. For details on how to set the number of patient sectors, see Section*12.3.3.3 Setting the Number of Patient Sectors*.**

3.5.4.1 Bed Number and Room Number Area

Selecting this area opens the drop-down list where you can perform a series of operations. Only items that may need special remarks are described here.

Menu Item	Description
Move Position to	Move a patient sector from the source patient sector to the target patient sector on the multibed screen. The source patient sector and the target patient sector swap positions.
	Reset alarms. For more information, see Section *5.7 Alarm Reset*.
	Pause alarms. For more information, see Section *5.6 Alarm Pause*.
	Standby mode. For more information, see Section *3.7.1 Standby Mode*.
Clear Overview Bed	Clear the patient sector. This function is used for clearing a patient sector where a monitoring device from a remote CentralStation is displayed. For details, see Section *11.4 Clearing an Overview Bed*.

3.5.4.2 Alarm Information Area

You can view all the alarm messages for a monitoring device by selecting the alarm information area.

This area displays the following items:

- Alarms and prompts from a monitoring device or from the CMS
- The operating mode and operation events of a defibrillator/monitor
- Alarm time for latched physiological alarms
- The highest priority alarms. When multiple highest priority messages exist, they scroll.

Alarms are displayed in different background colors to indicate alarm priorities:

- Red background indicates high priority alarms.
- Yellow background indicates medium priority alarms.
- Cyan background indicates low priority alarms.

You can set whether to let the background color of high priority and medium priority alarm messages flash. For details on how to perform this setting, see Section *12.4.3 Configuring Other Alarm-Related Items*.

3.5.4.3 Changing the Screen Setup for Patient Sector

To change the display settings for a patient sector, follow this procedure:

1. Select the bed number and room number area in the upper left corner of the desired patient sector on the multibed screen.
2. From the drop-down list, select **Screen Setup**.
3. Select the **Choose Screen** tab.
4. Set the desired screen. **Normal Screen** is most frequently used for patient monitoring. **Big Numerics** displays parameter numerics in big font size.
5. Select the **Tile Layout** tab.
6. Select a parameter area or waveform area, and then from the drop-down list select an element you want to display in this area. The parameters and waveforms selected are displayed in the patient sector.

NOTE

- **The list of parameters/waveforms under the drop-down list in the Tile Layout tab is the super set, not depending on a monitoring device. If a parameter/waveform is not supported by the monitoring device, it is not displayed in the patient sector.**
- **Changing the screen settings affects the patient sector only and does not affect the screen settings for the ViewBed screen.**

NOTE

- **Changes to the screen settings are only a temporary change. Once a patient is discharged from this patient sector, the screen settings configured will be cleared and the default department configurations will be automatically loaded to the patient sector.**

3.6 ViewBed Screen

You can have a more detailed view of a single patient's information, waveforms, and parameter numerics on the ViewBed screen. In addition, you can also perform additional operations such as configuring parameters, viewing multi-lead ECG waveforms, and recording waveforms for the selected patient.

3.6.1 Accessing the ViewBed Screen

Access the ViewBed screen in either of the following ways:

- Select the parameter area or waveform area in the desired patient sector on the multibed screen.
- Select the ViewBed symbol when it is displayed in the desired patient sector on the multibed screen.

NOTE

- **When the CMS is equipped with one display only, the ViewBed screen is displayed in the lower half of the screen or on the right of the screen depending on the display size.**
- **When the CMS is equipped with more than one display, the ViewBed screen can be displayed on a dedicated secondary display. For details on how to set the secondary display, see Section *12.3.3.4 Setting the Number of Primary Screens*.**

3.6.2 Example ViewBed Screen

The regular ViewBed screen is as shown in Figure 3-3.

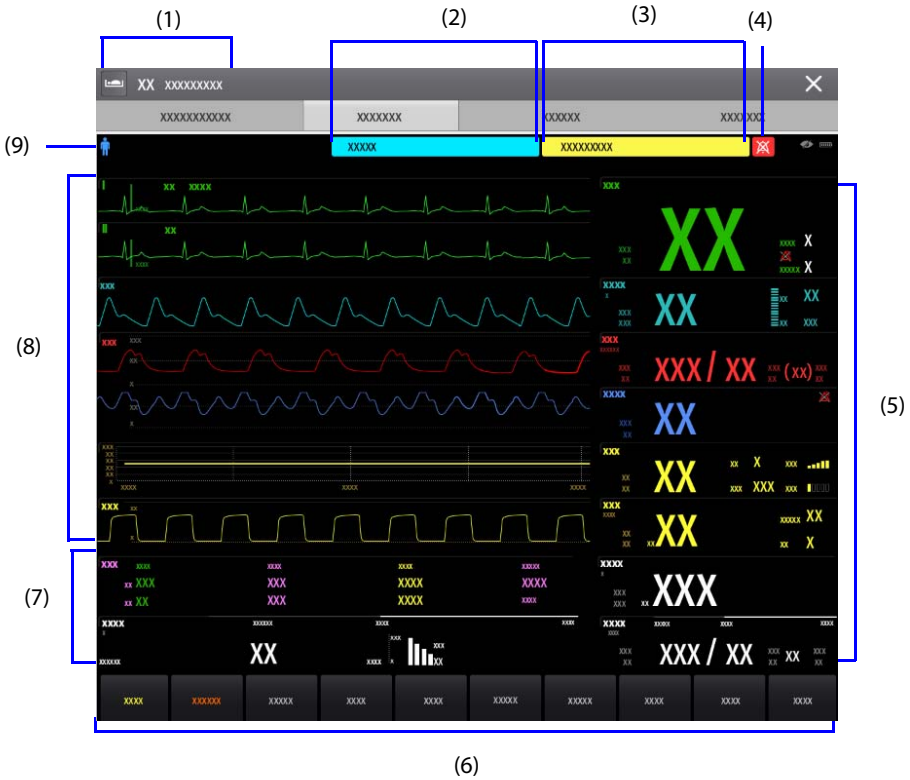


Figure 3-3 Example ViewBed Screen

(1)	Patient location information area: displays the bed number and room number.
(2)	Technical alarm information area: displays the highest priority technical alarm.
(3)	Physiological alarm information area: displays the highest priority physiological alarm.
(4)	Alarm status symbol: indicates current alarm status. For more information, see Section *3.5.2 Symbols on the Multibed Screen*.
(5)	Parameter area: displays parameter values, alarm limits, and alarm status. Selecting a parameter numeric block enters corresponding parameter menu. When a parameter alarm is turned off, the symbol is displayed in corresponding parameter area.
(6)	Quick key area: displays quick keysto access commonly used functions (keys are configurable)..
(7)	Parameter numerics/waveform area: displays parameter values and waveforms.
(8)	Waveform area: displays parameter waveforms. Selecting a waveform enters corresponding parameter menu.
(9)	Patient category symbol: displays current patient category. For more information, see Section *3.5.2 Symbols on the Multibed Screen*.

NOTE

- **The display layout of the CMS may not match that of a monitoring device. But the parameters/ waveforms displayed on the CMS are consistent with that on the monitoring device.**
- **If you change the display layout of parameters/waveforms on the monitoring device, the display layout of parameters/waveforms on the CMS will not change automatically and vice versa.**
- **When a module of a monitoring device is turned off, the waveform and numeric data for this module will be no longer displayed at the CMS.**
- **When a parameter is no longer being monitored, the waveform and numeric data for this parameter will be no longer displayed at the CMS.**

3.6.3 Changing ViewBed Screen Settings

To change ViewBed screen settings, follow this procedure:

1. Access the ViewBed screen setup menu in either of the following ways:
   - On the ViewBed screen,select the **Screen Setup** quick key from the quick key area.
   - Select the bed number and room number area in the upper left corner of the desired patient sector on the multibed screen → from the drop-down list select **Screen Setup** → select the **ViewBed** tab.
2. Select the **Choose Screen** tab and select the desired screen mode.
3. Select the **Tile Layout** tab.
4. Select a parameter area or waveform area, and then from the drop-down list select an element you want to display in this area. The selected parameters and waveforms are displayed on the ViewBed screen. The symbol indicates that this area can be set to display waveforms. Selecting this area changes this symbol to , indicating that this area can be set to display parameters.

NOTE

- **The list of parameters/waveforms under the drop-down list in the Tile Layout tab is the super set, not depending on a monitoring device. If a parameter/waveform is not supported by the monitoring device, it is not displayed on the ViewBed screen.**
- **Changing the ViewBed screen settings affects the ViewBed screen only and does not affect the screen settings for the patient sector.**
- **Changes to the screen settings are only a temporary change. Once a patient is discharged, the screen settings configured will be cleared and the default department configurations will be automatically loaded to the ViewBed screen.**

3.6.4 Quick Keys

The ViewBed screen provides quick keys for you to quickly access some functions. The **More** key is permanently located at the left bottom. Selecting the **More** quick key shows more quick keys. The quick keys displayed on the screen are configurable. When a function is not supported by a monitoring device, its quick key is inactive.

3.6.4.1 Available Quick Keys

The following table shows available quick keys.

Symbol	Label	Function	Symbol	Label	Function
	More	Show more quick keys		Screen Setup	Enter the **Screen Setup** menu
	Alarm Setup	Enter the **Alarm** menu		Alarm Reset	Acknowledge the ongoing alarms
	Alarm Pause	Pause the ongoing alarms		Main Screen	Return to the multibed screen
	Audio Pause	Pause alarm tone		OxyCRG	Enter the OxyCRG window
	NIBP Start/ Stop	Start an NIBP measurement or stop the ongoing NIBP measurement		Standby	Enter standby mode
	ECG Full-Screen	Enter the ECG full screen		Freeze	Freeze waveforms
	Load Config	Enter the **Load Config** menu		Minitrends	Enter the Minitrends window
	Parameters Setup	Enter the **Parameters Setup** menu		Send Message	Send a message to the TM80
	Print	Start printing a real-time report		Print Setup	Enter the **Print Setup** menu
	Privacy Mode	Enter privacy mode		Night Mode	Enter night mode
	Record Setup	Enters the **Record Setup** menu.		ECG 24h Sum	Views the 24-hour ECG summary.
	Record	Starts/Stops a recording.		Pace View	Enters the **Pace View** window.

Symbol	Label	Function	Symbol	Label	Function
	Discharge Patient	Enters the **Discharge Patients** dialog box.		Manual Event	Manually triggers and saves an event.

3.6.4.2 Configuring Quick Keys

To configure the desired quick keys, follow this procedure:

1. Access the quick keys setup menu in either of the following ways:
   - ◆ On the ViewBed screen select the **Screen Setup** quick key from the quick key area →select the **Quick Keys** tab.
   - ◆ Select the bed number and room number area in the upper left corner of the desired patient sector on the multibed screen → from the drop-down list select **Screen Setup** → select the **Quick Keys** tab.
2. Select the **Current** tab to configure the desired quick keys.
   - ◆ Add a quick key: on the top of **Quick Keys** tab, first select a key which you want to configure for a specific quick key function, next select the quick key function from the list of functions. For example, if you want to show the **Screen Setup** quick key in the first key, select the first key, and then select **Screen Setup** from the function list.
   - ◆ Clear a quick key: select the quick key to be cleared from the quick key and then select **Blank** from the function list.
3. Select the **More** tab to configure the quick keys to be displayed when the **More** quick key is selected.

3.6.5 Starting/Stopping NIBP Measurement

When the remote control functionality for NIBP measurement is enabled at the CentralStation or the WorkStation, some bedside monitors allow you to remotely to start an NIBP measurement or stop the ongoing measurement. For details on how to select this remote control functionality, see Section *12.12.8 Setting Remote Control*.

Start an NIBP measurementin either of the following ways:

- ■ On the ViewBed screen, select the **NIBP Start/Stop** quick key.
- ■ Select the bed number and room number area in the upper left corner of the desired patient sector on the multibed screen → from the drop-down list select **NIBP Start/Stop**.

In the measurement process, if you wish to stop the proceeding measurement, select **NIBP Start/Stop**.

WARNING

- **Before starting an NIBP measurement, make sure that the patient category on the bedside monitor is correct and the NIBP cuff size is appropriate for this patient.**

3.6.6 Freezing/UnFreezing Waveforms

During patient monitoring, you can freeze waveforms on the ViewBed screen to allow a closer review of the waveform data.

3.6.6.1 Freezing Waveforms

To freeze waveforms, follow this procedure:

1. On the ViewBed screen, select the **Freeze** quick key.
2. In the **Freeze** window, select the left or right arrow to move the waveforms backward or forward one second, to display the desired time.

Freezing waveforms does not affect:

- ■ Waveforms displayed in the minitrends window or parameter numerics on the ViewBed screen
- ■ Waveforms displayed on the multibed screen

- Alarm enunciation

3.6.6.2 Unfreezing Waveforms

To unfreeze waveforms, select the × button in the upper right corner of the **Freeze** window.

3.6.6.3 Printing Frozen Waveforms

Selecting the button in the upper left corner of the **Freeze** window prints frozen waveforms.

3.6.7 Loading Configurations for Telemetry Devices

You can load user defaults or factory defaults for the selected telemetry devices.

To load configurations, follow this procedure:

1. Select the **Load Config** quick key at the bottom of the ViewBed screen to enter the **Load Config** menu. If this quick key is not available, configure it by following steps in *3.6.4.2 Configuring Quick Keys*.
2. In the **Local** tab, select the desired option.
   - **User Default**: loads the saved user configuration.. For details about the configuration items, see Section *12.11.2 Setting Telemetry Configurations*.
   - **Factory Default**: loads factory default configuration.
3. Select the **Load** button.

3.6.8 Viewing Device Location

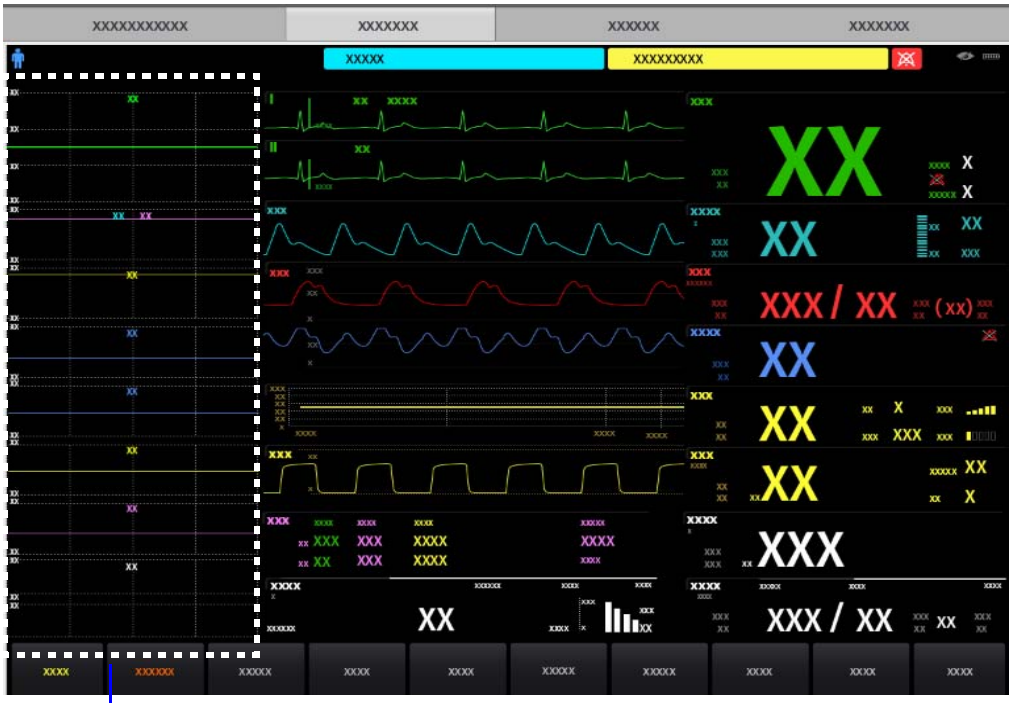
If the AP information has been imported into the CentralStation, you can view device location information for WiFi devices. For details on how to import AP information, see Section*12.9.8 Importing AP Information*.

To view device location, follow this procedure:

1. Select the symbol at the top of the ViewBed screen. The **Device Location** menu is displayed.
2. View the AP switch time in the Time column and the current location of devices in the **Location** column.
3. If you wish to lcoate a TM80 telemetry device, select the **Find Device** button.The TM80 will generate a continuous audible tone until it is acknowledged at the TM80.

3.6.9 Minitrends Window

You can split the normal screen so that the left hand side of the screen shows the minitrends window. In the minitrends window, graphic minitrends of parameters are displayed, as shown in the figure below. When external devices are connected to the monitor via the BeneLink module, you can also view the graphic minitrends of the external devices in the minitrends window. Parameter labels from external devices are prefixed with the plus sign (+).



Minitrends Window

3.6.9.1 Entering the Minitrends Window

Enter the Minitrends window in either of the following ways:

- On the ViewBed screen, select the **Minitrends** quick key.
- On the ViewBed screen, select the **Screen Setup** quick key →select **Choose Screen** →select **Minitrends**.

3.6.9.2 Configuring the Minitrends Window

To configure the minitrends window, follow this procedure:

1. In the minitrends window, select the minitrends area to enter the **Minitrend Setup** menu.
2. Configure parameters. If you wish to use the default parameters, select **Default Parameter**.
3. Configure **Minitrend Length**.

3.6.9.3 Exiting the Minitrends Window

To exit the Minitrends window, follow this procedure:

1. On the ViewBed screen,select the **Screen Setup** quick key.
2. Select the **Choose Screen** tab.
3. Select the desired screen mode.

3.6.10 OxyCRG Window

The OxyCRG functionality is applicable for neonatal monitoring only. The OxyCRG window provides realtime trend curves of HR, SpO_2, RR and compressed waveforms of CO_2 or Resp for patients being monitored by the CMS.

3.6.10.1 Entering the OxyCRG Window

To enter the OxyCRG window, follow this procedure:

1. Enter the ViewBed screen.
2. Enter the OxyCRG window in either of the following ways:

- Select the **OxyCRG** quick key at the bottom of the screen.
- Select the **Screen Setup** quick key at the bottom of the screen → select **Choose Screen** → select **OxyCRG**.

3.6.10.2 Setting the OxyCRG Window

To set the **OxyCRG window**, follow this procedure:

1. In the OxyCRG window, select **Setup** to enter the **OxyCRG Setup** menu.
2. Select the **Parameter** tab and set the desired parameters.
3. Select the **Scale** tab and set the desired scales. If you wish to use the default scales, select the × button to exit the setup menu. Then select **Auto Scale**.
4. Exit the setup menu.
5. Select the time period to view OxyCRG data in the **Zoom** field.

3.6.10.3 Printing an OxyCRG Report

To print an OxyCRG report, follow this procedure:

1. In the OxyCRG window, set the desired items.
2. Select the button in the upper left corner of the window.

3.6.10.4 Entering the OxyCRG Review Page

The OxyCRG review page provides up to 48-hour trend curves and compressed waveform data. To enter the OxyCRG review page, select **Review** in the OxyCRG window.

3.6.10.5 Exiting the OxyCRG Window

Exit the OxyCRG window in either of the following ways:

- Select the **Screen Setup** quick key at the bottom of the ViewBed screen → select **Choose Screen** →select the desired screen mode.
- In the OxyCRG window, select the × button.

3.6.11 ECG 24h Summary

When the CMS is connected with a BeneVision N series monitor, you can view the patient's 24-hour ECG summary from the CMS.

The ECG 24h Summary provides ECG statistics of the current patient over the latest 24 hours. You can view the following information through the ECG 24h Summary:

- Heart rate statistics
- Arrhythmia event statistics
- QT/QTc measurement statistics
- Maximum and minimum ST statistics of each lead
- Pacer statistics
- Typical ECG strips

To view the ECG 24h Summary, select the **ECG 24h Sum** quick key.

NOTE

- **The ECG 24h Summary is intended for the current patient. It is not intended for discharged patients.**
- **Pacer statistics is intended for paced patients.**
- **Patient data is saved, collected and displayed together in the ECG 24h Summary. Data displayed in the ECG 24h Summary is not recalculated.**
- **A license is required for the ECG 24h Summary function.**

3.6.12 Pace View

When the CMS is connected with a BeneVision N series monitor, you can access the patient's pace view from the CMS.

Pace View helps you view pace pulse details, including amplitude, width, shape, and duration.

To access Pace View, select the **Pace View** quick key.

NOTE

- **The Pace View function is intended for patients with implanted pacemaker. It is available only when Paced is set to Yes.**
- **A license is required for the Pace View function.**

3.7 Operating Mode

When monitoring devices support standby mode, night mode, or privacy mode, and remote control for these modes has been set at the CMS, you can control corresponding monitoring devices to enter or exit these modes either via the monitoring devices or via the CMS.

This section only describes how to put monitoring devices into or out of these modes via the CMS. For details on how to set remote control for these modes, see Section *12.12.8 Setting Remote Control*. For details on how to put monitoring devices into or out of these modes via the monitoring devices and how monitoring devices behave in these modes, see corresponding operator's manuals for the monitoring devices.

3.7.1 Standby Mode

When monitoring devices are connected to the CMS and are in the monitoring mode, if you need to temperately stop patient monitoring without switching off the devices, you can use standby mode.

NOTE

- **For the TM80 telemetry monitor, you can put them into or out of standby mode via the TM80 only.**
- **For TMS-6016, you can put them into or out of standby mode via the CMS only.**

3.7.1.1 Entering Standby Mode

To enter standby mode, follow this procedure:

1. Enter the standby dialog in either of the following ways:
   - ◆ Select the bed number and room number area in the upper left corner of the desired patient sector on the multibed screen → from the drop-down list select the symbol.
   - ◆ On the ViewBed screen, select the **Standby** quick key.
2. Set **Location** optionally to define where the patient is when the monitoring device is in standby mode.
3. Select **OK**.

NOTE

- **Options for Location are configurable. For more information, see Section *12.5.3 Setting Patient Location*.**

3.7.1.2 Changing the Patient Location in Standby Mode

If you need to change the patient's location in standby mode, select **Location** from the Standby screen.

3.7.1.3 Exiting Standby Mode

To exit standby mode, use one of the following ways:

- ■ Select the bed number and room number area in the upper left corner of the desired patient sector on the multibed screen → from the drop-down list select the symbol.

■ Select **Resume Monitor** in the Standby window.

3.7.2 Night Mode

The night mode is a special clinical monitoring mode. For how a patient monitor behaves in night mode, see the patient monitor's operator's manual.

3.7.2.1 Entering Night Mode

To enter night mode, follow this procedure:

1. Enter night mode dialog in either of the following ways:
   - ◆ Select the bed number and room number area in the upper left corner of the desired patient sector on the multibed screen → from the drop-down list select **Night Mode**.
   - ◆ On the ViewBed screen,select the **Night Mode** quick key.
2. Select whether to put current bedside monitor or all the bedside monitors into night mode.
3. Select **OK**. The bedside monitors selected will enter night mode. The symbol will be displayed in the upper right corner of the patient sectors for the bedside monitors which are in night mode.

3.7.2.2 Exiting Night Mode

To exit night mode, follow this procedure:

1. Enter the night mode dialog in either of the following ways:
   - ◆ Select the bed number and room number area in the upper left corner of the desired patient sector on the multibed screen → from the drop-down list select **Night Mode**.
   - ◆ On the ViewBed screen,select the **Exit Night Mode** quick key.
2. Select whether to put current bedside monitor or all the bedside monitors out of night mode.
3. Select **OK**.

NOTE

- **A bedside monitor must be connected to the CMS to be allowed to enter night mode. When a bedside monitor in night mode is disconnected from the CMS, it will automatically exit night mode.**

3.7.3 Privacy Mode

The privacy mode is a special clinical monitoring mode. In privacy mode, the bedside monitor does not display patient information and monitoring data. This provides controlled access to patient data and ensures confidentiality. In privacy mode, the bedside monitor continues monitoring the patient, but patient data is only visible at the CMS.

3.7.3.1 Entering Privacy Mode

To enter privacy mode, follow this procedure:

1. Enter privacy mode dialog in either of the following ways:
   - ◆ Select the bed number and room number area in the upper left corner of the desired patient sector on the multibed screen → from the drop-down list select **Privacy Mode**.
   - ◆ On the ViewBed screen, select the **Privacy Mode** quick key.
2. Select whether to put current bedside monitor or all the bedside monitors into privacy mode.
3. Select **OK**. The bedside monitors selected enter privacy mode. The symbol is displayed in the upper right corner of the patient sectors for these bedside monitors.

WARNING

- **In privacy mode, all audible alarms are suppressed and the alarm light is deactivated at the bedside monitor. Alarms are presented only at the CMS.**

3.7.3.2 Exiting Privacy Mode

To exit privacy mode, follow this procedure:

1. Enter the privacy mode dialog in either of the following ways:
   - Select the bed number and room number area in the upper left corner of the desired patient sector on the multibed screen → from the drop-down list select **Privacy Mode**.
   - On the ViewBed screen, select the **Privacy Mode** quick key.
2. Select whether to put current bedside monitor or all the bedside monitors out of privacy mode.
3. Select **OK**.

NOTE

- **If the bedside monitor is connected to the CMS, it automatically exitsprivacy mode when disconnected from the CMS.**

3.8 Defibrillator's Self-Test Reports

At the CentralStation, you can view and print a self-test report sent by a defibrillator. The self-test report contains information such as test time, battery power, and test result.

NOTE

- **Only the CentralStation provides the defibrillator self-test report functionality.**

3.8.1 Viewing Defibrillator's Self-Test Reports

To view a defibrillator's self-test report, follow this procedure:

1. Select the CentralStation's system menu area in the upper left corner of the screen.
2. From the drop-down list, select **Defibrillator Management**. The list of defibrillators and a self-test summary report for each defibrillator are displayed. The following symbols or texts may appear under the items of a summary report:
   - /: indicates that the self-test for this item does not start yet.
   - **Failed**: indicates that the self-test for this item failed.
   - **Pass:** indicates that the self-test for this item succeeded.
   - : indicates that current self-test for this defibrillator failed.
   - : indicates that current self-test for this defibrillator succeeded.
3. If you need to view more self-test details for the desired defibrillator, select the **More** button in the upper right corner of this defibrillator.

3.8.2 Printing Defibrillator's Self-Test Reports

You can manually print defibrillator's self-test reports or set the printer to start printing defibrillator's self-test reports automatically.

For details on how to manually print a self-test report, see *9.2.1.15 Printing Defibrillator's Self-Test Reports*. For details on how to automatically start printing a self-test report, see *9.2.2.3 Setting Automatic Printing for Defibrillator's Self-Test Reports*.

3.8.3 Confirming Defibrillator Self-Test Failure

In case of defibrillator self-test failure, the high priority technical alarm message "*****Defibrillator Test Failed (X)**" is displayed in the system alarm area at the top of the CentralStation screen. The "X" in the alarm message stands for the number of defibrillators whose self-test failed. For example, ***Defibrillator Test Failed (3) indicates that the self-test for three defibrillators failed.

If you already know that the self-test for some or all the defibrillators failed and do not need to be prompted by the technical alarm message any more, you can confirm the alarm message.

To confirm the alarm message, follow this procedure:

1. Select the CentralStation's system menu area in the upper left corner of the screen.
2. From the drop-down list, select **Defibrillator Management**.
3. Select the **Confirm** button in the upper right corner of the desired defibrillator. After confirmation, the **Confirm** button is not displayed.

When you confirm all the defibrillators whose self-tests failed, the alarm message "*****Defibrillator Test Failed (X)**" is not displayed in the system alarm area. But you can view this alarm in the **Technical Alarms** tab on the **Alarm List** screen. For details on how to access this screen, see *5.3.1 Viewing Alarms Triggered from All Monitoring Devices*.

For details on how to take corrective measures when this technical alarm occurs, see *B.2 Alarm Messages in the System Alarm Area*.

3.9 Viewing System Information

To view system information, follow this procedure:

1. Select the system menu area mindray in the upper left corner of the screen.
2. From the drop-down list, select **System Information**.

3.10 Turning Off the System

It is important to properly turn off the system. To turn off the system, follow this procedure:

1. Select the system menu area mindray in the upper left corner of the main screen.
2. From the drop-down list, select **Shutdown**. The system checks if any patients are being monitored:
   - If no patients are being monitored, go to the Step 3.
   - If patients are still being monitored, the message **N patients are being monitored. Are you sure you want to shut down??** (where N stands for the number of patients) is displayed. Select **Yes**.
3. Enter the shutdown password in the **Shutdown** dialog and select **OK**. The password is the same as that for accessing system setup tabs. Then, the CMS and the operating system will automatically shut down. Turn off the UPS if it is in use.
4. Disconnect the power cord of each device from the power outlet.

4 Patient Management

4.1 Patient Management Overview

The CentralStation, the WorkStation, and the ViewStation provide different patient management capabilities.

- The CentralStation and the WorkStation provide the following capabilities:
  - Obtaining patient information
  - Viewing patient information
  - Changing patient information
  - Searching for patients
  - Discharging patients
  - Transferring patients
- The ViewStation only provides the capability of viewing patient information.

4.2 Admitting Monitoring Devices at the CentralStation

The first step on monitoring a patient on the CMS is to admit monitoring devices. Only the CentralStation can admit monitoring devices. After admitting monitoring devices, the CentralStation can display the patient's physiological parameters, waveforms, alarm messages, and store the corresponding patient data.

The CentralStation can be set to automatically admit monitoring devices. Or monitoring devices can be manually admitted.

4.2.1 Automatically Admitting Monitoring Devices

For details on how to admit monitoring devices at the CentralStation, see Section *12.6.2.1 Automatically Admitting Monitoring Devices*.

4.2.2 Manually Admitting Monitoring Devices

A monitoring devices can be manually admitted in either of the following ways:

- Admit monitoring devices on the **Device Assignment** page. Using this method, a monitoring device is bound with a patient sector on the multibed screen. After discharging a patient, the patient's bed number is still displayed in the patient sector. When the monitoring device is online again, the patient data from this monitoring device will be displayed in the same patient sector. This method is appropriate for devices that are frequently used in a department and are less likely to be moved to another department.
- Admit monitoring devices on the multibed screen. Using this method, a monitoring device is not bound with a patient sector on the multibed screen. After discharging a patient, the patient sector is in idle status. You can select the + button to admit a monitoring device. This method is appropriate for devices that are used temporarily.

4.2.2.1 Admitting Monitoring Devices on the Device Assignment Page

For details on how to admit monitoring devices on the device assignment page, see Section *12.6.2.2 Manually Admitting Monitoring Devices on the Device Assignment Page*.

4.2.2.2 Admitting Monitoring Devices on the Multibed Screen

To admit monitoring devices on the multibed screen, follow this procedure:

1. Select a patient sector with the + symbol on the multibed screen to access the **Device Assignment** screen.
2. Select the button on the right of **Department** in the upper right corner of the screen.

3. From the drop-down list, select the desired department.
   - **Local**: displays monitoring devices in the local CentralStation.
   - Department where a remote CentralStation is located: displays monitoring devices from the remote CentralStation. If you wish to overview a bed that is currently being monitored by a remote CentralStation, you can select the desired department. For details on viewing other patients, see *11.3 Viewing Other Patients*.
4. Admit monitoring devices.
   - To assign a monitoring device to a specific patient sector, select the patient sector on the left side of this page. If a monitoring device has been assigned to this patient sector, disconnect this monitoring device from the CentralStation by selecting the ✕ button. Then select the + symbol on the right side of the desired monitoring device to admit this device.
   - To assign monitoring devices to the next available patient sector, select the + symbol on the right side of the desired monitoring devices to admit these devices. The devices will be assigned to idle patient sectors on the multibed screen in the order of left to right and from top to bottom.

NOTE

- **Within the LAN, a monitoring device can be admitted to any CentralStation, but can be admitted by one CentralStation only.**

4.3 Assigning Monitoring Devices from the Host CentralStation to the WorkStation/Viewstation

If you wish to perform patient management at the WorkStation or view patient information at the ViewStation, you need to assign monitoring devices from the host CentralStation to the WorkStation or the ViewStation first. You can assign monitoring devices manually or automatically.

For details on how to assign monitoring devices from the host CentralStation to the WorkStation/ViewStation, see Section *12.6.2.4 Assigning Monitoring Devices on the Device Assignment Page*.

NOTE

- **The WorkStation and the ViewStation must be connected to the host CentralStation and have bed authorization before they can admit a monitoring device from the host CentralStation. For information regarding connect ing the host CentralStation and granting bed authorization, see Section *12.9.6 Setting CentralStation Connection* and *12.9.7 Setting Bed Authorization*.**

4.4 Assigning an Overview Bed to a Patient Sector

An overview bed refers to the bed that is currently being monitored by a remote CentralStation. You can assign an overview bed to a patient sector on the local CentralStation/WorkStation/ViewStation. For details on how to assign an overview bed, see *11.3 Viewing Other Patients*.

4.5 Accessing the Patient Management Screen

After admitting or assigning monitoring devices, you can manage patients on the patient management screen at the CentralStation or the WorkStation.

Enter the patient management screen in either of the following ways:

- Select the patient information area in the desired patient sector on the multibed screen.
- On the ViewBed screen, the review screen, or the calculation screen, select the **Patient Management** tab.

4.6 Patient Information Management

You can enter patient information, edit patient information, and search a patient.

4.6.1 Entering Patient Information

If a monitoring device is admitted by the CentralStation, you can enter patient information either at the CentralStation or at the WorkStation.

Enter patient information on the patient management screen in one of the following ways:

- Load patient information from the hospital information system.
- Import discharged patient information.
- Scan patient information via a barcode reader.
- Enter patient information manually.

4.6.1.1 Loading Patient Information by ADT Query

When the CentralStation and the WorkStation are connected with a hospital information system, you can utilize the admit-discharge-transfer (ADT) query function to obtain patient information from the hospital information system and import the patient information to the CentralStation or the WorkStation.

To load patient information, follow this procedure:

1. On the patient management screen, select the **Find Patient** button.
2. Select the **ADT** tab.
3. Input query criteria.
4. Select the button.
5. Select the button at the end of the desired patient to import the patient information to the CentralStation or the WorkStation.

NOTE

- **The ADT tab is available only when the ADT feature is enabled in the system setup. For details on how to enable this feature, see Section *12.9.4.1 Setting ADT Query*.**
- **Loading patient information from the hospital information system updates only patient information in the monitoring device. The patient's monitoring data is not changed and the patient is not discharged.**

4.6.1.2 Importing Discharged Patient Information

To import discharged patient information, follow this procedure:

1. On the patient management screen, select the **Find Patient** button.
2. Select the **Discharged Patients** tab. The Discharged Patients will only be displayed within a tab when the ADT feature is enabled in the system setup. If the ADT feature is disabled, there is no tab to select. Proceed to the next step.
3. Input query criteria.
4. Select the button.
5. Select the button at the end of the desired patient.
6. When the **Readmit Patient** dialog appears, decide whether to import **History Data**.
   - Select **History Data** when you wish to merge the patient demographic information and historical data into current data for this patient.
   - Do not select **History Data** when you wish to merge only the patient demographic information into current data for this patient.
7. Select the **OK** button.

4.6.1.3 Manually Entering Patient Information

To manually enter patient information, follow this procedure:

1. On the patient management screen, enter or select the desired patient information fields. Only fields that may need special remarks are described here.
   - **Patient Category**: set the patient category corresponding to the patient being monitored.
   - **Paced**: set the patient's paced status. For paced patients, you must set **Paced** to **Yes**. If it is incorrectly set to **No**, the monitoring device could mistake a pace pulse for a QRS and fail to alarm when the ECG signal is too weak. Always keep pacermaker patients under close surveilance.
   - **Care Group**: after assigning a patient to a care group, the bed number and room number area in the upper left corner of the patient sector on the multibed screen is filled with the selected care group color. This allows a caregiver to quickly identify beds for which they are responsible. For details on how to customize the care group, see Section *12.5.5 Setting Care Group*.
   - **Patient Group**: after assigning a patient to a patient group, the starting portion of the patient information area on the multibed screen is filled with the selected patient group color. This allows you to quickly identify patient types. For details on how to customize the patient group, see Section *12.5.4 Setting Patient Group*.
   - **Notes**: you can enter additional information. The text entered is displayed in the notes area in the patient sector on the multibed screen.
2. When done, verify that all the patient information items are correct.
3. Select the **Save** button.

NOTE

- **The items displayed on the patient management screen are dependent on the system settings. For details on how to configure these items, see Section *11.5.2 Configuring Patient Fields*.**

4.6.2 Changing Patient Information

When you change the patient information at the CentralStation or the WorkStation, the information is also changed at the monitoring devices.

This section only describes how to change patient information at the CentralStation or the WorkStation. For details on how to change information at the monitoring devices, see corresponding operator's manuals for these devices.

To change patient information, follow this procedure:

1. On the patient management screen, change the patient information in the appropriate fields as necessary.
2. After finishing changing, verfify that all the patient information items are correct.
3. Select the **Save** button.

4.6.3 Searching Patient Information

You can search discharged patients in the HIS system or in a CentralStation. After finding the desired patient information, you can perform operations such as importing and reviewing the patient information. The searching patients feature is available at the CentralStation and the WorkStation.

4.6.3.1 Searching Patient Information by ADT

To search patient information in the HIS system by ADT, follow steps 1 to 4 in Section *4.6.1.1 Loading Patient Information by ADT Query*.

4.6.3.2 Searching Discharged Patients

You can search discharged patients on the patient management screen and on the discharged patients management screen. This section only describes how to search discharged patients on the patient management screen. For details on how to search discharged patients on the discharged patients management screen, see Section *4.9.2 Viewing Discharged Patient Information*.

To search discharged patients, follow steps 1 to 4 in Section *4.6.1.2 Importing Discharged Patient Information*.

4.6.4 Patient Information Synchronization

When monitoring devices are connected to the CentralStation/ WorkStation, any changes to the patient information at either the CentralStation/ WorkStation, or the monitoring devices will be synchronized to all displayed locations to ensure that the patient information is consistent.

When monitoring devices are disconnected from the CentralStation/ WorkStation and then reconnected, patient information from the monitoring devices will be updated to the CentralStation/ WorkStation.

4.7 Discharging A Patient

Before monitoring a new patient, discharge the previous patient. This ensures that data from a previous patient is not mixed with the data from the new patient.

This section only describes how to discharge patients at the CentralStation/ WorkStation. Patients can also be discharged from a bedside monitor or from a TD60 transmitter or a TM80 telemetry monitor. For details on how to discharge patients at the monitoring devices, see corresponding operator's manuals for these devices.

To discharge a patient,follow this procedure:

1. Access the **Discharge Patient** dialogin either of the following ways:
   - On the patient management screen, select the **Discharge Patient** button.
   - Select the bed number and room number area in the upper left corner of the desired patient sector on the multibed screen and then select **Discharge Patient** from the drop-down list.
2. If you need to print an end case report, select **Print End Case Report**.
3. Select whether to let this monitoring device enter standby mode after the patient is discharged.
4. Select the **OK** button.

4.8 Transferring Patient Data

You can transfer a patient from one bed (source bed)to another bed (destination bed) without losing patient data.

This section only describes how to transfer patients from the CentralStation/WorkStation. For details on how to transfer patient data at the bedside monitors, see corresponding operator's manuals for these devices.

To transfer patient data, follow this procedure:

1. On the patient management screen, select the **Transfer Patient** button to enter the **Transfer Patient** window
2. Select the destination unit under **Select the destination department:**.
   - **Local**: transfers the patient data to the destination bed in the current CentralStation/WorkStation.
   - **Other**: transfers the patient data to the destination bed in the target CentralStation. After selecting this option, select the target CentralStation name from the list box on the right of **Other**.
3. In the **Select the destination bed:** section, select the desired destination bed from the bed list.
4. Select the **OK** button.
5. In the **Transfer Patient** dialog box, select the **OK** button.
6. In the dialog box that prompts successful patient transfer, select the **OK** button.

Usually, data transfer takes about several minutes. The larger the amount of patient data, the longer time required by data transfer. Upon successful transfer, the source bed's patient sector displays **Discharged**. The **Transferring Data... xx** (where xx stands for the data transfer completion percentage) prompt message is displayed in the alarm information area of the destination patient sector, indicating merging patient data from the source bed to the destination bed is in progress. If the data transfer fails, the F**ailed to transfer the patient.** prompt message is displayed.

NOTE

- **When the system time at the source CentralStation/WorkStation and target CentralStation is not consistent, transferring a patient may cause patient data loss.**
- **Ensure that the same patient is monitored by both the source and destination beds. Otherwise, discharge the patient in the destination patient sector first.**

NOTE

- **When a patient is transferred from one bed to another bed, the source CentralStation/WorkStation will interrupt patient monitoring temporarily.**
- **If there is a version difference between two CentralStations, patient transfer may not be supported.**

4.9 Discharged Patients Management

On the discharged patients management screen, you can perform operations such as viewing the demographc information and historical data for discharged patients.

4.9.1 Accessing the Discharged Patients Management Screen

To access the discharged patients management screen, follow this procedure:

1. Select the system menu area ≡ mindray in the upper left corner of the screen.
2. From the drop-down list, select **Discharged Patients**.

4.9.2 Viewing Discharged Patient Information

To view discharged patient information, follow this procedure:

1. Select the target CentralStation from the button in the upper left corner of the discharged patients management screen.
   - **Local Central Station:** view discharged patients in current station.
   - Name of other central stations: view discharged patients in other central stations.
2. Input query criteria. If you need to specify more criteria, select the **Filter** button and input more criteria.
3. Select the **Search** button.
4. Select the **Detail** button in the desired patient information.After selecting this button, you can view the patient information and historical data in the **Patient Management** tab and **Review** tab respectively.

4.9.3 Deleting Discharged Patient Information

To view delete patient information, follow this procedure:

1. Select the target CentralStation from the button in the upper left corner of the patient management screen.
   - **Local Central Station:** view discharged patients in current station.
   - Name of other central stations: view discharged patients in other central stations.
2. Select the desired patient information. If you wish to select all the patient information, select the **Select All** button.
3. Select the **Delete** button.
4. In the dialog box that prompts whether to delete, select the **OK** button.

4.9.4 Printing Historical Data of Discharged Patients

After entering the review screen of discharged patients, you can print historical data of discharged patients.

For details on how to enter the review screen of discharged patients, see Section *6.2.2 Entering the Review Screen for Discharged Patients*.

For details on how to print historical data of discharged patients, see Chapter *6 Review*.

5 Alarms

5.1 Alarm Introduction

This chapter describes alarm functions and alarm settings.

5.2 Understanding the Alarms

Alarms triggered by a vital sign that appears abnormaly or due to technical problems identified by the monitoring devices are sent to the CMS. The CMS provides visual and audible alarm indications.

Most of the alarms come from monitoring devices. For alarms detected by the CMS, see Appendix *B CMS Alarm Messages*.

5.2.1 Alarm Categories

The CMS' alarms can be classified into two categories: physiological alarms and technical alarms.

- Physiological alarms, also called patient status alarms, are triggered by a monitored parameter value that violates set alarm limits or an abnormal patient condition. Physiological alarm messages are displayed in the physiological alarm area.
- Technical alarms, also called system status alarms, are triggered by a device malfunction or a patient data distortion due to improper operation or mechanical problems. Technical alarm messages are displayed in the technical alarm area.

Apart from the physiological and technical alarm messages, the CMS will show some messages telling the system status.

5.2.2 Alarm Priorities

By severity, the alarms are classified into the following priority levels:

- High priority alarms: indicate a life threatening situation or a severe device malfunction. High priority alarms require an immediate response.
- Medium priority alarms: indicate abnormal vital signs or a device malfunction. Medium priority alarms require a prompt response.
- Low priority alarms: indicate a discomfort condition, a device malfunction, or an improper operation. Low priority alarms require you to be aware of this condition.
- Prompts: provide additional information on the patient or the system status.

5.2.3 Alarm Indicators

When an alarm occurs, the CMS gives different visual or audible alarm indications as shown below.

Alarm Indicator		High Priority Alarm	Medium Priority Alarm	Low Priority Alarm	Message	Comments
Alarm tone pattern	ISO	Repeat pattern of 2 × 5 beep tones	Repeat pattern of 3-beep tones	1-beep tone	None	The alarm tones are issued at different frequencies.
	Mode 1	Repeat pattern of high-pitched 3-beep tones	Repeat pattern of 2-beep tones	Low-pitched 1-beep tone	None	
	Mode 2	Repeat pattern of high-pitched 3-beep tones	Repeat pattern of 2-beep tones	Low-pitched 1-beep tone	None	
Alarm message		White text inside a red box	Black text inside a yellow box	Black text inside a cyan box	White text	Alarm messages are displayed in the alarm information area on the ViewBed screen and in the patient sector on the multibed screen. You can set whether to let the background color of high priority and medium priority alarm messages flash. For details, see Section *12.4.3 Configuring Other Alarm-Related Items*.
Alarm priority indicator		***	**	*	None	The indicator is displayed in front of a corresponding alarm message.
Parameter value		Black text inside a flashing red box	Black text inside a flashing yellow box	Black text inside a flashing cyan box	None	None

NOTE

- **Alarm time is displayed for latched physiological alarms only.**
- **The audible and visual alarms given by the CMS comply with the IEC 60601-1-8 standard. The hospital or institution employing the use of the CMS should give adequate training to the operators.**
- **When multiple alarms of different priority levels occur simultaneously, the CMS selects the alarms of the highest priorities and provides alarm tones accordingly.**
- **When multiple alarms of the same priority levels occur simultaneously and are displayed in the same area, all the alarm messages will be scrolled in the display area.**
- **Some physiological alarms, such as asystole, are exclusive. They have identical alarm tones with normal high level physiological alarms, but their alarm messages are displayed exclusively. That is to say, when an exclusive physiological alarm and a normal high level physiological alarm are triggered simultaneously, only the alarm message of the exclusive physiological alarm is displayed.**

5.2.4 Alarm Status Symbols

Apart from the alarm indicators as described in *5.2.3 Alarm Indicators*, the following symbols are displayed on the screen to indicate the alarm status:

- : the alarm pause symbol, indicating that all the alarms are paused.
- : the alarm off symbol, indicating that individual measurement alarms are turned off or the system is in the alarm off status.

- : the audio pause symbol, indicating that audible alarm are paused.
- : the audio off symbol, indicating that audible alarm are turned off.
- : the alarm reset symbol, indicating that alarms are acknowledged and the alarm system is reset.

5.3 Viewing Alarms Triggered from Monitoring Devices

You can view physiological and technical alarms from all monitoring devices and from a single monitoring device that is being monitored by the CMS.

5.3.1 Viewing Alarms Triggered from All Monitoring Devices

You can view alarms from all the monitoring devices that occurred during a recent time period.

To do so, follow this procedure:

1. Select the alarm list area at the top of the main screen.
2. Select the time period in the upper right corner of the **Alarm List** screen.
3. Select the **Physiological Alarms** or **Technical Alarms** tab to view physiological alarms or technical alarms.
4. Select the **Detail** button on the right side of the desired alarm message to access the event review page and view the alarm details.

5.3.2 Viewing Alarms Triggered from A Single Monitoring Device

To view alarms from a single monitoring device, follow this procedure:

1. Open the alarm list window for a monitoring device in either of the following ways:
   - Select the alarm information area in the patient sector of this monitoring device on the multibed screen.
   - Select the physiological or technical alarm information area on the ViewBed screen of this monitoring device.
2. Select the desired alarm message to access the event review page and view the alarm details.

5.4 Clearing Alarms in the Alarm List

To clear physiological or technical alarms in the alarm list, follow this procedure:

1. Select the alarm list area at the top of the main screen.
2. Select the symbol in the upper right corner of the **Alarm List** screen. This clears all the physiological and technical alarms.

NOTE

- **Selecting the symbol only clears all the physiological and technical alarms in the alarm list window, but does not clear these alarms stored in the central monitoring system.**

5.5 Setting Alarm Properties

5.5.1 Setting Parameter Alarm Properties

You can set alarm properties for parameters collectively or individually.

5.5.1.1 Setting Parameter Alarm Properties Collectively

To set alarm properties for all the parameters collectively, follow this procedure:

1. Enter the **Alarm Setup** menu in either of the following ways:
   - Select the bed number and room number area in the upper left corner of the desired patient sector on the multibed screen → from the drop-down list select **Alarm Setup**.

- Select the **Alarm Setup** quick key at the bottom of the ViewBed screen. If this quick key is not available, configure it by following steps in *3.6.4.2 Configuring Quick Keys*.

2. Select the **Limits** tab.
3. Switch on or off alarms and set alarm limits, alarm priority, and alarm outputs.
   - When **Alarm Outputs** is disabled, alarm information for this parameter will not be printed automatically.
   - When **Alarm Outputs** is enabled and the parameter alarm is switched **on**, once an alarm is triggered, parameters and waveforms currently monitored at the alarm trigger time will be printed automatically. For details on how to configure the waveform printing duration, see *12.4.3 Configuring Other Alarm-Related Items*.

CAUTION

- **When changing the alarm limits of a monitoring device via the CentralStation or the WorkStation, make sure that the alarm limits settings are appropriate for your patient. Setting alarm limits to extreme values may cause the alarm system to become ineffective.**

NOTE

- **When Alarm Setup is selected on the remote control setup page, alarm settings can be changed either via the CentralStation/WorkStation or the monitor. Alarm changes made will be synchronized to all locations to ensure consistency. For detailed information on the remote control setup page, see Section*12.12.7 Setting Remote Control*.**
- **When Alarm Setup is not selected on the remote control setup page, alarm settings cannot be changed via the CentralStation/WorkStation. Besides, the symbol is displayed beside a parameter's alarm on/off option on the ViewBed screen.**
- **Alarm settings can only be viewed at the ViewStation. But they cannot be modified.**

5.5.1.2 Setting Parameter Alarm Properties Individually

To set alarm properties for a parameter, follow this procedure:

1. Access the ViewBed screen in either of the following ways:
   - Select the parameter area or waveform area in the desired patient sector on the multibed screen to access the ViewBed screen.
   - Select the ViewBed symbol when it is displayed in the desired patient sector on the multibed screen.
2. Select the parameter area or waveform area of the desired parameter.
3. Select the **Alarm** tab and perform settings.

5.5.2 Changing Arrhythmia Alarm Settings

To change arrhythmia alarm settings, follow this procedure:

1. Enter the arrhythmia alarm setup menu in one of the following ways:
   - Select the bed number and room number area in the upper left corner of the desired patient sector on the multibed screen → from the drop-down list select **Alarm Setup** → select the **Arrhythmia** tab.
   - Select the **Alarm Setup** quick key at the bottom of the ViewBed screen → select the **Arrhythmia** tab.
   - Select the ECG parameter area or waveform area on the ViewBed screen to enter the ECG setup menu → select the **Arrhythmia** tab.
2. Change alarm settings
   - Enable or disable arrhythmia alarms: you can enable arrhthmia alarms separately or collectively. If you need to enable these alarms collectively, select **All On**. If you need to disable all the arrhythmia alarms, select **All Off**. If you need to enable only the lethal arrhythmia alarms, select **Lethals Only**.
   - Set alarm priorities.
   - Set **Alarm Outputs**. For details on the options, see Section *5.5.1.1 Setting Parameter Alarm Properties Collectively*.

3. Select the **Threshold** tab and adjust as needed.

NOTE

- **If a bedside monitor is connected and the Lethal Arrhy Off option in the monitor's user maintenance menu is set to Disable, the All Off button in the arrhythmia alarm setup menu in the CMS is inactive.**
- **If a telemetry device is connected and the Lethal Arrhy Alarms Off button on the telemetry setup page is disabled, the All Off button in the arrhythmia alarm setup menu in the CMS is inactive.**
- **For information on the arrhyhtmia alarm settings and threshold settings, see operator's manuals for monitoring devices.**

5.5.3 Changing ST Alarm Settings

To change ST alarm settings, follow this procedure:

1. Enter the ST alarm setup menu in one of the following ways:
   - Select the bed number and room number area in the upper left corner of the desired patient sector on the multibed screen → from the drop-down list select **Alarm Setup** → select the **ST** tab.
   - Select the **Alarm Setup** quick key at the bottom of the ViewBed screen → select the **ST** tab.
   - Select the ECG parameter area or waveform area to enter the ECG setup menu → select the **ST** tab.
2. Change alarm settings.

5.5.4 Setting Alarm Volume

To set the alarm volume, follow this procedure:

1. Select the system menu area mindray in the upper left corner of the screen→ from the drop-down list select **System Setup** to access the **System Setup** menu.
2. Select the **General** tab to access the general setup page.
3. Select the **Volume** tab.
4. Set **Alarm Volume**. The optional alarm volume is between X to 10, in which X is the minimum volume, depending on the setting of minimum alarm volume, and 10 is the maximum volume. For more information, see Section *5.5.5 Setting Alarm Sound Pattern, Alarm Interval, and Alarm Volume Escalation*.
5. Set **High Alarm Volume**. The volume of the high priority alarm is higher than any other alarm volume. The volume of the high priority alarm is 10 even if it is set above 10.
6. Set **Reminder Volume**.

NOTE

- **Do not rely exclusively on the audible alarm system. Setting the alarm volume to a low level may result in a hazard to the patient.**
- **Exercise caution when setting Alarm Volume to zero. When Alarm Volume is set to zero, the High Alarm Volume is unconfigurable. Also, the alarm sound of the CMS is turned off. The ⊠ symbol is displayed at the top of the main screen. Even when a new alarm is triggered, the CMS will not provide an alarm sound but still provides other alarm indications such as an alarm message.**
- **Setting Alarm Volume to zero affects the CMS only and does not affect the alarm sound at the monitoring devices**

5.5.5 Setting Alarm Sound Pattern, Alarm Interval, and Alarm Volume Escalation

The CMS provides the function of automatically escalating the alarm tone volume. When this function is enabled, if an alarm is not reset within the designated delay time after the alarm occurs, the volume of the alarm tone increases automatically.

You can set the alarm volume escalation, alarm sound pattern, and alarm interval. For more information, see Section *12.4.2 Configuring Alarm Audio Properties*.

5.5.6 Setting Reminder Tones

The reminder tone is enabled by default. When the CMS is in silenced or audio off modeand if active alarms still exist in the system,the CMS issues a reminder tone at the designated reminder interval. For details on how to set alarm reminder, see Section *12.4.2 Configuring Alarm Audio Properties*.

5.5.7 Setting Alarm Latching

The latching setting for physiological alarms defines how alarm indicators behave if you do not reset the alarms. For details on how to reset alarms, see Section *5.8.1 Resetting Alarms*.

- If you do not "latch" physiological alarms, their alarm indications disappear when the alarm condition ends.
- If you "latch" physiological alarms, all visual and audible alarm indications remains until you reset the alarms. For latched alarms the time when the alarm is last triggered is displayed behind the alarm message.

You can set the physiological alarm latching conditions for telemetry devices at the CentralStation. For details, see Section *12.8.2 Setting Alarm Properties for Telemetry Devices*.

NOTE

- **When the alarm system is reset, latched physiological alarms are cleared. For details on how to reset alarms, see Section *5.8 Alarm Reset*.**

5.6 Alarm Pause

You can pause alarms from patient monitors remotely via the CentralStation or the WorkStation under the following conditions:

- The patient monitors support alarm pause via the monitors themselves and via the CentralStation or the WorkStation.
- The **Alarm Pause&Alarm Reset** function is enabled at the CentralStation or the WorkStation. For details on how to enable this function, see Section*12.12.8 Setting Remote Control*.

You can also pause alarms from telemetry devices at the CentralStation or the WorkStation.

5.6.1 Pausing Alarms

You have two ways to pause alarms.

5.6.1.1 Pausing Alarms in the Patient Sector

To pause alarms, follow this procedure:

1. Select the bed number and room number area in the upper left corner of the desired patient sector on the multibed screen.
2. From the drop-down list, select the symbol.

5.6.1.2 Pausing Alarms on the ViewBed Screen

To pause alarms, follow this procedure:

1. Access the ViewBed screen in either of the following ways:
   - Select the parameter area or waveform area in the desired patient sector on the multibed screen to access the ViewBed screen.
   - Select the ViewBed symbol when it is displayed in the desired patient sector on the multibed screen.
2. Select the **Alarm Pause** quick key at the bottom of the screen. If this quick key is not available, configure it by following steps in *3.6.4.2 Configuring Quick Keys*.

The alarm pause time for patient monitors needs to be configured on the monitors. The default alarm pause time for telemetry devices is two minutes. The alarm pause time for telemetry devices is configurable. For details, see Section *12.8.2 Setting Alarm Properties for Telemetry Devices*.

When the alarm pause time expires, the alarm paused status is automatically deactivated. You can also cancel the alarm paused status by selecting the **Alarm Pause** quick key.

5.6.2 System Responses after Pausing Alarms

When alarms are paused, the CentralStation or the WorkStation responds as follows:

- No physiological alarm issued by patient monitors will be presented.
- For technical alarms issued by patient monitors and telemetry devices, alarm sounds are paused, but alarm messages remain presented.
- The symbol, **Alarm Pause** text, and alarm pause countdown time are displayed in the patient sector and ViewBed screen. The background color of the **Alarm Pause** quick key remains red within the alarm pause time.

5.7 Alarm Audio Pause

You can pause the alarm sound issued by BeneVision N series patient monitors remotely via the CentralStation or the WorkStation under the following conditions:

- The alarm pause function is defined as the alarm audio pause at the N series patient monitors side. For details on how to define tha alarm audio pause function, see the operator's manual for N series patient monitors.
- The **Alarm Pause&Audio Pause&Alarm Reset** function is enabled at the CentralStation or the WorkStation. For details on how to enable this function, see *12.12.8 Setting Remote Control*.

5.7.1 Pausing Alarm Sound

To pause the alarm sound, follow this procedure:

1. Access the ViewBed screen in either of the following ways:
   - Select the parameter area or waveform area in the desired patient sector on the multibed screen to access the ViewBed screen.
   - Select the ViewBed symbol when it is displayed in the desired patient sector on the multibed screen.
2. Select the **Audio Pause** quick key at the bottom of the screen.

The audio pause time of N series patient monitors needs to be configured on the monitors. When the audio pause time expires, the audio paused status is automatically deactivated. You can also cancel the audio paused status by pressing the **Audio Pause** quick key.

5.7.2 System Responses after Pausing Alarm Sound

When alarm sound is paused, the CentralStation or the WorkStation responds as follows:

- The sound of all physiological alarms and technical alarms for the patient monitor are switched off but alarm messages are still displayed on the screen.
- The symbol is displayed in the patient sector and on the ViewBed screen. The background color of the **Audio Pause** quick key remains red within the audio pause time.

5.8 Alarm Reset

You can reset alarms from patient monitors remotely via the CentralStation or the WorkStation under the following conditions:

- The patient monitors support alarm reset via the monitors themselves and via the CentralStation or the WorkStation.
- The **Alarm Pause&Audio Pause&Alarm Reset** function is enabled at the CentralStation or the WorkStation. For details on how to enable this function, see *12.12.8 Setting Remote Control*.

You can also reset alarms from telemetry devices at the CentralStation or the WorkStation.

5.8.1 Resetting Alarms

You have two ways to reset alarms.

5.8.1.1 Resetting Alarms in the Patient Sector

Choose one of the following ways to reset alarms in the patient sector:

- Select the alarm information area in the desired patient sector on the multibed screen → select **Alarm Reset**.
- Select the Alarm Reset button in the desired patient sector on the multibed screen.
- Select the bed number and room number area in the upper left corner of the desired patient sector on the multibed screen → from the drop-down list select the symbol.

5.8.1.2 Resetting Alarms on the ViewBed Screen

To reset alarms on the ViewBed screen, follow this procedure:

1. Access the ViewBed screen in either of the following ways:
   - Select the parameter area or waveform area in the desired patient sector on the multibed screen to access the ViewBed screen.
   - Select the ViewBed symbol when it is displayed in the desired patient sector on the multibed screen.
2. Select the **Alarm Reset** quick key at the bottom of the screen. If this quick key is not available, configure it by following steps in *3.6.4.2 Configuring Quick Keys*.

5.8.2 System Responses after Resetting Alarms

When physiological alarms are reset, the following alarm indications are given:

- The alarm sound is silenced.
- The √ symbol appears before the alarm message, indicating that the alarm is acknowledged.
- The color of the parameter numeric background corresponds with the alarm priority, but the parameter numeric does not flash.

When technical alarms are reset, the following alarm indications are given:

- Some technical alarms are cleared and no alarm indications are given.
- Some technical alarms are changed to the prompt messages.
- For some technical alarms, the alarm is silenced and a √ appears before the alarm message, indicating that the alarm is acknowledged.

For details on technical alarms, see appropriate operator's manuals for monitoring devices.

NOTE

- **If a new alarm is triggered after the alarm system is reset, the alarm reset symbol will disappear and visual and audible alarm indications will be reactivated.**

5.9 CMS System Silence

The CMS system sounds include alarm tone, nurse call tone, event sound tone, and other tones. Silencing the CMS only silences the CMS system sounds. It does not affect the alarm sounds at monitoring devices. The CMS system silence feature is disabled by default.

5.9.1 Silencing CMS

Silence the CMS in either of the following ways:

- If the silence hot key is defined in the system setup screen, pressing this hot key silences the system sounds. For details on how to define the silence hot key, see Section *11.4.4 Configuring Other Alarm-Related Items*.

- Select the symbol at the top of the screen.

5.9.2 System Responses after Silencing CMS

When the system sounds are silenced, the following indications are given:

- The symbol is displayed at the top of the main screen.
- Audible alarm indications are not given, but other alarm indications, such as alarm messages, remain being presented.

5.9.3 Exiting CMS Silenced Status

When a new alarm occurs, the system silenced status will be automatically released. You can also press the silence hot key again or select the symbol to exit the system silenced status.

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6 Review

6.1 Overview

You can review historical data for online patients and discharged patients on the review screen. The review screen provides several tabs to display historical data from different aspects. The review tabs for the online patients and discharged patients are the same.

NOTE

- **Review tabs are dependent on the patient category and patient sector settings. For details on how to select the review tabs in the patient sector, see Section *12.3.4 Setting Patient Window*.**

6.2 Accessing the Review Screen

Methods for accessing the review screen for online patients and discharged patients are different.

6.2.1 Entering the Review Screen for Online Patients

Enter the review screen for online patients in either of the following ways:

- Select the bed number and room number area in the upper left corner of the desired patient sector on the multibed screen → from the drop-down list select **Review**.
- On the ViewBed screen, the patient management screen, or the calculation screen, select the **Review** tab.

6.2.2 Entering the Review Screen for Discharged Patients

To enter the review screen for discharged patients, follow this procedure:

1. Select the system menu area mindray in the upper left corner of the screen.
2. From the drop-down list, select **Discharged Patients**.
3. From the discharged patients list, select the **Detail** button at the end of the discharged patient.
4. Select the **Review** tab.

6.3 Example Review Page Structure

Select the ▼ button in the upper left corner of the review screen to enter appropriate review pages. The graphic trends review page is an example to introduce the items that may also be displayed on other review pages.

Figure 6-1 Example Review Page

(1)	Current review tab
(2)	Event type indicator: different color blocks match different types of events: ■ Red: high priority alarm event ■ Yellow: medium priority alarm event ■ Cyan: low priority alarm event ■ Green: manual event ■ White: operation-related event
(3)	Event area/current cursor time area: after you select an event from the event list, this event is displayed in this area. Selecting this area goes to the **Events** review page. If an event is not selected, this area displays current cursor time.
(4)	Scroll bar: indicates the position of currently displayed trend data in all the trend data.
(5)	Cursor: shows your current position in the trend page. When the cursor is moved across the timeline, the values measured at the cursor time are shown in the right column.
(6)	Parameter area: displays numeric values where the cursor stops. The color of numeric values is consistent with the color of parameter labels. The background color of numeric values matches the alarm priority.
(7)	Timeline: indicates the entire time length. ■ : indicates the time length of reviewable trend data. can be moved within this time length. ■ : indicates no patient monitoring. cannot be moved within this time length. ■ Different color blocks at the timeline indicate events of different types. See the color definition for the event type indicator.
(8)	Waveform area: displays trend curves. The color of trend curves is consistent with the color of parameter labels.

6.4 Symbols on Review Pages

The following table lists the symbols on review pages.

Symbol	Description
	Slider: indicates the position of current window time in the entire time length. Dragging the slider left or right enables you to locate the trend data at a specific time and also refreshes trend data in current window accordingly.
/	Go to the previous or next event.
	Event list: displays events in a chronological order. The most recent event is displayed at the top.The number of asterisk symbols before an event matches different alarm priorities. When a parameter label is prefixed with the plus sign (+), it indicates that the data is for external devices connected to the monitor via the BeneLink module.
	This symbol is displayed in the upper right corner of a review page for a landscape display greater than 19-inch. Selecting this symbol displays two review pages simultaneously.
	Print button
	Record button

6.5 Common Operations on Review Pages

This section describes common operations on review pages.

6.5.1 Browsing Trend Data

Browse trend data in one of the following ways:

- Move .
- Move the cursor.
- Move the scroll bar.

6.5.2 Viewing Events

You can view the following types of events:

- Manually triggered events
- Parameter-related operation events and alarm-related events, such as starting C.O. measurement
- Operation events not related to parameters

To view these events, follow this procedure:

1. Select the button in the upper left corner of the review screen.
2. From the drop-down list, select the desired review page.
3. View events in either of the following ways:
   - Select and select the desired event.
   - Select or to go to the previous or next event.

Events are displayed in a chronological order. The most recent event is displayed at the top. The number of asterisk symbols before an event matches the following alarm priority:

- ***: high priority alarm
- **: medium priority alarm
- *: low priority alarm

6.5.3 Displaying Two Review Pages Simultaneously

When the display size of the CMS is greater than 19-inch, you can display two review pages simultaneously.

To do so, follow this procedure:

1. Select the ▼ button in the upper left corner of the review screen.
2. From the drop-down list, select the desired review page.
3. Select the symbol located at the top right of the screen. A new review page will be displayed below the desired review page.

You can browse trend data, view events, select other review tabs on the two review pages. If you refresh the trend data by moving the cursor or on one review page, the trend data on the other review page will be updated accordingly.

NOTE

- **For 19-inch single display, two review pages can not be displayed simultaneously.**

6.6 Tabular Trends Review Page

The tabular trends review page displays trend data in a tabular form. When external devices are connected to the monitor via the BeneLink module, you can also view the trend data of the external devices on this review page. Parameter labels from external devices are prefixed with the plus sign (+).

6.6.1 Entering the Tabular Trends Review Page

To enter the tabular trends review page, follow this procedure:

1. Select the ▼ button in the upper left corner of the review screen.
2. From the drop-down list, select **Tabular Trends**.

6.6.2 Configuring Trends

To configure tabular trends, follow this procedure:

1. Enter the tabular trends review page.
2. Set **Trend Group** and select the desired option. For information on trend group settings, see Section *12.7.2.2 Renaming a Trend Group*.
3. Set **Interval** and select the desired option. Among these options, selecting **NIBP, EWS** (Early Warning Score), **GCS** (Glasgow Coma Scale), **TempIF**, or **C.O.** means to view the tabular trends when NIBP, EWS, GCS, infrared temperature,or C.O. measurements were acquired. The default for Interval in OR is **5 min** and in other departments is **30 min**.

NOTE

- **The options for Interval are dependent on the functions of a bedside monitor. For example, only when the bedside monitor supports the EWS function, EWS is displayed under Interval.**
- **The parameters displayed for Trend Group are dependent on the parameters configured at a bedside monitor.**

6.6.3 Printing a Tabular Trends Report

To print a tabular trends report, follow this procedure:

1. Enter the tabular trends review page.
2. Select the button in the upper right corner of this page. The **Print Setup** menu is displayed.
3. Set the desired options. Only items that may need special remarks are described here.

4. Select **Print**.

Menu Item	Description	Options
Period	Select the period during which a tabular trends report will be printed.	30 min, 1 hr, 2 hrs, 4 hrs, 8 hrs, 12 hrs, 24 hrs, 48 hrs, 72 hrs, 96 hrs, Auto, All **Auto**: print one page of a tabular trend report according to the interval and period selected.
Interval	Select the resolution of the tabular trends printed on a report.	5 sec, 30 sec, 1 min, 5 min, 10 min, 15 min, 30 min, 1 hr, 2 hrs, 3 hrs, NIBP, EWS, GCS, TempIF, C.O., Auto **NIBP, EWS**, **GCS**, **TempIF**, **C.O.**: print a tabular trends report when NIBP, EWS, GCS, infrared temperature, or C.O. measurements were acquired. Each NIBP, EWS, GCS, infrared temperature, or C.O. measurement will generate a column on the tabular trends review page. The values for the other measurements are added to the column to provide a complete dataset for the NIBP, EWS, GCS, infrared temperature, or C.O. measurement time. **Auto**: print a tabular trends report at the interval set in the **Interval** option in the upper right corner of the **Tabular Trends** review page.
Report Format	Select the printing principle.	**Parameter Oriented**: print one page of report by parameters when **Interval** is set to **Auto**. **Time Oriented**: print one page of report by timewhen **Interval** is set to **Auto**.

6.7 Graphics Trends Review Page

The graphic trends review page displays trend data in a graphic form. When external devices are connected to the monitor via the BeneLink module, you can also view the trend data of the external devices on this review page. Parameter labels from external devices are prefixed with the plus sign (+).

6.7.1 Entering the Graphic Trends Review Page

To enter the graphic trends review page, follow this procedure:

1. Select the ▼ button in the upper left corner of the review screen.
2. From the drop-down list, select **Graphic Trends**.

6.7.2 Setting Graphic Trends

To set graphic trends, follow this procedure:

1. Enter the graphic trends review page.
2. Set **Trend Group** and select the desired option. For information on trend group settings, see Section *12.7.2.2 Renaming a Trend Group*.
3. Set **Zoom**. Select the desired length of trend data displayed on the current screen.
   - **8 min:** the screen displays eight minutes of trend data. You can view the most recent one hour data.
   - **30 min, 1 hr, 2 hrs, 4 hrs**: the screen displays the selected length of trend data. You can view the most recent four hour data.
   - **8 hrs, 12 hrs, 24 hrs, 48 hrs**: the screen displays the selected length of trend data. You can view the most recent 240-hour data.
4. Select the desired number of **Trends**.

6.7.3 Printing a Graphic Trends Report

To print a graphic trends report, follow this procedure:

1. Enter the graphic trends review page.
2. Select the button in the upper right corner of this page. The **Print Setup** menu is displayed.
3. Set the desired options.
4. Select **Print**.

6.8 Full Disclosure Review Page

On the full disclosure review page, you can view the compressed waveforms and detail waveforms for each patient.

6.8.1 Entering the Full Disclosure Review Page

To enter the full disclosure review page, follow this procedure:

1. Select the button in the upper left corner of the review screen.
2. From the drop-down list, select **Full Disclosure**.

6.8.2 Selecting Waveforms

If you wish to review waveforms other than the default waveform on the full disclosure review page, you need to select the desired waveforms.

To do so, follow this procedure:

1. Enter the full disclosure review page.
2. Select **Setup**. The **Select Waveform** menu is displayed.
3. Select the **Storage** tab.
4. Configure the desired waveforms.
5. Select the **Display(Maximum: 3)** tab.
6. Enable the desired waveforms to be displayed on the review page.

In case of alarms, the background of compressed waveform block at the alarm time is marked with a special color:

- Red: high alarm priority
- Yellow: medium alarm priority
- Cyan: low alarm priority

6.8.3 Compressed Waveform

The full disclosure review page displays the compressed waveform.

6.8.3.1 Setting the Scale of Compressed Waveform

To set the size of the compressed waveform, follow this procedure:

1. Enter the full disclosure review page.
2. Set the ECG waveform size in either of the following ways:
   - Select the **Scale** button and then select the desired option.
   - Select the ECG lead label on the left side of the review page to enter the **Scale** menu. Then select the desired option.
3. Select other parameter labels on the left side of the review page to enter the **Scale** menu and set the scale for these parameters.

6.8.3.2 Setting the Duration of Compressed Waveform

To set the length of the compressed waveform, follow this procedure:

1. Enter the full disclosure review page.
2. Select **Duration**.
3. Configure the desired **Duration** time.

6.8.3.3 Marking Arrhythmia

In the compressed waveform area, the background of compressed waveform segments at the arrhythmia alarm trigger time is marked with a specific color automatically to indicate the arrhythmia category. For details on how to set the arrhythmia mark color, see Section *12.7.5 Setting Arrhythmia Mark Colors*.

6.8.3.4 Selecting Compressed Waveform Segments

You can select one or more compressed waveform segments and print these segments. For details on how to print these segments, see Section *6.8.5 Printing Compressed Waveforms or Detail Waveforms*.

To select compressed waveform segments, follow this procedure:

1. Enter the full disclosure review page.
2. Select compressed waveform segements in either of the following ways:
   - Using a mouse: In the compressed waveform area, hold down the right mouse button and drag the mouse to select the desired waveform segment. If you wish to select multiple waveform segments, release the right mouse button and then select the desired waveform segments. You can move the screen up and down to select more waveform segments.
   - Using a touch screen: Press the **Select** button and then touch the waveform. Hold and drag your finger to selec the desired waveform segment. If you wish to move the screen up and down, you need to re-select the **Select** button to exit the selected status.

If you wish to cancel current selection, select the ✕ button in the upper right corner of the waveform segment selection window.

6.8.4 Detail Waveform Window

In the detail waveform window, you can view the full size waveforms and parameter numerics.

Access the detail waveform window in either of the following ways:

- Select the **Detail** button in the compressed waveform window.
- Double click the desired compressed waveform.

In the detail waveform window:

- Timeline and waveforms are displayed in the upper part of the window.
- Numeric values are displayed in the lower part of the window. The background color of parameters indicate whether an alarm is triggered.
  - No color: no alarm is triggered.
  - Red: a high priority alarm is triggered.
  - Yellow: a medium priority alarm is triggered.
  - Cyan: a low priority alarm is triggered.

If you wish to close the detail waveform window, select the ✕ button in the upper right corner of the detail waveform window.

6.8.4.1 Setting ECG Gain

To set the ECG gain, follow this procedure:

1. Enter the detail waveform window.
2. Select the **ECG Gain button**.
3. Configure the desired ECG gain.

6.8.4.2 Setting Waveform Sweep Speed

To set the waveform sweep speed, follow this procedure:

1. Enter the detail waveform window.
2. Select the **Speed** button.
3. Configure the desired speed. The speed is applicable to all the ECG waveforms.

6.8.4.3 Caliper Measurement Overview

When full disclosure data is collected and stored in the CMS, horizontal calipers can be used to measure the PR, QRS, QT, and R-R waveform intervals, and vertical calipers can be used to meassure the ST waveform amplitude in the detail waveform window.

Horizontal caliper and vertical caliper are as shown in Figure 6-2.

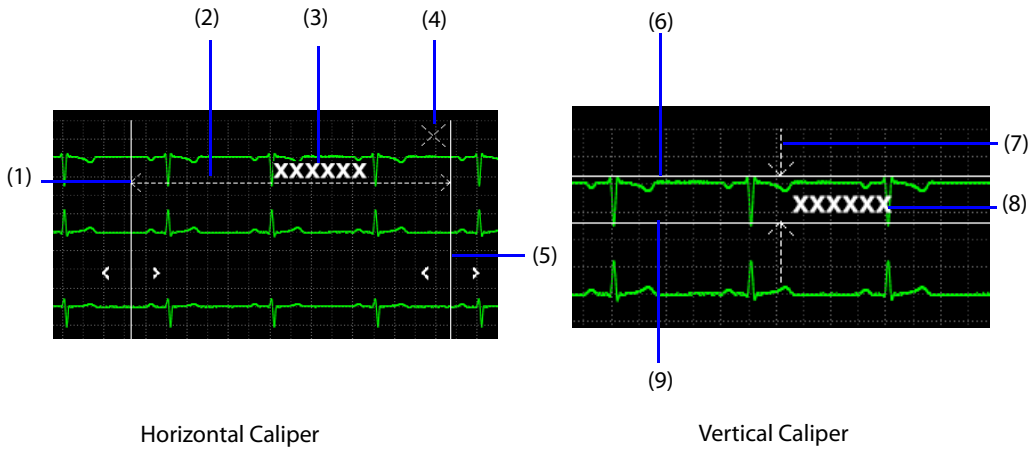

Figure 6-2 Horizontal and Vertical Calipers

(1)	Left caliper arm
(2), (7)	Measurement line
(4)	Close button: closes the caliper window without saving the measured value.
(5)	Right caliper arm
(3), (8)	Realtime measurement
(6)	Upper caliper arm
(9)	Lower caliper arm

Performing Measurements Using a Horizontal Caliper

To perform measurements using a horizontal caliper, follow this procedure:

1. Enter the detail waveform window.
2. Select **Caliper**. The **Caliper** menu is displayed.
3. Display a horizontal caliper in either of the following ways.
   - Using a mouse: left click the mouse at a place in the desired waveform area and then drag the cursor horizontally until your desired location is reached.
   - Using a touch screen: select a place in the desired waveform area and then drag your finger on the screen surface horizontally without losing the contact until the desired location is reached. If you need to adjust the caliper, slightly touch the left or right caliper arm and select the appropriate touchscreen control.
4. After finishing measurements, select **PR**, **QRS**, **RR**, and **QT** from the **Caliper** menu to obtain the measured values. The QTc value is calculated after RR and QT value have been acquired.
5. Select the desired option to **Save** or **Reset** measured values.

- **Save**: saves the caliper measurement values as a manual event in the event review page. In the **Please input new name:** field, enter the desired name of the event. The default name is **Saved ECG measurement**. In the **Please input comments:** field , enter the desired comments and then select **OK**. Up to 32 characters can be enterd for the name and the comments.The measured values are displayed above the first ECG waveform in the event detail window.
- **Reset**: clears the measured values.

Performing Measurements Using a Vertical Caliper

To perform measurements using a vertical caliper, follow this procedure:

1. Enter the detail waveform window.
2. Select **Caliper**. The **Caliper** menu is displayed.
3. Select **ST**.
4. Display a vertical caliper in either of the following ways.
   - Using a mouse: left click the mouse at a place in the desired waveform area and then drag the cursor vertically until your desired location is reached.
   - Using a touch screen: select a place in the desired waveform area and then drag your finger on the screen surface vertically without losing the contact until the desired location is reached. If you need to adjust the caliper, slightly touch the left or right caliper arm and select the appropriate touchscreen control.
5. After finishing measurements, select **ST** from the **Caliper** menu to obtain the measured value.
6. Select the desired option to **Save** or **Reset** measured values.
   - **Save**: saves the caliper measurement values as a manual event in the event review page. In the **Please input new name:** field, enter the desired name of the event. The default name is **Saved ECG measurement**. In the **Please input comments:** field , enter the desired comments and then select **OK**. Upto 32 characters can be enterd for the name and the comments. The measured values are displayed above the first ECG waveform in the event detail window.
   - **Reset**: clears the measured values.

Moving the Caliper

After displaying a caliper, you can also move calipers as shown below to measure the time difference between any two points:

- When the cursor is in the shape of , you can move the caliper horizontally.
- When the cursor is in the shape of , you can move the caliper vertically.
- When the cursor is in the shape of inside the caliper, you can move the entire caliper window.
- Select the button or the button in the horizontal caliper measurement area.
- Select the button or the button in the vertical caliper measurement area.

Cancelling Caliper Measurements

During caliper measurement, you can cancel a caliper measurement in either of the following ways:

- Select the **Caliper** button.
- Select anywhere in the waveform area.
- Select the button in the caliper window.

6.8.4.4 Setting Beat Annotation

When a bedside monitor and a telemetry device support the beat annotation functionality, you can enable or disable **Beat Annotation** to display or hide white beat labels above the first ECG waveform in the detail waveform window. The white beat labels indicate how the monitor or the telemetry device is classifying beats and may explain suspected, missed, or false arrhythmia calls.

- N = Normal
- V = Ventricular Ectopic
- S = Supraventricular Premature
- P= Paced
- L= Learning
- ?= Insufficient information to classify beats
- I= Inoperative (for example, Lead Off)
- M= Missed beat

When beat labels are displayed above the ECG waveform, they can be printed. If you enable or disable **Beat Annotation** in the detail waveform window, these beat labels will also be displayed or hidden in the event detail window, and vise versa.

CAUTION

- **Ensure that you have selected the optimal leads with the best waveform amplitude and the highest signal-to-noise ratio. Selecting the optimal leads is important for beat detection, beat classification, and V-Fib detection.**

6.8.4.5 Other Operations in the Detail Waveform Window

In the detail waveform window, you can save events and export waveforms to third party systems.

To do so, follow this procedure:

1. Enter the detail waveform window.
2. Select the button in the upper right corner of the window.
3. Select the desired option.
   - **Save As Event**: displays the **Saved Event** dialog box. The event can be saved with the desired **Name** and **Comments**. The default name for the event is **Saved Event**. The saved event is marked as a manual event on the event review page.
   - **Export to EMR**: export 20 seconds of waveform data after the start time of current waveform to third party systems.

NOTE

- **The Export to EMR option is available only when the CMS is configured with eGateway and the exporting to EMR feature is enabled on the system setup screen. For details on how to enable this feature, see Section *12.9.4.2 Setting Data Export*.**

6.8.5 Printing Compressed Waveforms or Detail Waveforms

To print the compressed waveform or detailed waveform, follow this procedure:

1. Enter the full disclosure review page.
2. Select the button in the upper right corner of this page. The **Print Setup** menu is displayed.
3. Select the desired options. Only items that may need special remarks are described here.
   - **Print Waveform Overview**: prints the compressed waveform displayed on the full disclosure review page.
   - **Print Overview of Selected Waveforms**: prints the selected compressed waveform segments. For details on how to select compressed waveform segments, see Section *6.8.3.4 Selecting Compressed Waveform Segments*.

- **Print Detail of Selected Waveforms**: prints the detail waveform for the selected compressed waveform segments.
- **Print Displayed Waveform Detail**: prints the detail waveform for the entire compressed waveform.

6.8.6 Recording Compressed Waveforms

To record the compressed waveform, follow this procedure:

1. Enter the full disclosure review page.
2. Select the button in the upper right corner of this page.

6.9 Events Review Page

On the events review page, you can view technical alarm events, physiological alarm events, manual events, and operational events.

When external devices are connected to the monitor via the BeneLink module, you can also view the data of the external devices on this review page. Parameter labels from external devices are prefixed with the plus sign (+).

6.9.1 Entering the Events Review Page

To enter the events review page, follow this procedure:

1. Select the button in the upper left corner of the review screen.
2. From the drop-down list, select **Events**.

6.9.2 Event List

In the event list:

- Events are displayed in descending chronological order. The most recent event is displayed at the top. The number of asterisk symbols before an event matches different alarm priorities.
  - ***: high priority alarm
  - **: medium priority alarm
  - *: low priority alarm
- Event lock status, trigger time, alarm priority, and event remarks if available are displayed for each event.
- Different color blocks are displayed on the left of each event to indicate different event types.
  - Red: high priority alarm event
  - Yellow: medium priority alarm event
  - Cyan: low priority alarm event
  - Green: manual event
  - White: operation-related event

6.9.2.1 Setting Filters

You can filter events by time, alarm priority, alarm category, and parameter group.

To filter events, follow this procedure:

1. Enter the event review page.
2. Enable **Filter**.
3. Select **Filter Setup** and set the desired filter criterion. The combination of selected filter criteria is used to filter events. Events after filtering will be displayed in the event list.

NOTE

- **When the filter is disabled, all the settings you made in the Filter Setup window will not take effect.**
- **BeneLink in the Parameter Group section of the Filter Setup window stands for parameters from external devices.**

6.9.2.2 Other Operations in the Event List

To perform other event-related operations, follow this procedure:

1. Enter the event review page.
2. Select the desired event from the event list.
3. Select the ⋮ button.
4. Select the desired option. Only items that may need special remarks are described here. The functions of locking and unlocking events is available in the CentralStation only.
   - **Lock**: lock the event manually. When the event is locked, the 🔒 symbol is displayed. The locked event needs to be unlocked before being deleted. Events can also be locked automatically. For details, see Section *12.7.4 Setting Events*.
   - **Rename**: you can rename a manual event or an arrhythmia alarm event.
   - **Note**: you can enter comments for the event. When comments are available, the 📝 symbol is displayed. You can also modify or delete comments in the event detail window. Up to 32 characters can be entered.

NOTE

- **If events are all locked, the CentralStation will automatically delete the oldest locked event with each addition of a new event when system storage reaches capacity.**

6.9.3 Event Detail Window

Enter the event detail window in either of the following ways:

- Double click or tap the desired event from the event list.
- Select the **Detail** button on the event review page.

In the event detail window:

- The upper half of the window displays:
  - Event name: displays in the upper left corner.
  - Event triggered time: displays in red font.
  - Waveform: 16 seconds of waveforms before and after the event triggered time.
- Numeric values are displayed in the lower part of the window. The background color of parameters indicate whether an alarm is triggered.
  - No color: no alarm is triggered.
  - Red: a high priority alarm is triggered.
  - Yellow: a medium priority alarm is triggered.
  - Cyan: a low priority alarm is triggered.

If you wish to close the event detail window, select the × button.

6.9.3.1 Configuring ECG Gain

To configure the ECG gain, follow this procedure:

1. Enter the event detail window.
2. Select **ECG Gain**.
3. Select the desired gain.

6.9.3.2 Configuring Waveform Sweep Speed

To configure the waveform sweep speed, follow this procedure:

1. Enter the event detail window.
2. Select the **Speed** button.
3. Configured the desired speed. The speed is applicable to all the ECG waveforms.

6.9.3.3 Caliper Measurement

For details on how to perform caliper measurement, see Section *6.8.4.3 Caliper Measurement Overview*.

6.9.3.4 Setting Beat Annotation

For details on how to set beat annotation,see Section *6.9.3.4 Setting Beat Annotation*.

6.9.3.5 Other Operations in the Event Detail Window

In the event detail window, you can perform operations such as renaming events.

To do so, follow this procedure:

1. Enter the event detail window.
2. Select the button in the right corner of the window.
3. Select the desired option. Among these options, selecting **Export to EMR** exports the event related data to third party systems.

NOTE

- **The Export to EMR option is available only when the CMS is configured with eGateway and the exporting to EMR feature is enabled on the system setup screen. For details on how to enable this feature, see Section *12.9.4.2 Setting Data Export*.**

6.9.4 Printing Events

To print the desired event list or event details, follow this procedure:

1. Enter the event review page.
2. Select the desired events from the event list.
3. Select the button in the upper right corner of this page. The **Print Setup** menu is displayed.
4. Select the desired options.
   - **Print All Event List**: print the entire event list.
   - **Print List of Selected Events**: print the selected events list.
   - **Print Detail of Selected Events**: print the details of selected events.
   - **Print Displayed Event Detail**: print the details of the selected event. If you wish to select this option, you need to open the event detail window first. If waveform data is available for the selected event, 16 seconds of waveforms before and after the alarm trigger time are printed by default. The actual waveform data printed depends on the option selected for **Printing Duration**. For details on how to set **Printing Duration**, see Section *12.10.9 Setting Other Print-Related Items*.

6.9.5 Recording Event Detail

To record the event detail, follow this procedure:

1. Enter the event detail page.
2. Select the button in the upper right corner of this page.

6.10 OxyCRG Review Page

The OxyCRG review functionality is applicable for neonatal monitoring only. In the OxyCRG review window, you can review the most recent 48-hour trend curves and compressed waveforms.

6.10.1 Entering the OxyCRG Review Page

To enter the OxyCRG review page, follow this procedure:

1. Select the button in the upper left corner of the review screen.
2. From the drop-down list, select **OxyCRG**.

6.10.2 Configuring the OxyCRG Review Page

On the OxyCRG review page, you can set the resolution of trend curves and select the desired compressed waveform.

To do so, follow this procedure:

1. Enter the OxyCRG review page.
2. Set **Zoom.**
3. Set **Waveform**.

6.10.3 Printing OxyCRG Review Reports

To print an OxyCRG review report, follow this procedure:

1. Enter the OxyCRG review page.
2. Select the button in the upper right corner of this page. The **Print Setup** menu is displayed.
3. Set the desired options.
4. Select **Print**.

6.11 12-Lead ECG Review Page

You can review 12-lead analysis events on the 12-lead ECG review page.

6.11.1 Entering the 12-Lead Review Page

To enter the 12-lead ECG review page, follow this procedure:

1. Select the button in the upper left corner of the review screen.
2. From the drop-down list, select **12-Lead ECG**.

6.11.2 Median Complex Window(For Glasgow Algorithm Only)

On the 12-lead ECG review page, selecting **Median Complex** enters the median complex window.

In the median complex window, the median complex template marks the start and end position of P-wave and QRS-wave and the end position of T-wave.

Selecting **Waveform** returns to the 12-lead ECG waveform review page.

6.11.3 Configuring 12-Lead ECG Waveforms

To configure 12-lead ECG waveforms on the review page, follow this procedure:

1. Enter the 12-lead ECG review page.
2. Configure **Speed** and **Gain**.
3. Configure **Layout**.
   - **12×1:** displays 12-lead ECG waveforms on one page in one column.
   - **6×2**: displays 12-lead ECG waveforms on one page in two columns, with 6 lines in each column.
   - **6×2+1**: displays 12-lead ECG waveforms on one page in two columns, with 6 lines in each column, and one rhythm lead waveform at the bottom.
   - **3×4+1**: displays 12-lead ECG waveforms on one page in 4 columns, with 3 lines in each column, and one rhythm lead waveform at the bottom.
   - **3×4+3**: displays 12-lead ECG waveforms on one page in 4 columns, with 3 lines in each column, and three rhythm lead waveforms at the bottom.

6.11.4 Caliper Measurement

Horizontal calipers can be used to measure the PR, QRS, QT, and R-R waveform intervals, and vertical calipers can be used to meassure the ST waveform amplitude in the median complex window.

To perform measurements using a horizontal or vertical caliper,follow this procedure:

1. Enter the 12-lead ECG review page.
2. Select the **Caliper** button.
3. From the drop-down list, select the desired option:
   - **Caliper**: perform measurement using a horizontal caliper.
   - **ST Caliper**: perform measurement using a vertical caliper.
4. Display a caliper.
5. Obtain measured values.

For details on how to display a horizontal or vertical caliper,how to obtain measured values, and how to move a caliper, see Section *6.8.4.3 Caliper Measurement Overview*.

NOTE

- **The measured caliper values on the 12-lead analysis review page are not saved on the event review page.**

6.11.5 Printing 12-Lead Analysis Reports

To print a12-lead ECG analysis report, follow this procedure:

1. Enter the 12-lead ECG review page.
2. Select the button in the upper right corner of this page. The **Print Setup** menu is displayed.
3. Set the desired options.
4. Select **Print**.

6.12 ST Review Page

You can review ST segments and values on the ST review page.

6.12.1 Entering the ST Review Page

To enter the ST review page, follow this procedure:

1. Select the button in the upper left corner of the review screen.
2. From the drop-down list, select **ST.**

6.12.2 Setting ST Reference Templates

You can set the desired ST template as a reference template.

To do so, follow this procedure:

1. Enter the ST review page.
2. Select the desired ST template.
3. Select **Set Reference**.

6.12.3 Displaying/Hiding ST Reference Templates

On the ST review page, select **Display Reference** or **Hide Reference**.

- **Display Reference**: the ST template storage time is displayed on the right of current segment and ST values are also displayed on the reference template.
- **Hide Reference**: hide the ST reference template.

6.12.4 Displaying/Hiding Markers

On the ST review page, select **Display Marker** or **Hide Marker**.

- **Display Marker**: a white vertical bar appears above each waveform, marking the positions of ISO, J point, and ST point.
- **Hide Marker**: the positions of ISO, J point, and ST point are not displayed.

6.12.5 Printing ST Segment Waveforms

To print ST segment waveforms, follow this procedure:

1. Enter the ST review page.
2. Select the button in the upper right corner of this page. The **Print Setup** menu is displayed.
3. Set the desired options.
4. Select **Print**.

6.13 Arrhythmia Statistics Page

On the arrhythmia statistics page, you can view the statistics about the HR value and the number of each arrhythmia category within a specific time period.

6.13.1 Entering the Arrhythmia Statistics Page

To enter the arrhythmia statistics page, follow this procedure:

1. Select the button in the upper left corner of the review screen.
2. From the drop-down list, select **Arrhy Statistics.**

6.13.2 Viewing Arrhythmia Statistics

Select **Zoom** in the upper right corner of the arrhythmia statistics page to view arrhythmia events within a specific time period. The default for **Zoom** is **24 hrs**.

6.13.3 Printing Arrhythmia Statistics Results

To print arrhythmia statistics results, follow this procedure:

1. Enter the arrhythmia statistics page.
2. Select the button in the upper right corner of this page. The **Print Setup** menu is displayed.
3. Set the desired options.
4. Select **Print**.

7 Calculation

7.1 Calculation Overview

The CMS provides calculation functions. The calculated values, which are not directly measured, are computed based on the values you provide. Any operation on a calculation page does not affect the patient monitored by the CMS.

The CMS can perform the following calculations:

- Drug calculations
- Hemodynamic calculations
- Oxygenation calculations
- Ventilation calculations
- Renal calculations

NOTE

- **The calculation functions that can be performed on the CMS are dependent on the system setup. For details on how to configure the calculation functions, see Section *12.3.4 Setting Patient Window*.**

7.2 Calculation Safety Information

WARNING

- **Decisions on the choice and dosage of drugs administered to patients must always be made by the physician in charge. The drug calculations are based on the values input, it does not check the plausibility of the calculation performed.**
- **Be sure to enter correct values and verify the correctness of the calculation results displayed on the screen before using them. We assume no responsibility for any consequences caused by wrong entries and improper operations.**
- **The calculations in the titration table are subject to the drug calculations. Therefore, always ensure the correctness of the drug calculations and confirm the reference, dose interval and dose type.**

7.3 Accessing the Calculation Screen

Access the calculation screen in either of the following ways:

- Select the waveform or parameter area in the desired patient sector on the multibed screen to enter the ViewBed screen → select the **Calculations** tab.
- Select the patient information area of the desired patient sector on the multibed screen to enter the patient management screen → select the **Calculations** tab.

7.4 Drug Calculations

On the drug calculation page, you can perform drug calculation, view calculation result, and print the titration table.

7.4.1 Performing Drug Calculations

To perform drug calculations, follow this procedure.

1. Select the ▼ button in the upper left corner of the calculation screen.
2. From the drop-down list, select **Drug** to access the drug calculation page.
3. Select the **Calculator** tab.
4. Set **Drug Name** and **Patient Category**. The dose calculation program has a library of commonly used drugs, of which Drug A through Drug E are user defined.
5. If the drug dose is weight dependent, enable **Weight Based** and input the patient's weight.
6. Enter the known values, for example **Drug Amount** and **Solution Volume**.
7. Select **Calculate**. The calculated values are indicated by red arrows.

NOTE

- **If available, the patient category and weight from the patient management screen are automatically entered when you first access the drug calculation page. You can change the patient category and weight. This will not change the patient category and weight stored on the patient management screen.**

7.4.2 Viewing a Titration Table

The titration table shows information on the currently used drugs. You can view the drug dose that will be applied to the patient at different infusion rates.

To access the titration table, follow this procedure.

1. Select the ▼ button in the upper left corner of the calculation screen.
2. From the drop-down list, select **Drug** to access the drug calculation page.
3. Select the **Titration Table** tab.
4. Select **Dose Type** to set the type of dose unit in the titration table.
5. Select **Interval** to set the interval between two adjacent titration table items.
6. If you need to sort the titration table, select the desired items at the top of the screen:
   - **Dose**: lists the drug doses in asending order.
   - **Infusion Rate**: lists the infusion rates in asending order.
   - **Exact Rate:** the resolution of the infusion rate can reach 0.01 so that you can display the infusion rate more accurately.

7.4.3 Printing a Titration Table

Select the button in the upper right corner of the titration table to print the calculation result.

7.4.4 Drug Calculation Formula

Description	Unit	Formula
Drug Amount	g series: mcg, mg, g unit series: Unit, KU, MU mEq series: mEq	Drug Amount =Dose × Infusion Time
Drug Amount (weight based)	g series: mcg, mg, g unit series: unit, kU, mU mEq series: mEq	Drug Amount (weight based) = Dose × Infusion Time × Weight
Solution Volume	ml	Volume = Infusion Rate × Infusion Time
Dose	Dose/hr Dose/min	Dose = Infusion Rate × Concentration
Dose (weight based)	Dose/kg/hr Dose/kg/min	Dose (weight based) = Infusion Rate × Concentration/Weight
Concentration	mcg/ml, mg/ml, g/ml, Unit/ml, KU/ml, MU/ml, mEq/ml	Concentration = Drug Amount/Solution Volume
Infusion Time	hr	Infusion time = Drug Amount/Dose
Infusion Time (weight based)	hr	Infusion Time (weight based) = Drug Amount/(Dose × Weight)
Infusion Rate	ml/hr	Infusion Rate = Dose/Concentration
Infusion rate (weight based)	g•ml/hr	Infusion Rate = Dose × Weight/Concentration

7.4.5 Titration Table Calculation Formula

Description	Unit	Formula
Infusion Rate	ml/hr	Infusion Rate = Dose/Concentration
Infusion Rate (weight based)	ml/hr	Infusion Rate = Weight × Dose/Concentration
Dose	Dose/hr Dose/min	Dose = Infusion Rate × Concentration
Dose (weight based)	Dose/kg/hr Dose/kg/min	Dose (weight based) = INF Rate × Concentration/ Weight

7.5 Hemodynamic Calculations

On the hemodynamic calculation page, you can perform hemodynamic calculation, view and print the calculation result.

7.5.1 Performing Hemodynamic Calculations

To perform hemodynamic calculation, follow this procedure.

1. Select the ▼ button in the upper left corner of the calculation screen.
2. From the drop-down list, select **Hemodynamics** to access the hemodynamic calculation page.
3. Enter values. For a patient who is being monitored, the currently measured values are automatically populated.
4. Select **Calculate**.

   The calculated value greater than the normal upper limit is indicated by an up arrow "↑". The calculated value lower than the normal lower limit is indicated by a down arrow "↓".

On the hemodynamic calculation page, you can also perform the following operations:

- Select **Range** to show the normal range of each parameter.
- Select **Range**, and then the **Range** button changes to **Unit**. Select **Unit** to show the unit of each parameter.

7.5.2 Viewing Hemodynamic Calculation Results

Upon completion of hemodynamic calculation, the most recent 100 calculation results are stored in the CMS automatically. The calculation time is displayed on the left of the hemodynamic calculation page. You can select the desired calculation time to view corresponding calculation result.

7.5.3 Printing Hemodynamic Calculation Results

Select the button in the upper right corner of the hemodynamic calculation page to print the calculation result.

7.5.4 Input Parameters for Hemodynamic Calculations

Input Parameter	Label	Unit
cardiac output	C.O.	L/min
heart rate	HR	bpm
pulmonary artery wedge pressure	PAWP	mmHg
artery mean pressure	PMAP	mmHg
pulmonary artery mean pressure	PA Mean	mmHg
central venous pressure	CVP	mmHg
end-diastolic volume	EDV	ml
height	Height	cm
weight	Weight	kg

7.5.5 Calculated Parameters and Formulas for Hemodynamic Calculations

Calculated Parameters	Label	Unit	Formula
cardiac index	C.I.	$L/min/m^2$	C.I. ($L/min/m^2$) = C.O. (L/min)/BSA (m^2)
body surface area	BSA	m^2	BSA (m^2) = $Wt^{0.425}$ (kg) × $Ht^{0.725}$ (cm) × 0.007184
stroke volume	SV	ml	SV (ml) = 1000× C.O. (L/min)/HR (bpm)
stroke index	SVI	ml/m^2	SVI (ml/m^2) = SV (ml)/BSA (m^2)
systemic vascular resistance	SVR	DS/cm^5	SVR (DS/cm^5) = 79.96 × [APMAP (mmHg) - CVP (mmHg)]/C.O. (L/min)
systemic vascular resistance index	SVRI	$DS•m^2/cm^5$	SVRI ($DS•m^2/cm^5$) = SVR (DS/cm^5) × BSA (m^2)
pulmonary vascular resistance	PVR	DS/cm^5	P VR (DS/cm^5) = 79.96 × [PAMAP (mmHg) - PAWP (mmHg)]/C.O. (L/min)
pulmonary vascular resistance index	PVRI	$DS•m^2/cm^5$	PVRI ($DS•m^2/cm^5$) = PVR (DS/cm^5)× BSA (m^2)
left cardiac work	LCW	kg•m	LCW (kg•m) = 0.0136 × APMAP (mmHg) × C.O. (L/min)
left cardiac work index	LCWI	$kg•m/m^2$	LCWI ($kg•m/m^2$) = LCW (kg•m)/BSA (m^2)

Calculated Parameters	Label	Unit	Formula
left ventricular stroke work	LVSW	g•m	LVSW (g•m) = 0.0136 × APMAP (mmHg) × SV (ml)
left ventricular stroke work index	LVSWI	g•m/m^2	LVSWI (g•m/m^2) = LVSW (g.m)/BSA (m^2)
right cardiac work	RCW	kg•m	R CW (kg•m) = 0.0136 × PAMAP (mmHg) × C.O. (L/min)
right cardiac work index	RCWI	kg•m/m^2	R CWI (kg•m/m^2) = RCW (kg.m)/BSA (m^2)
right ventricular stroke work	RVSW	g•m	R VSW (g•m) = 0.0136 × PAMAP (mmHg) × SV (ml)
right ventricular stroke work index	RVSWI	g•m/m^2	R VSWI (g•m/m^2) = RVSW (g•m)/BSA (m^2)
ejection fraction	EF	%	EF (%) = 100 × SV (ml)/EDV (ml)
End-diastolic volume index	EDVI	ml/m2	EDVI (ml/m^2) = EDV (ml)/BSA (m^2)
End-systolic Volume	ESV	ml	ESV (ml) = EDV (ml) –SV (ml)
End-systolic Volume index	ESVI	ml/m^2	ESVI (ml/m^2) = ESV (ml)/BSA (m^2)

7.6 Oxygenation Calculations

On the oxygenation calculation page, you can perform oxygenation calculation, view and print the calculation result.

7.6.1 Performing Oxygenation Calculations

To perform oxygenation calculations, follow this procedure.

1. Select the ▼ button in the upper left corner of the calculation screen.
2. From the drop-down list, select **Oxygenation** to access the oxygenation calculation page.
3. Enter values. For a patient who is being monitored, the currently measured values are automatically populated.
4. Select **Calculate**.

   The calculated value greater than the normal upper limit is indicated by an up arrow "↑". The calculated value lower than the normal lower limit is indicated by a down arrow "↓".

On the oxygenation calculation page, you can also perform the following operations:

- Select **OxyCont Unit**, **Hb Unit**, and **Pressure Unit**. Then corresponding parameter values will be automatically converted and updated accordingly.
- Select **Range** to show the normal range of each parameter.
- Select **Range**, and then the **Range** button changes to **Unit**. Select **Unit** to show the unit of each parameter.

7.6.2 Viewing Oxygenation Calculation Results

Upon completion of oxygenation calculation, the most recent 100 calculation results are stored in the CMS automatically. The calculation time is displayed on the left of the oxygenation calculation page. You can select the desired calculation time to view corresponding calculation result.

7.6.3 Printing Oxygenation Calculation Results

Select the button in the upper right corner of the oxygenation calculation page to print the calculation result.

7.6.4 Input Parameters for Oxygenation Calculations

Input Parameter	Label	Unit
cardiac output	C.O.	L/min
percentage fraction of inspired oxygen	FiO_2	%
partial pressure of oxygen in the arteries	PaO_2	mmHg, kPa
partial pressure of carbon dioxide in the arteries	$PaCO_2$	mmHg, kPa
arterial oxygen saturation	SaO_2	%
partial pressure of oxygen in venous blood	PvO_2	mmHg, kPa
venous oxygen saturation	SvO_2	%
hemoglobin	Hb	g/L, g/dl, mmol/L
respiratory quotient	RQ	None
atmospheric pressure	ATMP	mmHg, kPa
height	Height	cm, inch
weight	Weight	kg, lb

7.6.5 Calculated Parameters and Formulas for Oxygenation Calculations

Calculated Parameters	Label	Unit	Formula
body surface area	BSA	m^2	BSA (m^2) = $Wt^{0.425}$ (kg) × $Ht^{0.725}$ (cm) × 0.007184
oxygen consumption	VO_2	ml/min	VO_2 (ml/min) = C(a-v)O_2 (ml/L)× C.O. (L/min)
arterial oxygen content	CaO_2	ml/L, ml/dL	CaO_2 (ml/L) = 10× (0.0134 × Hb (g/dl) × SaO_2 (%) +0.031× PaO_2 (mmHg)
venous oxygen content	CvO_2	ml/L, ml/dL	CvO_2 (ml/L) = 10× (0.0134 × Hb (g/dl) × SvO_2 (%) +0.031 × PvO_2 (mmHg)
arteriovenous oxygen content difference	C(a-v)O_2	ml/L, ml/dl	C(a-v)O_2 (ml/L) = CaO_2 (ml/L) - CvO_2 (ml/L)
oxygen extraction ratio	O_2ER	%	O_2ER (%) = 100×C(a-v)O_2 (ml/L)/CaO_2 (ml/L)
oxygen transport	DO_2	ml/min	DO_2(ml/min) = C.O. (L/min) × CaO_2(ml/L)
partial pressure of oxygen in the alveoli	PAO_2	mmHg, kPa	PAO_2 (mmHg) = [ATMP (mmHg) - 47 mmHg] × FiO_2 (%)/100 - $PaCO_2$ (mmHg) × [FiO_2 (%)/100 + (1 - FiO_2 (%)/100)/RQ]
alveolar-arterial oxygen difference	$AaDO_2$	mmHg, kPa	$AaDO_2$ (mmHg) = PAO_2 (mmHg) - PaO_2 (mmHg)
capillary oxygen content	CcO_2	ml/L, ml/dl	CcO_2 (ml/L) = Hb (g/L) × 1.34 + 0.031 × PAO_2 (mmHg)
venous admixture	QS/QT	%	QS/QT (%) = 100× [1.34 × Hb (g/L) × (1 - SaO2 (%)/100) + 0.031 × (PAO2 (mmHg) - PaO2 (mmHg)]/[1.34 × Hb (g/L) × (1 - SvO2 (%)/100) + 0.031× (PAO2 (mmHg) - PvO2 (mmHg)]
oxygen transport index	DO_2I	ml/min/m^2	DO2I (ml/min/m^2) = CaO2 (ml/L) × (C.O. (L/min)/BSA (m^2))
oxygen consumption	VO_2I	ml/min/m^2	VO2I (ml/min/m^2) = C (a-v) O2 (ml/L) ×(C.O. (L/min)/BSA (m^2))

7.7 Ventilation Calculations

On the ventilation calculation page, you can perform ventilation calculation, view and print the calculation result.

7.7.1 Performing Ventilation Calculations

To perform ventilation calculations, follow this procedure.

1. Select the ▼ button in the upper left corner of the calculation screen.
2. From the drop-down list, select **Ventilation** to access the ventilation calculation page.
3. Enter values. For a patient who is being monitored, the currently measured values are automatically populated.
4. Select **Calculate**.

   The calculated value greater than the normal upper limit is indicated by an up arrow "↑". The calculated value lower than the normal lower limit is indicated by a down arrow "↓".

On the ventilation calculation page, you can also perform the following operations:

- Select **Pressure Unit**. Then corresponding parameter values will be automatically converted and updated accordingly.
- Select **Range** to show the normal range of each parameter.
- Select **Range**, and then the **Range** button changes to **Unit**. Select **Unit** to show the unit of each parameter.

7.7.2 Viewing Ventilation Calculation Results

Upon completion of ventilation calculation, the most recent 100 calculation results are stored in the CMS automatically. The calculation time is displayed on the left of the ventilation calculation page. You can select the desired calculation time to view corresponding calculation result.

7.7.3 Printing Ventilation Calculation Results

Select the button in the upper right corner of the ventilation calculation page to print the calculation result.

7.7.4 Input Parameters for Ventilation Calculations

Input Parameter	Label	Unit
percentage fraction of inspired oxygen	FiO_2	%
respiration rate	RR	rpm
partial pressure of mixed expiratory CO2	$PeCO_2$	mmHg, kPa
partial pressure of carbon dioxide in the arteries	$PaCO_2$	mmHg, kPa
partial pressure of oxygen in the arteries	PaO_2	mmHg, kPa
tidal volume	TV	ml
respiratory quotient	RQ	None
atmospheric pressure	ATMP	mmHg, kPa

7.7.5 Calculated Parameters and Formulas for Ventilation Calculations

Calculated Parameters	Label	Unit	Formula
partial pressure of oxygen in the alveoli	PAO_2	mmHg, kPa	PAO_2 (mmHg) = [ATMP (mmHg) - 47 mmHg] × FiO_2 (%)/100 - $PaCO_2$ (mmHg) × [FiO_2(%)/100 + (1 - FiO_2 (%)/100)/RQ]
alveolar-arterial oxygen difference	$AaDO_2$	mmHg, kPa	$AaDO_2$ (mmHg) = PAO_2 (mmHg) - PaO_2 (mmHg)
oxygenation ratio	Pa/FiO_2	mmHg, kPa	Pa/FiO_2(mmHg) = 100 × PaO_2 (mmHg)/FiO_2 (%)
arterial to alveolar oxygen ratio	a/AO_2	%	a/AO_2 (%) = 100 × PaO_2 (mmHg)/PAO_2 (mmHg)
minute volume	MV	L/min	MV (L/min) = [TV (ml) × RR (rpm)]/1000
volume of physiological dead space	Vd	ml	Vd (ml) = TV (ml) × [1 - $PeCO_2$ (mmHg)/$PaCO_2$ (mmHg)]
physiologic dead space in percent of tidal volume	Vd/Vt	%	Vd/Vt (%) = 100 × Vd (ml)/TV (ml)
alveolar volume	VA	L/min	VA (L/min) =[TV (ml) - Vd (ml)] × RR (rpm)/1000

7.8 Renal Calculations

On the renal calculation page, you can perform renal calculation, view and print the calculation result.

7.8.1 Performing Renal Calculations

To perform renal calculations, follow this procedure.

1. Select the ▼ button in the upper left corner of the calculation screen.
2. From the drop-down list, select **Renal** to access the renal calculation page.
3. Enter values.
4. Select **Calculate**.

   The calculated value greater than the normal upper limit is indicated by an up arrow "↑". The calculated value lower than the normal lower limit is indicated by a down arrow "↓".

On the renal calculation page, you can also perform the following operations:

- Select **Range** to show the normal range of each parameter.
- Select **Range**, and then the **Range** button changes to **Unit**. Select **Unit** to show the unit of each parameter.

7.8.2 Viewing Renal Calculation Results

Upon completion of renal calculation, the most recent 100 calculation results are stored in the CMS automatically. The calculation time is displayed on the left of the renal calculation page. You can select the desired calculation time to view corresponding calculation result.

7.8.3 Printing Renal Calculation Results

Select the button in the upper right corner of the renal calculation page to print the calculation result.

7.8.4 Calculated Parameters and Formulas for Renal Calculations

Input Parameter	Label	Unit
urine pstassium	URK	mmol/L
urinary sodium	URNa	mmol/L
urine	Urine	ml/24 hrs
plasm osmolality	Posm	mOsm/kgH_2O
urine osmolality	Uosm	mOsm/kgH_2O
serum sodium	SerNa	mmol/L
creatinine	Cr	µmol/L
urine creatinine	UCr	µmol/L
blood urea nitrogen	BUN	mmol/L
height	Height	cm
weight	Weight	kg

7.8.5 Calculated Parameters and Formulas for Renal Calculations

Calculated Parameters	Label	Unit	Formula
urine sodium excretion	URNaEx	mmol/24 hrs	URNaEx (mmol/24 hrs) = Urine (ml/24 hrs) × URNa (mmol/L)/1000
urine potassium excretion	URKEx	mmol/24 hrs	URKEx (mmol/24 hrs) = Urine (ml/24 hrs) × URK (mmol/L)/1000
sodium potassium ratio	Na/K	%	Na/K (%) = 100 × URNa (mmol/L)/URK (mmol/L)
clearance of sodium	CNa	ml/24 hrs	CNa (ml/24 hrs) = URNa (mmol/L) × Urine (ml/24 hrs)/SerNa (mmol/L)
creatinine clearance rate	Clcr	ml/min	Clcr (ml/min) = Ucr (µmol/L) × Urine (ml/24 hrs)/[Cr (µmol/L) × (BSA (m^2)/1.73) × 1440]
fractional excretion of sodium	FENa	%	FENa (%) = 100 × URNa (mmol/L) × Cr (µmol/L)/[SerNa (mmol/L) × Ucr (µmol/L)]
osmolar clearance	Cosm	ml/min	Cosm (ml/min) = Uosm (mOsm/kgH_2O) × Urine (ml/24 hrs)/(Posm (mOsm/kgH_2O) × 1440)
free water clearance	CH2O	ml/hr	CH2O (ml/hr) = Urine (ml/24 hrs) × [1 - Uosm (mOsm/kgH_2O)/Posm (mOsm/kgH_2O)]/24
urine to plasma osmolality ratio	U/P osm	None	U/P osm = Uosm (mOsm/kgH_2O)/Posm (mOsm/kgH_2O)
blood urea nitrogen creatinine ratio	BUN/Cr*	Mmol/L	BUN/Cr = 1000 × BUN (mmol/L)/Cr (µmol/L)
urine-serum creatinine ratio	U/Cr	None	U/Cr (mmol/L) = Ucr (µmol/L)/Cr (µmol/L)

*: BUN/Cr is a ratio at mol unit system.

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8 Blood Pressure Analysis Report

8.1 Overview

The CMS can integrate a third-party blood pressure analysis software. You can view blood pressure analysis reports in various presentation formats at the CMS and print these reports.

8.2 Accessing the Blood Pressure Analysis Screen

To access the blood pressure analysis screen, follow this procedure:

1. Select the system menu area in the upper left corner of the screen.
2. From the drop-down list, select **Blood Pressure Analyzer**.

8.3 Patient Management

You can add, change, search, import, export, and delete patient data.

8.3.1 Adding a New Patient

When a BP10 NIBP module (hereinafter referred to as the BP10) does not admit a patient yet and needs to be used for blood pressure measurement, you can add a new patient and enter the patient information on the blood pressure analysis screen. The patient information entered will be sent to the BP10.

Connect one end of a USB cable to the USB port on the BP10 and the other end to the USB port on the CMS' host computer. Then follow this procedure to add a new patient:

1. On the blood pressure analysis screen, select the **Desktop** tab.
2. In the **Patient List** section, select the **New** button. The **New Patient Information** menu is displayed.
3. Enter patient demographic information and configure blood pressure settings. Then select the **OK** button. The message **Write device settings successfully. Please plug-out the device**. is displayed. It means that the patient information entered has been sent to the BP10.
4. Disconnect the BP10 from the CMSand use the BP10 to measure patient's blood pressure.

After completing blood pressure measurement, you can import the blood pressure data from the BP10 to the CMS. For how to import the blood pressure data, see *8.3.4.1 Importing Patient Data from the CMS or the BP10*.

NOTE

- **This section describes how to add a patient at the CMS only. For how to add a patient at the BP10, see *BP10 NIBP Module Operator's Manual (P/N: 046-008269-00)*.**

8.3.2 Changing Patient Information

To change patient information, follow this procedure:

1. On the blood pressure analysis screen, select the **Desktop** tab.
2. In the **Patient List** section, select the desired patient from the patient list.
3. Select the **Patient** tab on the right of the screen.
4. Select the **Edit Patient** button or the **Edit Config** button and then modify the desired items.
5. Select the **OK** button.

8.3.3 Searching Patients

To search patients, follow this procedure:

1. On the blood pressure analysis screen, select the **Desktop** tab.
2. Select the **Patient Search** button.
3. Enter a patient name or ID and then select **By Name** or **By ID**. If you need to specify more search criteria, select **Advanced**.
4. Select the button.

8.3.4 Importing Patient Data

You can import the patient data from the current CMS, BP10, or a USB stick.

8.3.4.1 Importing Patient Data from the CMS or the BP10

If you wish to import data from the BP10, connect one end of a USB cable to the USB port on the BP10 and the other end to the USB port on the CMS' host computer before importing data.

To import the patient data, follow this procedure:

1. On the blood pressure analysis screen, select the **Desktop** tab.
2. In the **Patient List** section, select the **Import** button. The **Import** dialog box is displayed.
3. Select the source for data import: **Import Data From Central Monitoring Station or Import Data From Usb Device**.
4. Select the **OK** button.
5. If you have selected **Import Data From Central Monitoring Station**, you need to select the desired patient in the **Loading Patient** dialog box. If you have selected **Import Data From Usb Device**, you need to confirm the device information in the **Import** dialog box and then select the **OK** button.

Then the patient name, ID, and measured blood pressure data are imported into the blood pressure analysis software.

8.3.4.2 Importing CSV Patient Data Files from a USB Stick

When CSV patient data files are saved in a USB stick, you can import these files to the blood pressure analysis software. Before importing, connect the USB stick to the USB port on the CMS' host computer.

Then follow this procedure to import the patient data files:

1. On the blood pressure analysis screen, select the **Settings** tab→ select the **System** section → select the **Archive Reports** button.
2. Enter a patient name or ID and then select **By Name** or **By ID**. If you wish to display all the patients, select the button without entering a patient name or ID.
3. Select the desired patient.
4. Select the **Import** button.
5. Select the directory where the desired patient data file is stored and then select the **OK** button. If the patient data file contains blood pressure analysis reports, these reports are also imported.

8.3.5 Exporting Patient Data

You can export patient data to a USB stick. Before exporting patient data, connect the USB device to the CMS' host computer.

Then follow this procedure to export the patient data files:

1. On the blood pressure analysis screen, select the **Settings** tab→ select the **System** section → select the **Archive Reports** button.
2. Enter a patient name or ID and then select **By Name** or **By ID**. If you wish to display all the patients, select the button without entering a patient name or ID.
3. Select the desired patient.
4. Select the **Export** button.

5. Select the desired directory to save the file and then select the **OK** button. Exported patient data are saved in the CSV format.

8.3.6 Deleting Patients

To delete patients, follow this procedure:

1. On the blood pressure analysis screen, select the **Settings** tab → select the **System** section → select the **Archive Reports** button.
2. Enter a patient name or ID and then select **By Name** or **By ID**. If you wish to display all the patients, select the button without entering a patient name or ID.
3. Select the desired patient.
4. Select the symbol.
5. In the dialog prompting whether to delete the patient, select the **OK** button. The patient information and blood pressure analysis reports for this patient are deleted.

8.4 Blood Pressure Analysis Reports

You can view and print blood pressure analysis reports.

8.4.1 Viewing Blood Pressure Analysis Reports

On the blood pressure analysis screen, measured blood pressure data can be presented in these formats: Table, Trend, Histogram, Pie, Scatter, Compare, Curve, and Summary.

To view a blood pressure analysis report in the desired data presentation format, follow this procedure:

1. On the blood pressure analysis screen, select the **Desktop** tab.
2. In the **Patient List** section, select the desired patient.
3. Select the desired blood pressure measurement time for this patient.
4. Select the desired data presentation format tab on the right of the screen.

8.4.2 Printing Blood Pressure Analysis Reports

To print blood pressure analysis reports, follow this procedure:

1. On the blood pressure analysis screen, select the **Desktop** tab.
2. In the **Patient List** section, select the desired patient.
3. Select the desired blood pressure measurement time for this patient.
4. Select the **Print** tab in the upper left corner of the screen to enter the print review page.
5. Select the desired data presentation format on the left of the print review page. For a summary report, you can enter diagnosis comments. To do so, select the **Notes** button, enter the comments in the **Diagnosis** dialog box, and then select the **Submit** button.
6. Select the **Print** button in the lower right corner of the print review page.

8.5 Basic Settings

You can set blood pressure threshold, parameters, and measurement settings.

To do so, follow this procedure:

1. On the blood pressure analysis screen, select the **Settings** tab → select the **System** section → select the **Basic** button.
2. Set the desired items.
3. Select the **Submit** button.

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9 Printing

9.1 Supported Printers

The CMS can be connected to a printer to output reports on A4 or Letter paper. The printer has a separate power supply. The CMS supports the following printers:

- HP LaserJet 401n
- HP LaserJet M602
- HP LaserJet M202DW
- HP LaserJet M605n
- PDF printer

NOTE

- **For more details about the printer, see the document accompanying the printer. With product upgrades, the CMS may support more printers and no prior notice will be given. If you have any doubt about the printer you have purchased, contact Mindray.**
- **PDF reports can be output by PDFCreator only.**

9.2 Starting Printing

You can start a printing task manually or set the printer to start a printing task automatically.

9.2.1 Manually Starting Printing

You can manually start printing reports such as summary information, trends data, and calculation results.

9.2.1.1 Printing Summary Reports

When monitoring devices are being monitored by the CMS, you can print their summary reports which includes information such aspatient information, realtime data, and alarm data.

To print a summary report, follow this procedure.

1. Select the button in the upper right corner of the main screen.
2. Select **Print Summary**.

9.2.1.2 Printing Review Data

For details on how to print tabular trends, graphic trends, waveform review, waveform segment, event review, OxyCRG review,12-lead analysis review, ST segment, and arrhythmia statistics result, seeChapter *6 Review*.

9.2.1.3 CPrinting Calculation Results

For details on how to print a titration table, hemodynamic calculation result, oxygenation calculation result, ventilation calculation result, and renal calculation result, see Chapter *7 Calculation*.

9.2.1.4 Printing Realtime Waveforms

To print realtime waveforms for a single monitoring device, choose one of the following ways:

- Select the bed number and room number area in the upper left corner of the desired patient sector on the multibed screen and then select the **Print** button from the drop-down list.
- On the ViewBed screen, select the **Print** quick key.

- Select the **Print** button in the **Realtime Report** menu on the ViewBed screen. For details on how to access the **Realtime Report** menu, see Section *9.5.2.1 Setting Realtime Reports in the Normal Report Tab*.

9.2.1.5 Printing ECG Reports

When ECG monitoring with lead sets other than 3-lead is being performed, you can print corresponding ECG reports by following the procedure below.

1. Access the ViewBed screen in either of the following ways:
   - Select the parameter area or waveform area in the desired patient sector on the multibed screen to access the ViewBed screen.
   - Select the ViewBed symbol when it is displayed in the desired patient sector on the multibed screen.
2. Access the ECG full screen in one of the following ways:
   - On the ViewBed screen, select the **ECG Full-Screen** quick key.
   - On the ViewBed screen, select the **Screen Setup** quick key → select the **ViewBed** tab → select the **Choose Screen** tab → select **ECG Full-Screen**. If you are performing 12-lead ECG monitoring,select the **Choose Screen** tab → select **ECG 12-Lead**.
   - Select the ECG parameter area or waveform area to access the ECG menu → select the **Full-Screen** button in the lower left corner of the menu. If you are performing 12-lead ECG monitoring,select the 12-Lead.
3. Select the print button in either of the following ways:
   - On the ViewBed screen, select the **Print** quick key.
   - Select **Print** in the lower right corner of the **ECG Report** menu. For details on how to access this menu, see Section *9.5.2.2 Setting ECG Reports in the Normal Report Tab*.
4. Select the bed number and room number area in the upper left corner of the desired patient sector on the multibed screen and then select the **Print** button from the drop-down list.

NOTE

- **For details on how to set the ECG reports, see Section *9.5.2.2 Setting ECG Reports in the Normal Report Tab*.**

9.2.1.6 Printing EEG Reports (BIS Module)

To print EEG reports, follow this procedure:

1. Access the ViewBed screen in either of the following ways:
   - Select the parameter area or waveform area in the desired patient sector on the multibed screen to access the ViewBed screen.
   - Select the ViewBed symbol when it is displayed in the desired patient sector on the multibed screen.
2. Access the BIS menu in either of the following ways:
   - Select a BIS waveform or parameter area.
   - Select the **Parameter Setup** quick key → select **BIS**.
3. Select the **BIS Expand** button at the bottom of the menu.
4. Select the **EEG** tab.
5. After selecting desired settings, select the button in the upper left corner.

9.2.1.7 Printing EEG and CSA Reports (EEG Module)

To print EEG and the CSA reports, follow this procedure:

1. Access the ViewBed screen in either of the following ways:
   - Select the parameter area or waveform area in the desired patient sector on the multibed screen to access the ViewBed screen.
   - Select the ViewBed symbol when it is displayed in the desired patient sector on the multibed screen.
2. Access the EEG menu in either of the following ways:
   - Select the EEG waveform or parameter area.
   - Select the **Parameter Setup** quick key at the bottom of the screen → select **EEG**.
3. Select the **EEG Expand** button at the bottom of the menu.
4. Select the **EEG** tab.
5. After selecting desired settings, select the button in the upper left corner to print the EEG report.
6. Select the **CSA** tab.
7. After selecting desired settings, select the button to print the CSA report.

9.2.1.8 Printing CCO-Related Hemodynamic Parameters

To print CCO-related hemodynamic parameters, follow this procedure:

1. Access the ViewBed screen in either of the following ways:
   - Select the parameter area or waveform area in the desired patient sector on the multibed screen to access the ViewBed screen.
   - Select the ViewBed symbol when it is displayed in the desired patient sector on the multibed screen.
2. Access the CCO menu in either of the following ways:
   - Select the CCO parameter area.
   - Select the **Parameter Setup** quick key at the bottom of the screen → select **CCO**.
3. Select the **HemoSight** button at the bottom of the menu.
4. Select the **Diagnosis** tab.
5. Select the button in the upper left corner.

9.2.1.9 Printing ICG-Related Hemodynamic Parameters

To print ICG-related hemodynamic parameters, follow this procedure:

1. Access the ViewBed screen in either of the following ways:
   - Select the parameter area or waveform area in the desired patient sector on the multibed screen to access the ViewBed screen.
   - Select the ViewBed symbol when it is displayed in the desired patient sector on the multibed screen.
2. Access the ICG menu in either of the following ways:
   - Select the ICG waveform or parameter area.
   - Select the **Parameter Setup** quick key at the bottom of the screen → select **ICG**.
3. After selecting desired settings, select the **Hemodynamic Parameters** button at the bottom of the menu.
4. Select the button in the upper left corner.

9.2.1.10 Printing SvO_2/$ScvO_2$-Related Hemodynamic Parameters

To print SvO_2/$ScvO_2$-related hemodynamic parameters, follow this procedure:

1. Access the ViewBed screen in either of the following ways:

- Select the parameter area or waveform area in the desired patient sector on the multibed screen to access the ViewBed screen.
- Select the ViewBed symbol when it is displayed in the desired patient sector on the multibed screen.

2. Access the SvO_2 or $ScvO_2$ menu in either of the following ways:
   - Select the SvO_2 or $ScvO_2$ parameter area.
   - Select the **Parameter Setup** quick key at the bottom of the screen → select **SvO_2/$ScvO_2$**.
3. Select the **HemoSight** button at the bottom of the menu.
4. Select the **Diagnosis** tab.
5. Select the button in the upper left corner.

9.2.1.11 Printing QT Reports

When a bedside monitor or a telemetry device is performing QT analysis, you can print the QT report at the CMS.

To do so, follow this procedure:

1. Access the ViewBed screen in either of the following ways:
   - Select the parameter area or waveform area in the desired patient sector on the multibed screen to access the ViewBed screen.
   - Select the ViewBed symbol when it is displayed in the desired patient sector on the multibed screen.
2. Access **QT View** in one of the following ways:
   - Select the QT parameter area to access the QT menu→ select the **QT View** button at the bottom of the menu.
   - Select the ECG waveform or parameter area to access the **ECG** menu → select the **QT** tab → select the **QT View** button at the bottom of the menu.
   - Select the **Parameter Setup** quick key at the bottom of the screen → select **ECG**→ select the **QT** tab → select the **QT View** button at the bottom of the menu.
3. Select the button in the upper left corner.

9.2.1.12 Printing ST View

The ST View shows a complete QRS segment for each ST lead. To print ST View, follow this procedure:

1. Access the ViewBed screen in either of the following ways:
   - Select the parameter area or waveform area in the desired patient sector on the multibed screen to access the ViewBed screen.
   - Select the ViewBed symbol when it is displayed in the desired patient sector on the multibed screen.
2. Access **ST View** in one of the following ways:
   - Select ST segment waveform area.
   - Select the ST parameter area to access the **ST** menu → select the **ST View** button at the bottom of the menu.
   - Select the ECG waveform or parameter area to access the **ECG** menu → select the **ST** tab → select the **ST View** button at the bottom of the menu.
3. Select the button in the upper left corner.

9.2.1.13 Printing Frozen Waveforms

For details on how to print frozen waveforms, see Section *3.6.6.3 Printing Frozen Waveforms*.

9.2.1.14 Printing Alarm Setup

To print alarm setup, follow this procedure:

1. Access the ViewBed screen in either of the following ways:

- Select the parameter area or waveform area in the desired patient sector on the multibed screen to access the ViewBed screen.
- Select the ViewBed symbol when it is displayed in the desired patient sector on the multibed screen.

2. Select the **Alarm Setup** quick key at the bottom of the screen.
3. After selecting the desired settings, select the button in the upper left corner.

9.2.1.15 Printing Defibrillator's Self-Test Reports

You can print the self-test reports for one or all the defibrillators.

To print the self-test report for one defibrillator, follow this procedure:

1. Select the CentralStation's system menu area in the upper left corner of the screen.
2. From the drop-down list, select **Defibrillator Management**.
3. Select the **More** button in the upper right corner of the desired defibrillator.
4. Select the button in the upper left corner. The **Print** window is displayed.
5. If you need to print self-test reports within a specific time period, set **Start Time** and **End Time**.
6. Select **Print**.

To print the self-test reports for all the defibrillators, follow this procedure:

1. Select the CentralStation's system menu area in the upper left corner of the screen.
2. From the drop-down list, select **Defibrillator Management**.
3. Select the button in the upper left corner. The most recent selft-test reports for all the defibrillators are printed.

9.2.2 Automatically Starting Printing

You can set a printer to start printing reports automatically.

9.2.2.1 Setting Scheduled Printing

You can set scheduled printing of tabular trends reports. For details, see Section *9.5.3 Setting Scheduled Reports* and *12.10.3 Setting Scheduled Reports*.

9.2.2.2 Enabling Print on Alarm

When a parameter alarm switch is set to on and an alarm is triggered for this parameter, you can set a printer to start alarm printing automatically.

To enable automatic printing via the printer when an alarm is triggered, make sure that **Printer on Alarm** is set to **Printer**. For detail, refer to *12.10.9 Setting Other Print-Related Items*.

To enable the printer to automatically start printing on alarm, follow this procedure:

1. Access alarm related tabs such as the **Alarm** tab for a parameter in one of the following ways:
   - Select the **Alarm Setup** quick key at the bottom of the screen.
   - Select the parameter or waveform area of the desired parameter → select an alarm related tab.
   - Select the **Parameter Setup** quick key at the bottom of the screen → select the desired parameter → select an alarm related tab.
2. Enable **Alarm Outputs** for a parameter. The configured length of waveform data for the parameter will be printed automatically.

9.2.2.3 Setting Automatic Printing for Defibrillator's Self-Test Reports

To set a printer to automatically print defibrillator's self-test reports, follow this procedure:

1. Select the CentralStation's system menu area in the upper left corner of the screen.
2. From the drop-down list, select **Defibrillator Management**.
3. Select the button in the upper left corner. The **Set** window is displayed.

4. Enable **Auto Print**.

Upon completion of settings, when a defibrilaltor sends a self-test report to the CentralStation, the self-test report will be printed automatically. When multiple defibrillators send self-reports, the most recent self-test result for each defibrillator will be printed automatically in chronological order according to the time when self-test reports are sent.

9.3 Stopping Printing

To stop a printing task, follow this procedure:

1. Select the button in the upper right corner of the CMS main screen.
2. Select **Print Queue**.
3. Select one printing task and then select **Delete** to stop this printing task; select all the printing tasks and then select **Delete All** to stop all the printing tasks.

9.4 Viewing Printer Statuses

You can view the status of the most recent printing tasks in the **Print Queue** window. To view the status of printing tasks, follow this procedure:

1. Select the button in the upper right corner of the CMS main screen.
2. Select **Print Queue**.

9.5 Configuring Reports

You can configure report related contents such as patient information items on a report. For details, see Section *12.10 Print Tab*. This section only describes how to set normal reports and end case reports for a single monitoring device in the **Print Setup** menu.

9.5.1 Accessing the Print Setup Menu

To access the **Print Setup** menu, follow this procedure:

1. Access the ViewBed screen in either of the following ways:
   - Select the parameter area or waveform area in the desired patient sector on the multibed screen to access the ViewBed screen.
   - Select the ViewBed symbol when it is displayed in the desired patient sector on the multibed screen.
2. Select the **Print Setup** quick key at the bottom of the screen to enter the **Print Setup** menu. If this quick key is not available, configure it by following steps in *3.6.4 Quick Keys*.

9.5.2 Setting Normal Reports

In the **Normal Report** tab, you can set realtime reports and ECG reports.

9.5.2.1 Setting Realtime Reports in the Normal Report Tab

To set realtime reports, follow this procedure:

1. In the **Print Setup** menu, select the **Normal Report** tab.
2. Select the **Realtime Report** tab.
3. Set **Speed**.
4. Select waveforms to be printed.
   - **Current Waveforms**: print the realtime report for current waveforms.
   - **Selected Waveforms**: print the realtime report for the selected waveforms. If you wish to select this option, you need to further select **Select Waveform** and select the desired waveforms in the **Select Waveform** menu.

9.5.2.2 Setting ECG Reports in the Normal Report Tab

When ECG monitoring with lead sets other than 3-lead is being performed, you can set ECG reports.

To do so, follow this procedure:

1. In the **Print Setup** menu, select the **Normal Report** tab.
2. Select the **ECG Report** tab.
3. Set the desired options. Only items that may need special remarks are described here.

Menu item	Description	Options	Default
Auto Interval	Defines the spacing between the ECG waveforms on a printout	**On:** automatically adjusts the space between waveforms to avoid overlapping. **Off**: each waveform area has the same sizeon a printout.	Off
	Note: This setting is only relevant when **12×1** is selected for **12-Lead Format**.		
12-Lead Format	Select the format of 12-lead ECG waveforms on a printout.	**12×1:** displays 12-lead ECG waveforms on one page in one column. **6×2**: displays 12-lead ECG waveforms on one page in two columns, with 6 lines in each column. **6×2+1**: displays 12-lead ECG waveforms on one page in two columns, with 6 lines in each column, and one rhythm lead waveform at the bottom. **3×4+1**: displays 12-lead ECG waveforms on one page in 4 columns, with 3 lines in each column, and one rhythm lead waveform at the bottom. **3×4+3**: displays 12-lead ECG waveforms on one page in 4 columns, with 3 lines in each column, and three rhythm lead waveforms at the bottom.	3×4+1
Rhythm Lead 1 Rhythm Lead 2 Rhythm Lead 3	Select the lead that will be used as Rhythm Lead 1, 2, or 3.	I, II, III, aVR, aVL, aVF, V1, V2, V3, V4, V5, V6	Rhythm Lead 1: II Rhythm Lead 2: V2 Rhythm Lead 3: V5,
	Note: This setting is only relevant when **6×2+1**,**3×4+1**, or **3×4+3** is selected for **12-Lead Format**.		
Format sequence	Select the recording method of ECG report generated by auto measurement	**Sequential**: 12-lead ECG data are recorded sequentially and displayed in 3 lines and 4 columns with 2.5 seconds of ECG data for each column. **Simultaneous**: Record simultaneous 12-lead ECG data.	Sequential

9.5.3 Setting Scheduled Reports

You set scheduled printing of tabular trends reports in the **Print Setup** menu only when **Scheduled Report Switch** is enabled in the **System Setup** menu.

The scheduled report settings in the **System Setup** menu define some conditions for printing tabular trends reports. If you need to further set more conditions, you can set them in the **Print Setup** menu. For details on how to set scheduled printing of tabular trends reports in the **System Setup menu**, see Section *12.10.3 Setting Scheduled Reports*.

To set scheduled printing of tabular trends reports, follow this procedure:

1. In the **Print Setup** menu, select the **Scheduled Report** tab.
2. Select the Tabular Trends Report tab.

Set the desired options. For meanings of each option, see Section *6.6.3 Printing a Tabular Trends Report*.

9.5.4 Setting End Case Reports

The report types under the **End Case Report** tab are dependent on the report types selected in the system setup menu. If no report types are selected in the system setup menu, the **End Case Report** tab is not available. For details, see Section *12.10.4 Setting End Case Reports*.

This section assumes that ECG Report, Realtime Report, Tabular Trends Report, and Graphic Trends Report have been selected in this system setup menu and describes how to set the four types of reports in the **End Case Report** tab.

9.5.4.1 Setting ECG Reports in the End Case Report Tab

To set ECG reports as end case reports, follow this procedure:

1. In the **Print Setup** menu, select the **End Case Report** tab.
2. Select the **ECG Report** tab.
3. Set the desired options. For details on how to set the options, see Step 3 of Section *9.5.2.2 Setting ECG Reports in the Normal Report Tab*.

9.5.4.2 Setting Realtime Reports in the End Case Report Tab

To set realtime reports as end case reports, follow this procedure:

1. In the **Print Setup** menu, select the **End Case Report** tab.
2. Select the **Realtime Report** tab.
3. Set the desired options. For details on how to set the options, see Step 3 and Step 4 of Section *9.5.2.1 Setting Realtime Reports in the Normal Report Tab*.

9.5.4.3 Setting Tabular Trends Reports in the End Case Report Tab

To set tabular trends reports as end case reports, follow this procedure:

1. In the **Print Setup** menu, select the **End Case Report** tab.
2. Select the **Tabular Trends Report** tab.
3. Set the desired options. Only items that may need special remarks are described here.

Menu Item	Description	Options	Default
Period	Select the period during which a tabular trends report will be printed.	30 min, 1 hr, 2 hrs, 4 hrs, 8 hrs, 12 hrs, 24 hrs, 48 hrs, 72 hrs, 96 hrs, Auto, All **Auto**: print one page of a tabular trend report according to the interval and period selected.	Auto
Interval	Select the resolution of the tabular trends printed on a report.	5 sec, 30 sec, 1 min, 5 min, 10 min, 15 min, 30 min, 1 hr, 2 hrs, 3 hrs, NIBP, EWS, GCS, TempIF, C.O., Auto **NIBP, EWS**, **GCS**, **TempIF**, **C.O**.: print a tabular trends report when NIBP, EWS, GCS, infrared temperature, or C.O. measurements were acquired. Each NIBP, EWS, GCS, infrared temperature measurement will generate a column on the tabular trends review page. The values for the other measurements are added to the column to provide a complete dataset for the NIBP, EWS, GCS, infrared temperature, measurement time. **Auto**: print a tabular trends report at the interval set in the **Interval** option in the upper right corner of the **Tabular Trends** review page.	Auto

Menu Item	Description	Options	Default
Report Format	Select the printing principle.	**Parameter Oriented**: print one page of report by parameters when **Interval** is set to **Auto**. **Time Oriented**: print one page of report by timewhen **Interval** is set to **Auto**.	Parameter Oriented

9.5.4.4 Setting Graphic Trends Reports in the End Case Report Tab

To set graphic trends reports as end case reports, follow this procedure:

1. In the **Print Setup** menu, select the **End Case Report** tab.
2. Select the **Graphic Trends Report** tab.
3. Set the desired options.

9.6 Printer Out of Paper

When the printer runs out of paper, the print request will not be acknowledged. If there are too many print jobs that are not acknowledged a printer error may occur. In this case, install paper and then re-send the print request. Restart the printer if necessary.

To avoid a printer error condition, ensure that there is enough paper in the printer before sending a print request.

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10 Recording

10.1 Supported Recorders

The CMS can be equipped with to a thermal recorder. Supported recorders are TR6-B and TR6-F. The thermal recorder records patient information, measurement numerics and waveforms. The thermal recorder has a separate power supply. It can be connected to the CMS via a RS232 serial port.

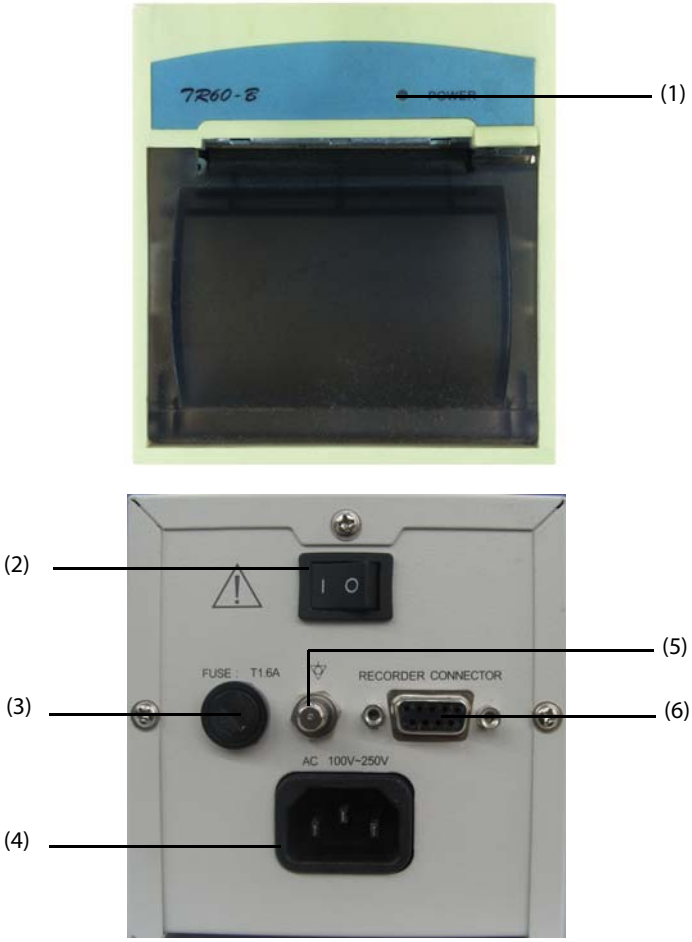


Figure 10-1 TR6-B Recorder

(1) Power indicator

- On: when the recorder works correctly.
- Off: when the recorder is powered off.
- Flashes: if an error occurred to the recorder.

(2) Power input (3) Fuse

(4) Power switch (5) Equipotential Grounding Terminal

(6) RS232 serial port

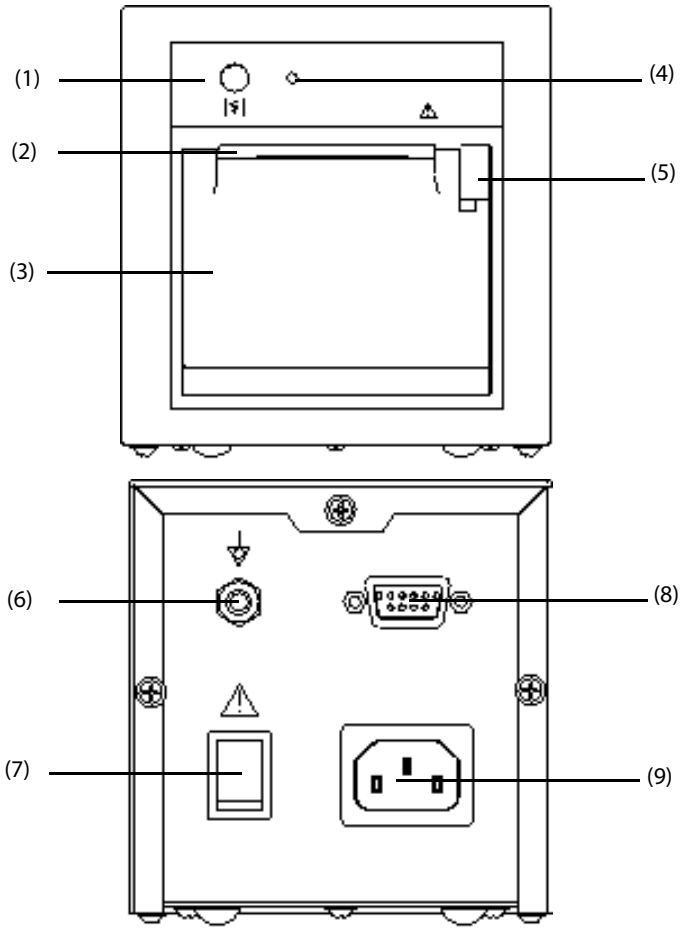


Figure 10-2 TR6-F Recorder

(1) hardkey

(2) Paper outlet

(3) Recorder door

(4) Power indicator

- On: when the recorder works correctly.
- Off: when the recorder is powered off.
- Flashes: if an error occurred to the recorder.

(5) Latch: pull it outwards to open the recorder door.

(6) Equipotential Grounding Terminal

(7) Power switch

(8) RS232 serial port

(9) Power input

NOTE

- **When connected to the CMS, pressing the hardkey on the front panel of the TR6-F recorder cannot start a recording, but can manually stop a recording.**

10.2 Loading Paper

WARNING

- **Use the standard (50mm) thermal recorder paper for the recording. Otherwise, the recording may be in poor quality, unusable, or the thermal printer head may become damaged.**
- **Be careful to avoid damaging the thermal printer head when loading the paper. Never pull the paper bail at the upper left corner of the recorder except if you plan to replace the recorder paper or resolve an issue.**

10.2.1 Loading Paper for the TR6-B Recorder

Figure 10-3

Figure 10-4

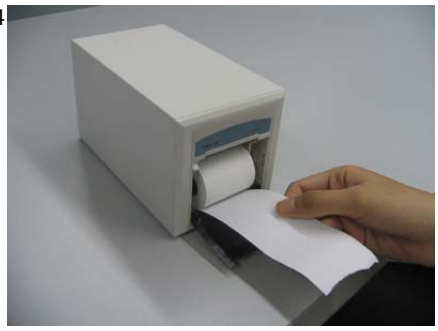


Figure 10-5

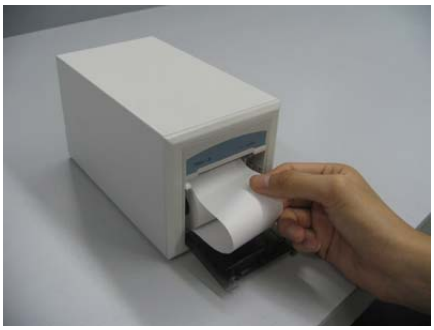


Figure 10-6

Figure 10-7

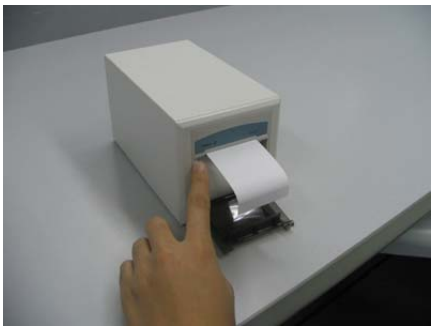
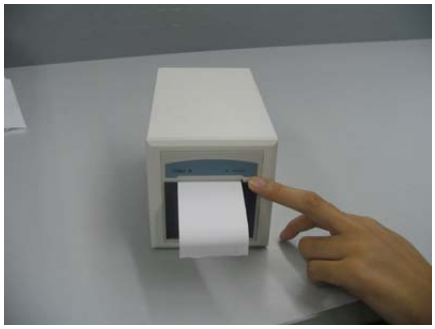


Figure 10-8



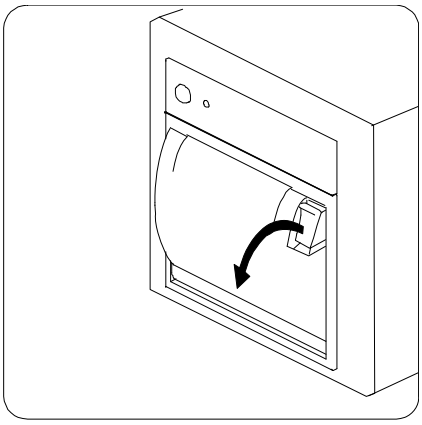
To load paper for the TR6-B recorder, follow this procedure referring to Figure 10-3 to Figure 10-8 as shown above:

1. Use the latch at the upper right of the recorder door to pull the door open, and then pull the paper bail at the upper left of the recorder door.
2. Insert a new roll into the compartment.
3. Feed the paper through the paper outlet.
4. Pull some paper out from the top of the roller and adjust it.
5. Push down the paper bail.
6. Close the recorder door.

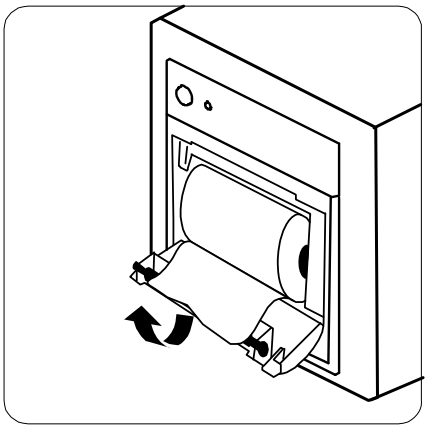
10.2.2 Loading Paper for the TR6-F Recorder

To load paper for the TR6-F recorder, follow this procedure:

1. Use the latch at the upper right of the recorder door to pull the door open.



2. Insert a new roll into the compartment as shown below. Feed the paper through and pull some paper out from the top of the roller.



3. Close the recorder door.
4. Check if paper is loaded correctly and the paper end is feeding from the top.

10.3 Setting the Recorder

To set the recorder, follow this procedure:

1. Select the **Record Setup** quick key at the bottom of the ViewBed screen to enter the **Record Setup** menu.
2. Select the desired waveform for **Waveform 1**, **Waveform 2** and **Waveform 3** in turn. The recorder can record up to 3 waveforms at a time.

For more information on setting the recorder, see *12.10.8 Setting the Recorder* and *12.10.9 Setting Other Print-Related Items*.

10.4 Starting Recordings

You can start a recording task manually or set the recorder to start a recording task automatically.

10.4.1 Manually Starting Recordings

You can manually start a recording task, such as the recording for realtime waveforms, event and waveform reviews by selecting the button.

10.4.1.1 Recording Realtime Waveforms

To manually start recording realtime waveforms for a single monitoring device, follow this procedure:

1. Access the ViewBed screen in either of the following ways:
   - Select the parameter area or waveform area in the desired patient sector on the multibed screen to access the ViewBed screen.
   - Select the ViewBed symbol when it is displayed in the desired patient sector on the multibed screen.
2. Select the desired quick key at the bottom of the ViewBed screen to start a recording task.
   - Select the **Record** quick key.
   - Select the **Record Setup** quick key → select **Record**.

10.4.1.2 Recording Event and Waveform Reviews

You can manually start the recording task on the full disclosure review screen and the events review screen. For details, see *6.8.6 Recording Compressed Waveforms* and *6.9.5 Printing Event Detail*.

10.4.2 Automatically Starting Recordings

You can set the recorder to start the recording task automatically.

10.4.2.1 Enabling Scheduled Recording

To enable scheduled recordings, follow this procedure:

1. Select the **Record Setup** quick key at the bottom of the ViewBed screen to enter the **Record Setup** menu.
2. Set **Interval**. The recorder can automatically starts the recording task at the configured interval.

10.4.2.2 Setting Recording on Alarm

When a parameter alarm switch is set to on and an alarm is triggered for this parameter, you can set the recorder to start alarm recording automatically.

Printer on Alarm is set to **Printer** by default. To enable automatic recording via the recorder when an alarm is triggered, make sure that **Printer on Alarm** is set to **Recorder**. For detail, refer to *12.10.9 Setting Other Print-Related Items*.

To set the recorder to automatically start recording on alarm, follow this procedure:

To set recording on alarm, follow this procedure:

1. Access alarm related tabs such as the Alarm tab for a parameter in one of the following ways:
   - Select the **Alarm Setup** quick key at the bottom of the screen.
   - Select the parameter or waveform area of the desired parameter → select an alarm related tab.
   - Select the **Parameter Setup** quick key at the bottom of the screen → select the desired parameter → select an alarm related tab.
2. Enable **Alarm Outputs** for a parameter. The set length of waveform data for the parameter will be recorded automatically.

10.5 Stopping Recordings

Recordings can be stopped manually or automatically.

10.5.1 Manually Stopping Recordings

To manually stop a recording, choose one of the following ways:

- Access the **Record Setup** menu → select **Clear All Tasks**.
- Access the **System Setup** menu → select the **Print** setup page → select the **Recorder** tab → select **Reset RecordService**.
- Press the hardkey on the front panel of the TR6-F recorder.

10.5.2 Automatically Stopping Recordings

Recordings stop automatically when:

- A recording is completed.
- The recorder runs out of paper.
- The recorder has an alarm condition.

11 Viewing Other Patients

11.1 Overview

When there are multiple CentralStations within a LAN, you can observe and review patients at a remote CentralStation through the other patients feature.

You can also use a CMS viewer or a mobile viewer to view patients from a remote CentralStation. For more information on the CMS viewer, see *BeneVision Central Monitoring System Viewer Operator's Manual (P/N: 046-010296-00)*. For more information on the mobile viewer, see *BeneVision Mobile Viewer Installation and Use Guide (P/N: 046-010298-00)*.

11.2 Network Settings and Bed Authorization for Remote View

To view patients at a remote CentralStation, you need to connect the local CentralStation/WorkStation/ViewStation to the remote CentralStation. For details, see Section *12.9.6 Setting CentralStation Connection*.

Besides, you also need to authorize the beds that can be viewed by the local CentralStation/WorkStation/ViewStation. For details, see Section *12.9.7 Setting Bed Authorization*.

In addition, you also need to grant permissions for the local CentralStation/WorkStation/ViewStation to access the remote CentralStation. For details on how to grant permissions, see Section *12.9.5 Setting CentralStation Authorization*.

- **Full Control**: after a bed is authorized, you can view patient data, change measurement controls such as alarm limits,admit, discharge, or transfer a patient at the local CentralStation/WorkStation.
- **Read Only**: you can view the patient data, but you cannot change measurement controls, admit, discharge, or transfer a patient.

11.3 Viewing Other Patients

You can view both real-time and historical patient data for patients monitored by a remote CentralStation in either of the following ways:

- Overview a bed in a patient sector on the CentralStation/WorkStation/ViewStation. For details, see Section *11.3.1 Assigning an Overview Bed to a Patient Sector*.
- View a bed temporarily through the bed list. For details, see Section *11.3.2 Viewing a Bed Temporarily through the Bed List*.

For details on how to view the real-time data, the historical data, and the calculation result of a remote monitoring device, see Section *3.5 ViewBed Screen*, Chapter *6 Review*, and Chapter *7 Calculation*.

11.3.1 Assigning an Overview Bed to a Patient Sector

Patient sectors that do not currently have a bed or equipment assigned are available for overview. If the patient sector already has an overview bed, you must clear the sector before assigning another bed to overview. For how to clear an overview bed, see Section *11.4 Clearing an Overview Bed*.

To assign a bed currently being monitored by a remote CentralStation to a patient sector, follow this procedure:

1. Select a patient sector with the + symbol on the multibed screen of the curent CentralStation/WorkStation/ViewStation to access the **Device Assignment** screen.
2. Select the button on the right of **Department** in the upper right corner of the screen.
3. From the drop-down list, select the department where the desired remote CentralStation is located.

NOTE

- **Do not select Local. Because Local means the local CentralStation/WorkStation/ViewStation.**

4. Assign monitoring devices.
   - To assign a monitoring device to a specific patient sector, select the patient sector on the left side of this page.
   - To assign monitoring devices to the next available patient sector, select the + symbol on the right side of the desired monitoring devices to admit these devices. The devices will be assigned to idle patient sectors on the multibed screen in the order of left to right and from top to bottom.

After remote monitoring devices are assigned to the local CentralStation, the symbol is displayed in the patient sectors of these devices, indicating that these devices are from a remote CentralStation.

11.3.2 Viewing a Bed Temporarily through the Bed List

To view other patients temporarily at the local CentralStation, follow this procedure:

1. Access the ViewBed screen of local CentralStation/WorkStation/ViewStation in either of the following ways:
   - Select the parameter area or waveform area in the desired patient sector on the multibed screen to access the ViewBed screen.
   - Select the ViewBed symbol when it is displayed in the desired patient sector on the multibed screen.
2. Select the symbol in the upper left corner of the screen.
3. Select the button.
4. From the drop-down list, select the department where the desired remote CentralStation is located. A bed list is displayed under the CentralStation.

NOTE

- **Do not select Local. Because Local means the local CentralStation/WorkStation/ViewStation.**

5. From the bed list, select the desired bed. A colored block is displayed before bed numbers to indicate different statuses:
   - Red: indicates that a high priority alarm is triggered.
   - Yellow: indicates that a medium priority alarm is triggered.
   - Cyan: indicates that a low priority alarm is triggered.

11.4 Clearing an Overview Bed

When you do not need to observe a monitoring device from a remote CentralStation on the multibed screen, follow this procedure:

1. Select the bed number and room number area in the upper left corner of the desired patient sector on the multibed screen.
2. From the drop-down list, select **Clear Overview Bed**.

NOTE

- **If the patient sector is only for overview, clearing that sector only clears the sector. It does not discharge the patient from the remote CentralStation or clear the patient sector on the remote CentralStation.**

12 System Setup

12.1 System Setup Overview

The CMS comes with factory defaults. You can make changes to the configuration items to accommodate the needs of your unit. Changes made to the CentralStation, the WorkStation, and the ViewStation are independent of each other.

WARNING

- **Changing configuration items may alter the way the CMS performs when monitoring patients. Do not change anything unless you are aware of the possible consequences, especially if you are monitoring a patient while in clinical settings.**

NOTE

- **Many settings in the CentralStation, the WorkStation, and the ViewStation are password protected. It is recommended to change the default passwords and keep the passwords safe. Passwords should only be changed by authorized personnel. Contact your department manager or biomedical engineering department for the passwords used at your facility.**

12.2 General Tab

In the **General** tab, you can adjust alarm volumes, view the help information, and calibrate the touchscreen. This tab is also available at the WorkStation and the ViewStation, and it functions in the same manner as it does at the host CentralStation.

12.2.1 Accessing the General Setup Page

To access the general setup page, follow this procedure:

1. Select the system menu area mindray in the upper left corner of the screen → select **System Setup** to access the **System Setup** menu.
2. Select the **General** tab.

12.2.2 Adjusting Alarm Volumes

In the **Volume** tab, you can set the alarm volume, high alarm volume, and reminder volume. For details on how to set the volumes, see Section *5.5.4 Setting Alarm Volume*.

12.2.3 Viewing System Help Information

In the **Help** tab, you can view the system help information such as the system software version, system configuration information, and system status by selecting the **Help** button.

12.2.4 Calibrating the Touchscreen

To calibrate the touchscreen, follow this procedure:

1. On the general setup page, select the **Cal Touchscreen** tab.
2. Touch and hold the + symbolareas where indicated.
3. Release the area when indicated.

If the CMS system is configured for multiple touchscreens, perform the steps above for the rest of the touchscreens.

12.3 Display Tab

In the **Display** tab, you can set display-related items such as parameter colors. This tab is also available at the WorkStation and the ViewStation, and it functions in the same manner as it does at the host CentralStation.

12.3.1 Accessing the Display Setup Page

To access the display setup page, follow this procedure:

1. Select the system menu area ≡ mindray in the upper left corner of the screen → select **System Setup** to access the **System Setup** menu.
2. Select the **Display** tab → enter the required password → select the **OK** button.

12.3.2 Setting Parameter Colors

Parameter-related information such as realtime waveforms and parameter numerics are displayed in the same color on the CMS screen.

To set the desired parameter colors, follow this procedure:

1. On the display setup page, select the **Param Color** tab.
2. Select the color box on the right side of the desired parameter.
3. From the drop-down list, select the desired color.

12.3.3 Screen Setup

You can set the screen for the following items:

- Screen size
- Display layout
- The number of patient sectors and primary screens
- The position for the Viewbed Screen
- Whether to enable auto close the Viewbed screen and the desired closing time.

12.3.3.1 Setting Screen Size

To set the screen size, follow this procedure:

1. On the display setup page, select the **Screen** tab.
2. Select **Screen Size** and then select the desired option. If your screen size is not available, select **Others** and select the + or − button on the rulers to adjust the screen size.
3. Select **OK**.

NOTE

- **Restart the system after making a change in the screen size.**
- **The CentralStation, WorkStation, or ViewStation each can connect up to four displays and will check the number of displays every time it starts. When more than one display is used simultaneously,they should be the same resolution. The displays support two resolutions: 1280 x 1024 pixels and 1920 x 1080 pixels.**

12.3.3.2 Setting Display Layout

When the CentralStation, the WorkStation, or the ViewStation is equipped with four displays, you can select **Display Layout** and change the cursor moving mode. The display layout selected should be consistent with the appearance of your displays.

To select the display layout, follow this procedure:

1. On the display setup page, select the **Screen** tab.
2. Set **Display Layout**.

- **1 x 4**: select this option when four displays are arranged in a line. You can move the cursor from the current display to its neighboring diplays, as shown in Figure 11-1.

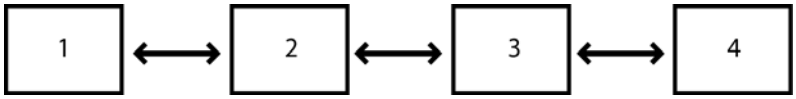


Figure 12-1 Cursor Moving Mode for 1 x 4

- **2 x 2**: select this option when two displays are stacked on the other two displays. You can move the cursor from current display to its neighboring displays, as shown in Figure 11-2.

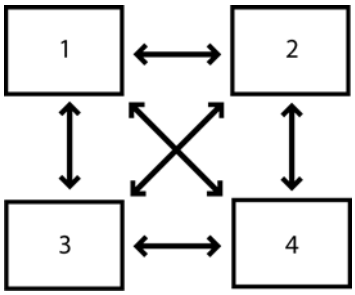


Figure 12-2 Cursor Moving Mode for 2 x 2

12.3.3.3 Setting the Number of Patient Sectors

You can set the desired number of patient sectors on the multibed screen.

To do so, follow this procedure:

1. On the display setup page, select the **Screen** tab.
2. Set **Patient Sector Number**.

12.3.3.4 Setting the Number of Primary Screens

The **Primary Screen Number** option is displayed only when more than one display is available. You can select the desired number of displays where the multibed screen is shown.

To set the number of primary screens, follow this procedure:

1. On the display setup page, select the **Screen** tab.
2. Set **Primary Screen Number**. Select the desired number of displays. For example, if you select 3, the multibed screen will be shown on three displays.

12.3.3.5 Setting the Position of ViewBed Screen

The **ViewBed Screen Position** option is displayed only when more than one display is available. You can select the desired display where the ViewBed screen is shown.

To set the position of ViewBed screen, follow this procedure:

1. On the display setup page, select the **Screen** tab.
2. Set **ViewBed Screen Position**. Select the desired display number. For example, if you select 3, then the ViewBed screen will be shown on the third display.

12.3.3.6 Enabling Auto Close ViewBed Screen

To enable closing the Viewbed screen automatically and set the desired closing time, follow this procedure:

1. On the display setup page, select the **Screen** tab.
2. Set **Auto Close ViewBed Screen**. Select any option from the list expect **Never** to enable closing the Viewbed screen automatically.

3. Configure the closing time. If you do not perform any operation after accessing the ViewBed screen within the configured time, the CMS automatically closes the ViewBed screen and accesses the multibed screen.

12.3.4 Setting Patient Window

You can select the desired items displayed on the review and calculation screens. To do so, follow this procedure:

1. On the display setup page, select the **Patient Window** tab.
2. Select the desired items.

12.3.5 Setting Other Display Items

You can set the waveform sequence for multi-lead ECG and whether to enable soft keyboard. To do so, follow this procedure:

1. On the display setup page, select the **Other** tab.
2. Set **ECG Lead Sequence**. After changing the ECG lead sequence, the waveform sequence will be updated in the 12-lead ECG printouts and on the 12-lead ECG screen.
   - **Normal**: I, II, III, aVR, aVL, aVF, V1, V2, V3, V4, V5, V6
   - **Cabrera**: aVL, I, -aVR, II, aVF, III, V1, V2, V3, V4, V5, V6
3. Enable or disable **Soft Keyboard**. When this option is enabled, a soft keyboard displays when a field that requires manual input is selected.

12.4 Alarm Tab

In the **Alarm** tab, you can set alarm-related properites. This tab is also available at the WorkStation and the ViewStation, and it functions in the same manner as it does at the host CentralStation.

12.4.1 Accessing the Alarm Setup Page

To access the alarm setup page, follow this procedure:

1. Select the system menu area mindray in the upper left corner of the screen → select **System Setup** to access the **System Setup** menu.
2. Select the **Alarm** tab → enter the required password → select the **OK** button.

12.4.2 Configuring Alarm Audio Properties

To configure alarm audio properties, follow this procedure:

1. On the alarm setup page, select the **Audio** tab.
2. Set the desired options. Only items that may need special remarks are described here.
   - **Minimum Alarm Volume**: ranges from 0 to10.
   - **Alarm Sound**: three sound patterns (**ISO**, **Mode 1**, and **Mode 2**) are available. Only in **ISO** mode, **High Alarm Interval**, **Med Alarm Interval**, and **Low Alarm Interval** are configurable.
   - **Auto Increase Volume**: **2 Steps** and **1 Step** mean that if an alarm is not reset within the designated delay time after the alarm occurs, the alarm volume automatically increases by two levels or by one level. **Off** means that if an alarm is not reset within the designated delay time after the alarm occurs, the alarm volume does not change.
   - **Increase Volume Delay**: set the delay time of alarm volume escalation.
   - **Alarm Reset Reminder**: when this option is enabled, If alarms are not acknowledged before the system is silenced, the alarm reset reminder tone is issued at the designated reminder interval after the system is silenced. After alarms in a patient sector are reset, if unacknowledged alarms still exist, the alarm reset reminder tone is issued at the designated reminder interval.
   - **Alarm Off Reminder**: when this option is enabled and alarms for a monitoring device are turned off, a reminder tone is issued at the designated reminder interval.
   - **Reminder Interval**: when **Alarm Reset Reminder** and/or **Alarm Off Reminder** is enabled, you can set the reminder interval.

NOTE

- **The alarm volume escalation function is not applied to the latched alarms.**

12.4.3 Configuring Other Alarm-Related Items

To configure other alarm-related items, follow this procedure:

1. On the alarm setup page, select the **Other** tab.
2. Set the desired options:
   - **Global Silence Hotkey**: set the hotkey that can be used to silence the system.
   - **Global Silence**: it is defaulted to **Disable**. When disabled, the global silence feature is not available. When it is set to **Enable**, pressing the silence quick key or selecting the symbol silences the system.
   - **Offline Alarm Priority**: set the alarm priority when a monitoring device, the WorkStation, the ViewStation, or eGateway is disconnected.
   - **Flashing Alarm Bar**: when this option is enabled, the background color of high priority and medium priority alarm messages on the mulitbed and ViewBed screen flashes.
   - **No offline alarm if patient discharged**: when this option is enabled, after the patient is discharged the offline alarm will not be triggered if the monitor is disconnected.
   - **eGateway Communication Lost**: when this option is enabled and eGateway is disconnected from the CentralStation, the WorkStation, or the ViewStation, the eGateway Communication Lost alarm message will be displayed in the system alarm area at the top of the screen.

12.4.4 Exporting the Alarm Log

To export the alarm log, follow this procedure:

1. On the alarm setup page, select the **Other** tab.
2. Select **Export** and set the location where the log file is saved. To save the alarm log to the preset location, select **One Key Export**. The alarm logs are saved at c:\Log by default.

You can use the Mindray Alarm Statistics Tool (PN: 110-005516-00) to analyze the alarm log and create a report. To do so, follow this procedure:

1. Open the Mindray Alarm Statistics Tool.
2. Select the folder symbol under **Source File Path** to find where the log file is saved.
3. Select the folder symbol under **Report File Path** to specify where the report is saved.
4. Set other items as desired.
5. Select **Analyze** to create a report.

NOTE

- **You can only export alarm logs for patients that triggered alarms.**
- **The statistics reports are Microsoft Excel files. Make sure that you have installed Microsoft excel before using the Mindray Alarm Statistics Tool.**

12.5 Patient Management Tab

In the **Patient Management** tab, you can set patient related items.

12.5.1 Accessing the Patient Management Page

To access the patient management screen, follow this procedure:

1. Select the system menu area mindray in the upper left corner of the screen → select **System Setup** to access the **System Setup** menu.
2. Select the **Patient Management** tab → enter the required password → select the **OK** button.

12.5.2 Configuring Patient Fields

In the **Field** tab, you can set the patient fields that are displayed on the patient management screen. This tab is also available at the WorkStation and the ViewStation, and it functions in the same manner as it does at the host CentralStation.

To configure patient fields, follow this procedure:

1. On the patient management screen**,** select the **Field** tab.
2. Select the desired items.
3. If necessary, select the customized fields and input names for these fields.

12.5.3 Setting Patient Location

In the **Location** tab, you can set the patient location options displayed on the patient sector and ViewBed screen after a monitoring device enters standby mode. This tab is available at the CentralStation Only.

To set a patient location, follow this procedure:

1. On the patient management screen**,** select the **Location** tab.
2. Select the desired locations.

12.5.4 Setting Patient Group

In the **Patient Group** tab, you can set the name of patient groups displayed under the **Patient Group** field on the patient managment screen. This tab is available at the CentralStation Only.

To set the patient group, follow this procedure:

1. On the patient management screen**,** select the **Patient Group** tab.
2. Select the patient group with desired background color.
3. Select the button and then enter the desired patient group name.

12.5.5 Setting Care Group

In the **Care Group** tab, you can set the name of care groups displayed under the **Care Group** field on the patient managment screen. This tab is available at the CentralStation Only.

To set the care group, follow this procedure:

1. On the patient management screen**,** select the **Care Group** tab.
2. Select the care group with desired background color.
3. Select the button and then enter the desired care group name.

12.5.6 Setting Other Patient Information Items

In the **Other** tab, you can set other items related to patient information. This tab is also available at the WorkStation and the ViewStation, and it functions in the same manner as it does at the host CentralStation.

To set the items, follow this procedure:

1. On the patient management screen**,** select the **Other** tab.
2. Set **Display Information in Patient Sector.**
   - **Patient Name**: displays the patient name in the patient sector on the multibed screen.
   - **Patient ID**: displays the patient ID in the patient sector on the multibed screen.
   - **None**: displays nothing in the patient sector on the multibed screen.
3. Set the desired **Patient Name Display Hotkey**.

12.6 Device Management Tab

In the **Device Management** tab of the CentralStation,you can admit monitoring devices. This tab contains two subtabs: **Device Assignment** and **Auto Admit Device.**

In the **Device Management** tab of the WorkStation and the ViewStation, you can assign monitoring devices from the host CentralStation. This tab contains two subtabs: **Device Assignment** and **Auto Admit Bed**.

12.6.1 Accessing the Device Management Page

To access the device management page, follow this procedure:

1. Select the system menu area mindray in the upper left corner of the screen → select **System Setup** to access the **System Setup** menu.
2. Select the **Device Management** tab → enter the required password → select the **OK** button.

12.6.2 Device Management Tab at the CentralStation

The CentralStation can be set to automatically admit monitoring devices. Or, monitoring devices can be manually admitted .

NOTE

- **Within the LAN, a monitoring device can be admitted to any CentralStation, but can be admitted by one CentralStation only.**

12.6.2.1 Automatically Admitting Monitoring Devices

To automatically admit monitoring devices, follow this procedure:

1. Select the system menu area in the upper left corner of the screen → select **System Setup** to access the **System Setup** menu.
2. Select the **Device Management** tab → enter the required password → select the **OK** button to access the device management screen..
3. Select the **Auto Admit Device** tab to access the device admission page.
4. Set the desired condition for auto admission.
   - To admit monitoring devices within an IP address range, enter the IP address into the **From** and **To** text boxes respectively. Then select **Add**.
   - To admit monitoring devices within a particulardepartment, enter the department name into the **Department** text box. Then select **Add**. The department entered must match the department entered at the monitoring devices.

Once the eligible monitoring devices are detected, they will be automatically admitted and assigned to the CentralStation's multibed screen.

If you wish to cancel an auto admission condition, select the desired condition on the device admission page, and then select **Delete**.

A monitoring device automatically admitted is not bound with a patient sector on the multibed screen. After discharging a patient, the patient sector is in idle status.

12.6.2.2 Manually Admitting Monitoring Devices on the Device Assignment Page

To manually admit monitoring devices on the device assignment page, follow this procedure:

1. Select the system menu area mindray in the upper left corner of the screen → select **System Setup** to access the **System Setup** menu.
2. Select the **Device Management** tab→ enter the required password → select the **OK** button.
3. Select the **Device Assignment** tab to access the device assignment page.
   - On the left side of this page: select the ◀ or ▶ button to toggle between available displays and view corresponding patient sectors on the multibed screen. If monitoring devices have been admitted, related information such as their department and bed numbers are displayed in

the patient sectors. Selecting the ✕ button in the upper right corner of a patient sector disconnects the monitoring device from the CentralStation.

- The right side of this page displays the list of all the monitoring devices. The − symbol indicates that this device has been admitted by current CentralStation. Selecting the − symbol disconnects this device from the CentralStation. Selecting the + symbol admits this device to the current CentralStation. Selecting the ⓘ symbol displays information such as the device model. Clicking the heading of any column or selecting the ▲▼ button sorts this column in ascending or descending order. Selecting the ↻ button refreshes the monitoring device list.

4. Admit monitoring devices.
   - To assign a monitoring device to a specific patient sector, select the patient sector on the left side of this page. If a monitoring device has been assigned to this patient sector, disconnect this monitoring device from the CentralStation by selecting the ✕ button. Then select the + symbol on the right side of the desired monitoring device to admit this device.
   - To assign monitoring devices to the next available patient sector, select the + symbol on the right side of the desired monitoring devices to admit these devices. The devices will be assigned to idle patient sectors on the multibed screen in the order of left to right and from top to bottom.

12.6.2.3 Automatically Assigning Monitoring Devices from the Host CentralStation

To automatically assign monitoring devices from the host CentralStation, follow this procedure:

1. Select the WorkStation/ViewStation's system menu area ≡ mindray in the upper left corner of the screen → select **System Setup** to access the **System Setup** menu.
2. Select the **Device Management** tab→ enter the required password → select the **OK** button to access the device management screen.
3. Select the **Auto Admit Bed** tab.
4. Select the **Admit** button at the end of the desired CentralStation.
5. Set the desired conditions for auto assignment.
   - If you need to assign all the monitoring devices from the host CentralStation, select **Auto admit all beds in this CentralStation**.
   - If you need to assign some monitoring devices from the host CentralStation, do not select **Auto admit all beds in this CentralStation**. Instead, enter the desired numbers under **Room No** and **Bed No**, and then select the **Add** button on the right of **List of Beds Admitted Automatically**.

Once the eligible devices are detected, they will be automatically assigned to idle patient sectors on the multibed screen of the WorkStation or the ViewStation in the order of left to right and from top to bottom.

A monitoring device automatically assigned is not bound with a patient sector on the multibed screen. After discharging a patient, the patient sector is in idle status.

12.6.2.4 Assigning Monitoring Devices on the Device Assignment Page

To manually assign monitoring devices from the host CentralStation, follow this procedure:

1. Select the WorkStation/ViewStation's system menu area in the upper left corner of the screen → select **System Setup** to access the **System Setup** menu.
2. Select the **Device Management** tab→ enter the required password → select the **OK** button.
3. Select the **Device Assignment** tab to access the device assignment page. The device assignment page of the WorkStation or the ViewStation is similar to that of the CentralStation. For details about the device assignment page, see Step 3 in Section *12.6.2.2 Manually Admitting Monitoring Devices on the Device Assignment Page*.
4. Select the ▾ button on the right of **Department** in the upper right corner of the screen.
5. From the drop-down list, select the desired CentralStation.
6. Assign monitoring devices.
   - To assign a monitoring device to a specific patient sector, select the patient sector on the left side of this page. If a monitoring device has been assigned to this patient sector, disconnect this monitoring device from the WorkStation or the ViewStation by selecting the ✕ button. Then select the + symbol on the right side of the desired monitoring device.

- To assign monitoring devices to the next available patient sector, select the + symbol on the right side of the desired monitoring devices. The devices will be assigned to idle patient sectors on the multibed screen in the order of left to right and from top to bottom.

Using this method, a monitoring device is bound with a patient sector on the multibed screen. After discharging a patient, the patient's bed number is still displayed in the patient sector. When the monitoring device is online again, the patient data from this monitoring device will be displayed in the same patient sector. This method is appropriate for devices that need to be monitored frequently.

12.7 Review Tab

In the **Review** tab, you can set trends related items.

12.7.1 Accessing the Review Setup Page

To access the review setup page, follow this procedure:

1. Select the system menu area mindray in the upper left corner of the screen → select **System Setup** to access the **System Setup** menu.
2. Select the **Review** tab → enter the required password → select the **OK** button.

12.7.2 Setting Trend Groups

In the **Trends** tab, you can set a trend group. A trend group defines the trend displayed on the trend review page and printed in trend reports and recordings. The **Trends** tab is available the WorkStation and the ViewStation, and it functions in the same manner as it does at the host CentralStation.

You can perform the following operations:

- Add/Delete parameters
- Overlap/Unoverlap parameters
- Move parameters
- Rename trend groups

NOTE

- **In the Group Setup menu, HR is always displayed in the first row. It cannot be deleted or moved.**

12.7.2.1 Setting Trend Parameters

To add, delete, overlap, unoverlap,or move a parameter, follow this procedure:

1. On the review setup page, select the **Trends** tab.
2. Select the **Group Setup** button to enter the **Group Setup** screen.
3. Select the desired tab and then the desired parameter.
4. Select the desired buttons. Only buttons that may need special remarks are described here.
   - **Overlap**: selecting two parameters in the right column and then selecting the **Overlap** button overlaps these parameters. When parameters are overlapped, the symbol displays on the right side of overlapped parameters. The numeric values and trend curves of two parameters are displayed in one row on the screen.
   - **Unoverlap**: after two parameters are overlapped, selecting the **Unoverlap** button cancels the overlapping relation.

12.7.2.2 Renaming a Trend Group

To rename a trend group, follow this procedure:

1. On the review setup page, select the **Trends** tab.
2. Select the **Group Setup** button to enter the **Group Setup** screen.
3. Select the desired trend group tab.
4. Select the symbol in the upper left corner of the screen.

5. Input the desired name. The new trend group is displayed in the list of options for **Trend Group** on the tabular trends review page and the graphic trends review page.

12.7.3 Setting Full Disclosure

In the **Full Disclosure** tab, you can set the waveform saving accuracy. This tab is available at the CentralStation only. If the hard-disk partition is less than 450 G, full disclosure waveforms are saved in low accuracy by default. If the hard-disk partition is greater than or equal to 450 G, full disclosure waveforms are saved in medium accuracy by default.

To change the waveform saving accuracy, follow this procedure:

1. On the review setup page, select the **Full Disclosure** tab.
2. Select the desired accuracy for **Save Waveform**.

12.7.4 Setting Events

In the **Event** tab, you can set the conditions for locking and renaming events. This tab is available at the CentralStation only.

To set events, follow this procedure:

1. On the review setup page, select the **Event** tab.
2. In the **Lock** column, select the alarm priority of the events that need to be locked. When alarms of the selected priority are triggered, corresponding events will be locked in the event review page automatically.
3. Enable or disable **Rename Event**. When this option is enabled, you can rename events in the event review page.

12.7.5 Setting Arrhythmia Mark Colors

In the **Arrhy Mark** tab, you can set colors for arrhythmia marks that display in the compressed waveform. This tab is also available at the WorkStation and the ViewStation, and it functions in the same manner as it does at the host CentralStation.

To do so, follow this procedure:

1. On the review setup page, select the **Arrhy Mark** tab.
2. Select the desired arrhythmia category.
3. Select the color box on the right side of the arrhythmia category.
4. From the drop-down list, select the desired color.

12.8 Telemetry Tab

In the **Telemetry** tab, you can set items for telemetry devices. This tab is available at the CentralStation only.

12.8.1 Accessing the Telemetry Setup Page

To access the telemetry setup page, follow this procedure:

1. Select the system menu area mindray in the upper left corner of the screen → select **System Setup** to access the **System Setup** menu.
2. Select the **Telemetry** tab → enter the required password → select the **OK** button.

12.8.2 Setting Alarm Properties for Telemetry Devices

To set the alarm properties for telemetry devices, follow this procedure:

1. On the telemetry setup page, select the **Alarm** tab.
2. Set **Alarm Delay**. If an alarm condition is resolved within the delay time, the CentralStation does not present the alarm. The alarm delay setting affects some physiological alarms.
3. Set **ST Alarm Delay**.
4. Set the alarm priority for **ECG Lead Off, SpO2 Sensor Off**, and **No RF Signal**.

5. Set **Alarm Reset** and **Alarm Pause**.
6. Enable or disable **Lethal Arrhys Off**.
   - **Enable**: all the lethal arrhythmia alarms are configurable. The **All Off** button in the **Arrhythmia** menu is active.
   - **Disable**: all the lethal arrhythmia alarms are disabled and not configurable. The **All Off** button in the **Arrhythmia** menu is inactive.
7. Select the desired alarm priorities under **Latching** on the right side of the screen to latch physiological alarms. You can separately latch visual indications or simultaneously latch the visual and audible indications.
   - **Lethal, High**, **Med**, and **Low**: indicate alarm priorities.
   - **Visible**: latches visual indications only. When an alarm condition is resolved, visual indications, including alarm message and its background color remain. The time when the alarm is last triggered is displayed behind the alarm message.
   - **Audible**: when this option is selected, **Visible** will be selected automatically. All the visual and audible alarm indications remain until you reset the alarms. Besides, the time when the alarm is last triggered is displayed behind the alarm message.

NOTE

- **Changing alarm priorities may affect the latching status of corresponding alarms. Determine if you need to reset the alarm latching status after changing alarm priorities.**
- **When the alarm system is reset, latched physiological alarms are cleared.**
- **Selecting alarms of lower priority simultaneously latches higher priority alarms.**

12.8.3 Setting ECG Related Items for Telemetry Devices

To set the notch filter, QTc formula, and analysis lead, follow this procedure:

1. On the telemetry setup page, select the **ECG** tab.
2. Set **Notch Frequency** to **50 Hz** or **60 Hz** according to the power line frequency. The default is **60Hz**.
3. Enable or disable **Analysis Lead**. After this option is enabled, when ECG monitoring with lead sets other than 3-lead is being performed, the **Analysis Lead** option is displayed in the ECG menu. You can select the desired lead as the analysis lead. Then, the CentralStation uses the analysis lead to detect beats, compute heart rate and arrhythmia alarms and so on.

12.8.4 Setting Nurse Call Properties

To set nurse callfucntions, follow this procedure:

1. On the telemetry setup page, select the **Nurse Call** tab.
2. Set **Nurse Call**.
   - **Enable**: enables the nurse call function and displays the **Nurse Call** field on the patient management screen.
   - **Disable**: disables the nurse call function.
3. Set **Nurse Call Audio Mode**.

WARNING

- **Do not rely exclusively on the nurse call system for alarm notification. Remember that the most reliable alarm notification combines audible and visual alarm indications with the patient's clinical condition.**

12.8.5 Frequency Setup

In the **Frequency Setup** tab , you can perform operations such as setting or changing the IP address of receivers. This tab is applicable for the TMS-6016 Telemetry Monitoring System. Operations in this tab should be performed by authorized service personnel.

12.9 Network Tab

In the **Network** tab, you can set communication conditions between a CMS and other central monitoring systems, between a CMS and monitoring devices, between a CMS and eGateway, between CMS and the MLDAP server.

12.9.1 Accessing the Network Setup Page

To access the network setup page, follow this procedure:

1. Select the system menu area mindray in the upper left corner of the screen → select **System Setup** to access the System Setup menu.
2. Select the **Network** tab → enter the required password → select the **OK** button.

12.9.2 Setting General Communication Conditions

In the **General** tab, you can set central monitoring network, bedside monitoring network, remote view, and multicast parameters.This tab is available at the CentralStation, the WorkStation, and the ViewStation. But you can only set the central monitoring network at the WorkStation and ViewStation.

To configure the settings, follow this procedure:

1. On the network setup page, select the **General** tab.
2. In the **Central Monitoring Network Setup** section, set the desired options:
   - **Master Server IP Address**: enter the IP address of the CentralStation that is used as the master server. There can only be one master in the network.
   - **Local IP address**: from the drop-down list, select current CMS IP address. This IP address is used for communication between the CentralStation and external systems such as Gateway, WorkStation, ViewStation, and remote CentralStations.
3. In the **Bedside Monitoring Network Setup** section, set **Local IP address**. From the drop-down list, select the IP address of the CentralStation network adapter which is used for internal communication between the CentralStation and monitoring devices.
4. In the **Remote View** section, set the desired options:
   - **Support Monitor Remote View**: after bedside monitors and telemetry devices are admitted by the CentralStation, if the bedside monitors support the remote view functionality, you can view other bedside monitors or telemetry devices in the bedside monitors' remote view window.
   - **CMS Sending Broadcast Data**: this function is used when you view telemetry data on the bedside monitor screen. For the bedside monitor which only supports broadcast mode, you must select the check box before **CMS Sending Broadcast Data**.
5. In the **Multicast Setup** section, set **Multicast Address** and **Multicast TTL** values. These values are used for communication between a CentralStation and monitoring devices, between different Centralstations, and between a CentralStation and WorkStation, ViewStation, or CMS Viewer.
6. In the **Encryption Connection Type** section, set data encryption type when connecting devices:
   - **Only Private Encryption**: Mindray private encryption is used to encrypt the transmitted data. You cannot connect devices supporting SSL (secure sockets layer) encryption.
   - **SSL Encryption Priority**: for devices supporting SSL encryption, SSL encryption is used when connecting the devices. For devices not supporting SSL encryption, private encryption is used when connecting the devices.

12.9.3 Setting Results Service

In the **Results Service** tab, you can configure the CentralStation with a Patient Data Share (hereinafter called PDS) Gateway to communicate with external systems such as clinical information sytems and electronic medical record systems. This tab is available at the CentralStation Only.

For more information about the PDS Gateway, see *Mindray Patient Data Share Solution Guide (P/N: 0010-20-43061)*.

The CentralStation provides the unsolicited mode and solicited mode:

- In unsolicited mode, the CentralStation acts as the server side of TCP socket and listens to the external systems' connection. Once the connection is established, it will send data unsolicitedly at the configured interval.
- In solicited mode, the CentralStation acts as the server side of TCP socket and listens to the external systems' connection. Once the connection is established, CentralStation waits for results requests. Upon receiving a request, it sends the current results data to the requesting external system.

12.9.3.1 Setting Unsolicited Mode

To set unsolicited mode, follow this procedure:

1. On the network setup page, select the **Results Service** tab.
2. In the **Unsolicited Mode** section, set the desired options:
   - **Port:** it refers to the TCP listening port. The port number defaults to 4602. If you need to modify the port number, select the button.
   - **Interval**: select the desired interval to send data.
3. In the **Contents to be sent** section, enable or disable the desired items. When items are enabled, these types of data will be sent at the configured interval.

12.9.3.2 Setting Solicited Mode

To set solicited mode, follow this procedure:

1. On the network setup page, select the **Results Service** tab.
2. In the **Solicited Mode** section, set the desired options:
   - **Port:** it refers to the TCP listening port. The port number defaults to 4603. If you need to modify the port number, select the button.
   - **Mode**: select the desired mode. When **Mode 1** is selected, a packet including all beds' data is sent. When **Mode 2** is selected, packets are sent one by one. Each packet includes only one bed's data.

NOTE

- **Restart the system after changing the port of solicited mode or unsolicited mode.**

12.9.4 Configuring eGateway

In the **eGateway** tab, you can configure the settings when the CentralStation, the WorkStation, and the ViewStation communicate with an eGateway.

12.9.4.1 Setting ADT Query

When a CMS is connected with the HIS, you can use the admit-discharge-transfer (ADT) server to obtain patient information from the HIS and export the patient formation to the CMS. The ADT query function is also available at the WorkStation and the ViewStation, and it functions in the same manner as it does at the host CentralStation.

To set ADT query, follow this procedure:

1. On the network setup page, select the **eGateway** tab.
2. In the **ADT Query Setup** section, enable **ADT Query**.
3. Set **IP Address** and **Port**. The IP address and port should be consistent with that for the eGateway.

After setting ADT query, the **ADT** tab is displayed in the **Find Patient Window**.

12.9.4.2 Setting Data Export

The data export function is available at the CentralStation only. To set data export, follow this procedure:

1. On the network setup page, select the **eGateway** tab.
2. In the **Data Export Setup** section, enable or disable **Export Waveform Data**.

- Enabled: the **Export to EMR** button is displayed on the detail waveform review window and event review page. You can export waveform segments and events to third party systems.
- Disabled: you cannot export waveform segments or events to third party systems.

3. Enable or disable **Export 12-lead ECG Data**.
   - Enabled: the CentralStation can receive the 12-lead ECG analysis result from the bedside monitor,automatically generate a 12-lead ECG report, and then send the report to third party systems.
   - Disabled: 12-lead ECG report will not be generated.
4. Set **IP Address** and **Port**. The IP address and port must be consistent with that for the eGateway.

12.9.5 Setting CentralStation Authorization

In the **CentralStation Authorization** tab, you can set access types of the remote CentralStation, WorkStation, the ViewStation, and the CMS Viewer to control current CentralStation. This tab is available at the CentralStation only.

To set CentralStation authorization, follow this procedure:

1. On the network setup page, select the **CentralStation Authorization** tab.
2. Set **Access Control**.
   - **Full Control:** the WorkStation has full control over this CentralStation.
   - **Read Only**: the WorkStation, the ViewStation,and the CMS Viewer can view this CentralStation only.
   - **Off**: the remote CentralStation, WorkStation, the ViewStation,and the CMS Viewer cannot access this CentralStation.
3. Decide whether to enable password protection for access to the CentralStation. If you need to enable password protection, select **A password is required to access this CentralStation**. You can enter the desired password. If you need to change the password, enter,and then confirm the new password in the **Reset Password** section and then select **Save**.

NOTE

- **It is recommended that the password be changed by authorized personnel.**

12.9.6 Setting CentralStation Connection

In the **CentralStation Connection** tab, you can connect your current station (the CentralStation, the WorkStation, or the ViewStation) to the desired CentralStation. This tab is also available at the WorkStation and the ViewStation, and it functions in the same manner as it does at the host CentralStation.

To connect the desired CentralStation, follow this procedure:

1. On the network setup page, select the **CentralStation Connection** tab.
2. Select the desired CentralStation from the list of CentralStations.
3. Select the **Connect** button. If password protection is enabled for the chosen CentralStation, you need to enter a password.
4. If you need to disconnect from a CentralStation, select the desired station, and select the **Disconnect** button.

12.9.7 Setting Bed Authorization

In the **Bed Authorization** tab, you can set whether to allow beds at current CentralStation to be viewed by external systems such as the WorkStation. This tab is available at the CentralStation only.

To set bed authorization, follow this procedure:

1. On the network setup page, select the **Bed Authorization** tab.
2. Select the desired beds.
3. If you need to allow the beds that are newly admitted by the CentralStation to be viewed by default, select **Default authorization to be enabled for all beds**.

12.9.8 Importing AP Information

In the **Import AP Information** tab, you can import the desired AP information into the CentralStation. This tab is available at the CentralStation only. The text file contains the AP information must be in this format: AP Name, Mac address, Location, Detail. AP Name, Mac address, Location, and Detail are customizable.

To import AP information, follow this procedure:

1. On the network setup page, select the **AP Management** tab.
2. Select **Import AP Information**.
3. Select the desired file.
4. Select **OK**.

12.9.9 MLDAP

MLDAP refers to Mindray LDAP (Lightweight Directory Access Protocol). It is an independent process which can be installed on eGateway or other application server (Windows). MLDAP provides user identity and authentication.

The MLDAP server is connected with the hospital LDAP server. All monitoring devices are connected to the MLDAP server to implement identity and authentication for the following operations:

12.9.10 Setting MLDAP

To access the MLDAP server, you should set your CMS as follows:

1. On the network setup page, select the **MLDAP** tab.
2. Set IP address and port of the MLDAP server.

12.9.11 Testing MLDAP Server Connection

To test whether the CMS is properly connected with the MLDAP server, follow this procedure:

1. On the network setup page, select the **MLDAP** tab.
2. Select **Network Test**.

12.10 Print Tab

In the **Print** tab, you can set items related to a printer or reports. This tab is also available at the WorkStation and the ViewStation, and it functions in the same manner as it does at the host CentralStation.

12.10.1 Accessing the Print Setup Page

To access the print setup page, follow this procedure:

1. Select the system menu area ≡ mindray in the upper left corner of the screen → select **System Setup** to access the System Setup menu.
2. Select the **Print** tab → enter the required password → select the **OK** button.

12.10.2 Setting a Printer

In the **Printer** tab, you can set printer-related items.

To set a printer, follow this procedure:

1. On the print setup page, select the **Printer** tab.
2. Set **Printer**, **Paper Size**, and **Resolution**.

NOTE

- **PDF reports can be output by PDFCreator only.**

12.10.3 Setting Scheduled Reports

In the **Scheduled Report** tab, you can set whether to enable scheduled printing of tabular trends reports.

To enable scheduled printing, follow this procedure:

1. On the print setup page, select the **Scheduled Report** tab.
2. Enable **Scheduled Report Switch**.
3. Set the desired options:
   - **Time**: set the time to start printing automatically.
   - **Scheduled Report Interval**: select the desired printing interval.
   - **Tabular Trends**: select whether to print tabular trends. Only when this option is seleced, can scheduled printing of tabular trends reports be performed.
   - **Print Now**: print the report now regardless of the set printing start time.

12.10.4 Setting End Case Reports

In the **End Case Report** tab, you can select the desired report type. When you discharge a patient, if you select **Print End Case Report**, the selected end case report for this patient will be printed automatically.

To set end case reports, follow this procedure:

1. On the print setup page, select the **End Case Report** tab.
2. Select the desired report types.

12.10.5 Setting Report Layout

In the **Report Layout** tab, you can set the patient information displayed on the reports other than ECG reports. For details on how to set ECG reports, see Section *12.10.6 Setting ECG Reports*.

To set the report layout, follow this procedure:

1. On the print setup page, select the **Report Layout** tab.
2. Select the desired items under **Report Name**. **N/A** indicates that this item is not displayed on a report.

12.10.6 Setting ECG Reports

In the **ECG Report** tab, you can set the patient information displayed on ECG reports.

To set ECG reports, follow this procedure:

1. On the print setup page, select the **ECG Report** tab.
2. Select the desired items.

12.10.7 Setting PDF File Name

In the **PDF File Name** tab, you can set the naming rule for PDF reports.

To set PDF file name, follow this procedure:

1. On the print setup page, select the **PDF File Name** tab.
2. Select the desired items under **PDF File Name**.

12.10.8 Setting the Recorder

In the **Recorder** tab, you can change the recording related settings.

To set the recorder, follow this procedure:

1. On the print setup page, select the **Recorder** tab.
2. Enable **Recorder Switch**.
   - **Recorder COM Port**: selects the desired port for the recorder.
   - **Reset RecordService**: stops all the current recordings and resets the recording task.

12.10.9 Setting Other Print-Related Items

In the **Other** tab, you can make the printing and recording settings on alarm.

To set other print-related items, follow this procedure:

1. On the print setup page, select the **Other** tab.
2. Set the desired option.
   - **Print on Alarm**: select the output device when an alarm is triggered. When an alarm is triggered for a parameter and **Alarm Outputs** is enabled, the CMS can start a printing or recording task automatically.
   - **Printing Duration on Alarm**: sets the desired length of waveforms that will be printed or recorded automatically when **Alarm Outputs** for a parameter is enabled and an alarm is triggered. 10 sec, 20 sec, and 30 sec means printing the waveforms 5 seconds, 10 seconds, and 15 seconds before and after the alarm triggered time respectively.
   - **Monitor Remote Print**: if you enable this option, the CMS automatically starts a printing task after the printing task is started at the patient monitor.
   - **Second Mark (Printer)**: if you enable this option, second marks will be displayed on the report printout.
   - **Recording Duration**: sets the recording duration of the recorder.
   - **Recorder Paper Speed**: sets the paper speed of the recorder.

12.11 Configuration Tab

In the **Configuration** tab, you can set configuration items for telemetry devices and bedside monitors.

12.11.1 Accessing the Configuration Setup Page

To access the configuration setup page, follow this procedure:

1. Select the system menu area mindray in the upper left corner of the screen → select **System Setup** to access the **System Setup** menu.
2. Select the **Configuration** tab → enter the required password → select the **OK** button.

12.11.2 Setting Telemetry Configurations

In the **Telemetry Configuration** tab, you can save the measurement settings and alarm settings of the selected bed as configuration items. These configuration items can be loaded to other telemetry devices. For details on how to load these configuration items, see Section *3.6.7 Loading Configurations for Telemetry Devices*.This tab is available at the CentralStation only.

To save telemetry configuration items, follow this procedure:

1. On the configuration setup page, select the **Telemetry Configuration** tab.
2. Select the desired bed for **Select Bed**.
3. Select the **Save Telemetry Configuration** button.

12.11.3 Setting Department Configurations

In the **Department Configuration** tab, you can use the display, print, and history settings of the selected bed as configuration items. These configuration items can be loaded for bedside monitors automatically when the monitors are newly admitted by the CentralStation. This tab is also available at the WorkStation and the ViewStation, and it functions in the same manner as it does at the host CentralStation.

To set department configurations, follow this procedure:

1. On the configuration setup page, select the **Department Configuration** tab.
2. Select the desired bed for **Select Bed**.
3. Select the **Save Department Configuration** button.
4. If you need to load the department configurations for all beds, select **Load Department Configuration for All Beds**.

12.12 Other Tab

In the **Other** tab, you can set items such as unit of measurement. This tab is available at the CentralStation, the WorkStation, and the ViewStation.

12.12.1 Accessing the Other Setup Page

To access the other setup page, follow this procedure:

1. Select the system menu area mindray in the upper left corner of the screen → select **System Setup** to access the **System Setup** menu.
2. Select the **Other** tab → enter the required password → select the **OK** button.

12.12.2 Setting Units

To set units, follow this procedure:

1. On the other setup page, select the **Unit** tab.
2. Set the desired units.

12.12.3 Setting Units for Integrated Devices

When a bedside monitor is integrated with an external device such as a ventilator via the BeneLink module, you can set the units for integrated devices at the CMS.

To set units, follow this procedure:

1. On the other setup page, select the **Integrated Devices Unit** tab.
2. Set the desired units.

12.12.4 Configuring Module

In the **Module** tab, you can set barometric pressure and ECG standard at the CentralStation. You can set the ECG standard only at the WorkStation and ViewStation.

To set the ECG standard,follow this procedure:

1. On the other setup page, select the **Module** tab.
2. Set **Barometric Pressure**
3. Set **QTc Formula**. The CentralStation uses the Hodges correction formula by default to correct the QT interval for heart rate.
   - Hodges: $QTc = QT + 1.75 \times (HearRate - 60)$
   - Bazett: $QTc = QT \times \left(\frac{HearRate}{60}\right)^{\frac{1}{2}}$
   - Fridericia: $QTc = QT \times \left(\frac{HeartRate}{60}\right)^{\frac{1}{3}}$
   - Framingham: $QTc = QT + 154 \times \left(1 - \frac{60}{HeartRate}\right)$
4. Set **ECG Standard**.

12.12.5 Setting Time

WARNING

- **When the CMS software is running, do not change the operating system time. If the operating system time must be changed, do not change the operating system time directly. Instead, change the CMS system time according to instructions in this section.**

NOTE

- **After the CentralStation time is changed, the networked monitor system time will be synchronized to match the CentralStation system time.**
- **When a monitor is connected to the CentralStation, the monitor system time will be synchronized to match the CentralStation system time.**
- **The CentralStation will synchronize the time of the networked monitors at start of every hour.**

To change time manually or automatically at the CMS, follow this procedure:

1. On the other setup page, select the **Time** tab.
2. Set **Time Synchronization**.
   - **Master Server**: synchronizes time with the master server.
   - **NTP Server**: synchronizes time with the NTP server. After selecting this option, you need to futher set **Time Server**.
   - **Manual**: sets the time manually. You need to set **Date**, **Time**, **Date Format**, and **24-Hour Time** in the **Set System Time** section. When **24-Hour Time** is enabled, the 24-hour clock system is used. When **24-Hour Time** is disabled, the 12-hour clock system is used.

12.12.6 Setting Passwords

12.12.6.1 Selecting Password for User Authentication

You can select what password is used when accessing the **System Setup** menu. To do so, follow this procedure:

1. On the other setup page, select the **Password** tab.
2. Set **User Maintenance**.
   - **Local Password**: the CMS's password for accessing the **System Setup** menu is required.
   - **User Password**: the user name and password saved in the MLDAP server are required.

12.12.6.2 Setting the Password Timeout Period

If you use the password saved in the MLDAP server to access the **System Setup** menu, alarm settings and arrhythmia settings, you can set the password timeout period. If the timeout period is reached, you need to re-enter the password. To do so, follow this procedure:

1. On the other setup page, select the **Password** tab.
2. Set **Retention Time**.

12.12.6.3 Changing the Local Password for Accessing System Setup

To change the CMS's password required to access the system setup tabs other than the **Factory Maintenance** tab, follow this procedure:

1. On the other setup page, select the **Password** tab.
2. Select **Modify Local Password** and respectively enter the old password and new password.
3. Select the **Save** button.

NOTE

- **It is recommended that the password be set by authorized personnel.**

12.12.7 Setting Remote Control

In the **Remote Control** tab, you can set the permissions to control telemetry devices or bedside monitors to perform certain operations at the CentralStation or the WorkStation.

You can control telemetry devices or bedside monitors to perform relevant operations remotely via the CentralStation or the WorkStation under the following conditions:

- Permissions for these operations have been set to **Enable** or **Password Protection** in the **Remote Control** tab.
- These remote control functionalities are supported by the telemetry devices or bedside monitors.

To set remote control, follow this procedure:

1. On the other setup page, select the **Remote Control** tab.
2. Set the desired options:
   - **Read Only**: on the CMS, you can only view information from the monitors or telemetry devices.
   - **Enable**: on the CMS, you can control monitors or telemetry devices remotely. No password is required.
   - **Local Password**: on the CMS, changing the settings of alarms from the monitors and telemetry devices is password protected. This includes alarm settings, arrhythmia settings, pausing, and resetting alarms. The CMS's password for changing alarm settings is required.
   - **User Password**: on the CMS, changing the settings of alarms from the monitors and telemetry devices are password protected. This includes alarm settings, arrhythmia settings, pausing, and resetting alarms. The user name and password saved in the MLDAP server are required.

You can select **Change Local Password** to changed the CMS's password for accessing alarm settings, arrhythmia settings, pausing, and resetting alarms.

12.12.8 Setting Language

In the **Language** tab, you can select the desired language. When the CMS starts, it is displayed in the language selected at the time of installation.

To change a language, follow this procedure:

1. On the other setup page, select the **Language** tab.
2. Select the desired language. It is recommended to select the language supported by the operating system.
3. Restart the system.

NOTE

- **If the language you selected is inconsistent with that of the operating system, unrecognizable characters may appear on the CMS. If this occurs, change the operating system language and region settings by following the operating system operator's manual.**

12.12.9 Setting A Device Location

To set the the facility name and department where the CMS is located, follow this procedure:

1. On the other setup page, select the **Device Location** tab.
2. Set **Facility Name**. The facility name should be no more than 128 characters.
3. Set **Department**. The department name should be no more than 8 characters.

12.12.10 Defining the Night Time

You can define the nighttime for heart rate statistics on the CentralStation. To do so, follow this procedure:

1. On the other setup page, select the **ECG 24h Summary** tab.
2. Select **From** and **To** to define the beginning and end of the nighttime.

12.13 Factory Maintenance Tab

Items in the **Factory Maintenance** tab are used by service personnel. This tab is available at the CentralStation, the WorkStation, and the ViewStation.

13 Maintenance

13.1 Overview

Regular maintenance is essential to ensure that the CMS functions properly.

13.2 Maintenance Safety Information

WARNING

- **Failure on the part of the responsible individual hospital or institution using the CMS to implement a recommended maintenance schedule may cause undue equipment failure and possible health hazards.**
- **The safety checks or maintenance involving any disassembly of the CMS should be performed by professional service personnel. Otherwise, undue equipment failure and possible health hazards could result.**
- **The service personnel must be properly qualified and thoroughly familiar with the operation of the CMS.**
- **Turn off the CMS if no patients are to be centrally monitored.**
- **Restart the CMS every three months. Long time operation of the system may lead to a failure of the operating system.**
- **When the CMS is restarting, patient data will not be stored. To prevent any data loss, only restart your system when patients are not monitored.**
- **Disinfection or sterilization may cause damage to the equipment. Therefore, when preparing to disinfect or sterilize the equipment, consult your hospital's Infection Control Officer or Epidemiologist.**
- **Check the equipment after disinfection. If there is any sign of damage, remove it from use.**

CAUTION

- **If needed, contact Mindray for information concerning the repair of the CMS.**
- **All servicing and future upgrades must be carried out by the service personnel trained and authorized by Mindray.**
- **Make sure that the CMS operating environment meets the specific requirements. Otherwise, unexpected consequences, e.g. damage to the equipment, could result.**
- **When disposing of the packaging material, be sure to observe the applicable waste control regulations and keep it out of children's reach.**
- **At the end of its service life, the CMS must be disposed of in compliance with the guidelines regulating the disposal of such products. If you have any questions concerning disposal of the equipment, please contact Mindray.**

13.3 General Inspection

Whenever your system is first installed, repaired, upgraded or has been used for 6 to 12 months, a thorough inspection should be performed by qualified service personnel to ensure its reliability.

Follow these guidelines:

- Inspect the equipment and its accessories for mechanical damage.
- Make sure that the environment and power supply meet the specific requirements.
- Inspect all power cords and signal lines for fraying or other damages, and ensure that they are properly connected and insulated.
- Ensure that the sound system functions normally.
- Ensure that each function of the system is in good condition.

In case of any damage or abnormality, do not use the CMS. Contact your hospital biomedical engineers or Mindray service personnel immediately.

13.4 Cleaning

Your equipment should be cleaned on a regular basis. If you are in an area that is heavily polluted or dusty, the equipment should be cleaned more frequently.

The equipment to be cleaned includes the host, displays, printer, keyboard, and mouse.

CAUTION

- **Shut down the system and disconnect all power cords from the outlet before cleaning the equipment.**
- **Take extra care when cleaning the screen because it is more sensitive to rough cleaning method than the housing.**
- **When cleaning the mouse, keyboard, or other peripheral devices, disconnect them from the CMS.**
- **Never immerse any part of the CMS in liquids or allow liquid to enter the interior.**
- **Do not pour or spray any liquid directly on the equipment or accessories or permit fluid to seep into the casing, connectors, switches, or any ventilation openings. If you spill liquid on the equipment or accessories, disconnect the power supply, dry the equipment, and contact your service personnel.**
- **Check the equipment after cleaning. If there is any sign of damage, remove it from use.**

For the best performance, it is recommended that the CMS touch screen be kept clean. Observe the following precautions when cleaning the touch screen:

- Always remember to use a cloth or towel to apply glass cleaner to the touch screen.
- Any standard glass cleaner can be used to clean the touch screen. DO NOT use abrasive cleaning materials to clean a touch screen. DO NOT use alcohol or solvents containing chlorinated hydrocarbons.
- Remove fingerprints and stains by using a liquid lens cleaner and a soft cloth.
- Use a fine soft-hair brush to carefully brush away dust and dirt particles.

CAUTION

- **To avoid potential system issues, deactivate the touch screen before cleaning.**
- **DO NOT spray glass cleaner directly on a display as it could possibly leak inside a non-sealed unit and cause damage.**
- **Follow your hospital protocol for handling of blood and body fluids.**

A Technical Specifications

A.1 Requirements

The computer of the CMS should be highly reliable and stable. The recommended configurations are as follows:

Components	Requirements
System	Meet the IEC60950 requirements defined for ITE equipment, and comply with CE low voltage directives (LVD) and EMC directives.
Host	CentralStation, WorkStation, and ViewStation ■ CPU: 4 cores and 2.9 GHz minimum ■ Memory: 4GB minimum ■ Hard disk: 500 GB minimum, ■ Network adapter: 100M (minimum) self-adapting, Ethernet 802.3 ■ USB ports: two or more ■ Serial ports: one or more
Display	■ 19-inch or above display with 1280×1024 resolution ■ 21-inch or above widescreen display with 1920×1080 resolution ■ Support up to four displays with 1280×1024 resolution and up to four displays with 1920×1080 resolution ■ Support touchscreen
Operating System	■ Support Windows® 7 Professional SP1 ■ Support Windows® Server 2008 ■ Support Windows® Server 2012 R2
Antivirus	Support McAfee Application Control
Graphic card	Support dual or multiple displays
Printer	Support A4 or Letter paper size
Recorder	Thermal array, serial port
Speaker	■ Built in the computer or the display ■ Give alarm tones (45 to 85 dB) and alarm tones comply with IEC 60601-1-8.

NOTE

- **The configurations above are for reference only.**

A.2 Audio Signals

Alarm tone	600 Hz, ISO pattern
Self-test tone	650 Hz, a short beep
Alarm volume adjustment tone	650 Hz, a short beep
Nurse call tone	Three beeps

A.3 Maximum Number of Monitoring Devices

Item	Maximum Number of Monitoring Devices That Can Be Connected
CentralStation	64
WorkStation	64
ViewStation	64

A.4 Maximum Number of Patient Sectors on the Multibed Screen

Display resolution	Maximum Number of Patient Sectors on the Multibed Screen of a Single Display
1280×1024	16
1920×1080	24

A.5 Wired Network

Network structure	Ethernet 802.3
Transmission rate	100 Mbps or above
Max alarm time delay	≤ 3s

A.6 Review

Minitrend	Most recent 8 hours of minitrends for all parameters
Trend review	Most recent 240 hours of tabular trends and graphic trends for all parameters
Full disclosure	Most recent 240 hours of full-disclosure waveforms and compressed waveforms
C.O. review	Most recent 720 C.O. measurements
NIBP review	Most recent 1000 NIBP measurements
Event review	Most recent 1000 events, including the parameter name and 16-second waveform before and after an alarm is triggered
OxyCRG review	Most recent 48 hours of OxyCRG curve
Historic review	More than 200 discharged patients' data
12-lead Review	Most recent 720 12-lead analysis results, 12 analysis waveforms for each analysis result
ST review	Most recent 240 hours of ST segments

A.7 Calculation

The CMS supports the following calculations:

- Drug calculation
- Hemodynamic calculation, supporting reviewing 100 calculation results
- Oxygenation calculation, supporting reviewing 100 calculation results
- Ventilation calculation, supporting reviewing 100 calculation results
- Renal calculation, supporting reviewing 100 calculation results

A.8 Print

Paper size	A4 or Letter
Outputs	Patient information, drug calculation, hemodynamic calculation, oxygenation calculation, ventilation calculation, renal calculation, graphic trend, tabular trend, waveform review, full disclosure, waveform segment report, event review, 12-lead review, ST waveforms, OxyCRG review, real-time report, 12-lead ECG report, multi-lead ECG report, Arrhy statistics result, EEG report, Hemosight parameters report, CSA report, Rescue report, ECG 24h summary, Typical strips, Paging report, alarm settings report, real-time alarm report, OxyCRG graph, QT report, defibrillator self-test report

A.9 Recorder

Type	Thermal recorder
Port	RS232 serial port
Paper width	50 mm
Paper speed	25 mm/s, 50 mm/s
Horizontal resolution	16 dots/mm (25 mm/s paper speed)
Vertical resolution	8 dots/mm
Number of waveform channels	Maximum 3
Outputs	Realtime waveforms, realtime alarms, full disclosure review, event detail review

A.10 Data Export

Data format	Standard xml file [1]
ECG waveform sampling characteristics	
Frequency	Export in full disclosure and event review:80 Hz minimum
	Export in 12 lead analysis review: 500 Hz
Amplitude resolution	19.5313 uV/LSB
Significant bits	12 lead analysis review: 16 bits Full disclosure and event review:8 bits

Note:

[1]: For more information, please contact Mindray.

B CMS Alarm Messages

B.1 Overview

This chapter only lists technical alarms coming from the CentralStation, the WorkStation, or the ViewStation. For physiological and technical alarm messages from monitoring devices, see appropriate operator's manuals for these devices.

B.2 Alarm Messages in the System Alarm Area

The following table lists alarm messages in the system alarm area at the top of the screen.

Alarm Message	Default Priority	Cause and Solution
Network is disconnected. Please check.	High	Check the network.
Sound Card Error	High	Replace the computer with a known good one.
***Defibrillator Test Failed (X)	High	Take measures as prompted in the **Solution** column on a self-test report. Note: The "X" in the alarm message stands for the number of defibrillators whose self-test failed.
The patient data storage space is nearly full. Please delete some discharged patients.	Medium	This alarm is available at the CentralStation only. The discharged patient storage space is nearly full. Please delete some discharged patients.
Station (Department) , Network Disconnected	Low	Check the connection between the WorkStation and the CentralStation or between the ViewStation and the CentralStation. Note: The "(Department)" in the alarm message refers to the department where the WorkStation or the ViewStation is located. The alarm priority is configurable in the **System Setup** Menu. For details, see Section *12.4.3 Configuring Other Alarm-Related Items*.
eGateway Communication Lost	Low	Check the connection between the eGateway and the CentralStation. Note: The alarm priority is configurable in the **System Setup** Menu. For details, see Section *12.4.3 Configuring Other Alarm-Related Items*.
Time Server Unavailable	Low	Check the time synchronization settings and check that the external time server works properly.
No bed is online. Please check network connection.	Prompt	This alarm is available at the CentralStation only. Check whether the patient monitor is connected to the network.
Bed No. conflicts. Please check.	Prompt	Check the bed numbers of all the monitoring devices. If the same bed number is found, change it to a different bed number.
Recorder Head Hot: Please Wait	Prompt	Wait till the thermal printer head cools down and this prompt disappears. Then the recorder can work properly.
Recorder Unavailable	Prompt	Power off the recorder and restart the recorder. Then select **Reset RecordService** on the CMS. For details, see *12.10.8 Setting the Recorder*.
Recorder Initializing	Prompt	
Recorder Comm Error	Prompt	Check the connection cable of the recorder serial port.
Recorder Out Of Paper	Prompt	Load the paper.

B.3 Alarm Messages in Patient Sectors

The following table lists alarm messages in the patient sectors on the multibed screen.

Alarm Message	Default Priority	Cause and Solution
Offline	Low	The monitoring device is disconnected from the CentralStation network. Please check.
No RF Signal	Low	The telemetry device is disconnected from the CentralStation network. Please check.

C Default Settings

C.1 Overview

This chapter only describes default settings in the **System Setup** menu at the CentralStation, the WorkStation, and the ViewStation. For the default settings of monitoring devices monitored at the CMS, see appropriate operator's manuals for these devices. For details on how to access the **System Setup** menu, see Chapter *12 System Setup*.

"All" in the "Available At" column of the following tables indicates that this function is available at the CentralStation, the WorkStation, and the ViewStation.

"/" in the following tables indicates that this item is not configurable.

C.2 General Tab

Subtab	Item	Default	Available At
Volume	Alarm Volume	2	All
	High Alarm Volume	Alarm Volume+2	
	Reminder Volume	5	

C.3 Display Tab

Subtab	Item	Default	Available At
Screen	Patient Sector Number	16	All
	Auto Close ViewBed Screen	Never	
Patient Window	OxyCRG	Selected	
	12-Lead ECG		
	ST		
	Arrhy Statistics		
	Drug Calculation		
	Hemo Calculation		
	Oxygenation Calculation		
	Ventilation Calculation		
	Renal Calculation		
Other	ECG Lead Sequence	Normal	
	Soft Keyboard	Off	

C.4 Alarm Tab

Subtab	Item	Default	Available At
Audio	Minimum Alarm Volume	1	All
	Alarm Sound	ISO	
	High Alarm Interval	10 sec	
	Med Alarm Interval	20 sec	
	Low Alarm Interval	20 sec	
	Auto Increase Volume	2 Steps	
	Increase Volume Delay	20 sec	
	Alarm Reset Reminder	Off	
	Alarm Off Reminder	Off	
	Reminder Interval	1min	
Other	Silence Hotkey	None	All
	Silence	DIsable	
	Offline Alarm Priority	Low	
	Flashing Alarm Bar	On	
	No offline alarm if patient discharged	On	
	eGateway Communication Lost	Off	

C.5 Patient Management Tab

Subtab	Item	Default	Available At
Field	Visit Number	Unselected	All
	Room No		
	Middle Name		
	Race		
	Age		
	Facility		
	Department		
	Patient Group		
	Patient ID	Selected	
ADT Query	Facility	Unselected	
	Department		
	Room No		
	Bed No		
	Visit Number		
	Patient ID		
	Patient Name	Selected	

Subtab	Item	Default	Available At
Location	Location 1	Cathlab	CentralStation
	Location 2	X-Ray	
	Location 3	MRI	
	Location 4	CT Scan	
	Location 5	Ultrasound	
	Location 6	Hemodislysis	
	Location 7	OR	
	Location 8	Therapy	
Patient Group	Group 1 to 16	Selected	
Care Group	Group 1 to 16	Selected	
Other	Display Information in Patient Sector	Patient Name	All
	Patient Name Display Hotkey	F9	

C.6 Review Tab

Subtab	Item	Default	Available At
Full Disclosure	Save Waveform	Low Accuracy (when the hard-disk partition is less than 450G) Medium Accuracy (when the hard-disk partition is greater than or equal to 450G)	CentralStation
Event	Lethal	On	
	Rename Event	On	

Subtab	Item	Default	Available At
Arrhy Mark	Asystole	Selected	All
	V-Fib		
	V-Fib/V-Tach		
	V-Tach		
	Vent Brady		
	Extreme Tachy		
	Extreme Brady		
	PVCs/min High		
	Vent Rhythm		
	R on T		
	Multiform PVC		
	Pauses/min High		
	Run PVCs		
	Couplet		
	PVC		
	Bigeminy		
	Trigeminy		
	Tachy		
	Brady		
	Pacer Not Capture		
	Pacer Not Pacing		
	Missed Beats		
	Nonsus V-Tach		
	Pause		
	A-Fib		
	Irr Rhythm		

C.7 Telemetry Tab

Subtab	Item	Default	Available At
Alarm	Alarm Delay	6 sec	CentralStation
	ST Alarm Delay	30 sec	
	ECG Lead Off	Low	
	SpO2 Sensor Off	Low	
	No RF Signal	Med	
	Alarm Reset	Permanent	
	Alarm Pause	2 min	
	Lethal Arrhys Off	Disable	
ECG	Notch Frequency	60Hz	
	Analysis Lead	Off	
Nurse Call	Nurse Call	Enable	
	Nurse Call Audio Mode	Standard	

C.8 Network Tab

Subtab	Item	Default	Available At
General	Support Monitor Remote View CMS Sending Broadcast Data Port (Unsolicited Mode)	Off	CentralStation
eGateway	ADT Query	Off	All
	Export Waveform Data		CentralStation
	Export 12-lead ECG Data		
CentralStation Authorization	Access Control	Full Control	
	A password is required to access this CentralStation	Unselected	
Bed Authorization	Default authorization to be enabled for all beds.	Selected	

C.9 Print Tab

Subtab	Item	Default	Available At
Printer	Printer	Output PDF	All
	Paper Size	A4	
	Resolution	300 dpi	
Scheduled Report	Scheduled Report Switch	Off	
End Case Report	Tabular Trends Report	Selected	
Report Layout	Patient Name	Displayed	
	Patient Category		
	DOB		
	Gender		
	Patient ID		
	Bed No		
	Height/Weight		
	Paced		
ECG Report	Patient ID	Selected	
	Visit Number		
PDF File Name	Patient ID_Title_Time	Displayed	
Recorder	Recorder Switch	Disabled	
Other	Print on Alarm	Printer	
	Printing Duration on Alarm	20 sec	
	Monitor Remote Print	Enabled	
	Second Mark (Printer)	Enabled	
	Recording Duration	8 sec	
	Recording Speed	25 mm/s	

C.10 Other Tab

Subtab	Item	Default	Available At
Unit	Height Unit	cm	All
	Weight Unit	kg	
	ST Unit	mV	
	Hb Unit	g/dl	
	CVP Unit	cmH_2O	
	ICP Unit	mmHg	
	CO_2 Unit		
	O_2 Unit	%	
	Temp Unit	°C	
	Pressure Unit	mmHg	
	SVR Unit	DS/cm^5	
Integrated Devices Unit	Pressure Unit	cmH2O	
	CO_2 Unit	mmHg	
	$TcpCO_2/tcpO_2$ Unit		
Module	Barometric Pressure	760mmHg	CentralStation
	QTc Formula	Hodges	
	ECG Standard	AHA	All
Time	Time Synchronization	Master Server	
	Date Format	yyyy-mm-dd	
	24-Hour Time	On	
Password	Retention Time	20 sec	
	User Maintenance	Local Password	

Subtab	Item	Default		Available At
Remote Control		For Telemetry	For Monitor	
	Alarm Setup	Enable	Read Only	CentralStation
		Read Only		WorkStation
	Arrhythmia	Enable	Read Only	CentralStation
		Read Only		WorkStation
	Alarm Pause&Audio Pause&Alarm Reset	Enable	Enable	CentralStation
		Read Only	Read Only	WorkStation
	Privacy Mode	/	Read Only	CentralStation, WorkStation
	Night Mode			
	Modify Patient Demographics	Enable	Enable	CentralStation, WorkStation
	Discharge Patient			
	Standby			
	NIBP Start/Stop	Enable	Enable	CentralStation,
		Read Only	Read Only	WorkStation
	NIBP Interval	/	Enable	CentralStation,
			Read Only	WorkStation
	NIBP Clock Measure	/	Enable	CentralStation,
			Read Only	WorkStation
	ST Analysis	Enable	Enable	CentralStation
		Read Only	Read Only	WorkStation
	QT Analysis	Enable	Enable	CentralStation
		Read Only	Read Only	WorkStation
	QRS Threshold	Enable	/	CentralStation
		Read Only		WorkStation
	Pacemaker Rate	Enable	Enable	CentralStation
		Read Only	Read Only	WorkStation
	Pacer Reject	Enable	Enable	CentralStation
		Read Only	Read Only	WorkStation
	ECG Relearn	Enable	Enable	CentralStation
		Read Only	Read Only	WorkStation
	Send Data to EMR/Export EMR	Enable	Enable	CentralStation
		Enable	Enable	WorkStation
	Care Group Assignment	Enable	Enable	CentralStation
		Enable	Enable	WorkStation
ECG 24h Summary	Nighttime From	22：00		CentralStation
	Nighttime End	06：00		

D Units, Symbols and Abbreviations

D.1 Units

Abbreviation	Full Form
µA	microampere
µV	microvolt
µs	Microsecond
A	ampere
Ah	ampere hour
bpm	beat per minute
bps	bit per second
°C	centigrade
cc	cubic centimeter
cm	centimeter
cmH_2O	cmH_2O
dB	decibel
DS	dyne second
°F	fahrenheit
g	gram
GB	giga bytes
GHz	gigahertz
GTT	gutta
h	hour
hPa	hPa
Hz	hertz
"	inch
k	kilo
kg	kilogram
kPa	kilopascal
L	litre
lb	pound
m	meter
mAh	milliampere hour
mbar	mbar
M	mega
MB	mega byte
MHz	mega hertz

Abbreviation	Full Form
mcg	microgram
mEq	milli-equivalents
mg	milligram
min	minute
ml	milliliter
mm	millimeter
mmHg	millimeters of mercury
ms	millisecond
mV	millivolt
mW	milliwatt
MΩ	megaohm
nm	nanometer
rpm	breaths per minute
s	second
V	volt
VA	volt ampere
Ω	ohm
W	watt

D.2 Symbols

Symbol	Explanation
–	minus
-	negative
%	percent
/	per; divide; or
~	to
+	plus
=	equal to
<	less than
>	greater than
≤	less than or equal to
≥	greater than or equal to
±	plus or minus
×	multiply
©	copyright

D.3 Abbreviations

Abbreviation	In Full
AC	alternating current
ADT	Admission\Discharge\Transfer
Adu	adult
AG	anaesthesia gas
AgAwRR	Anaesthesia Gas Air-way Respiration Rate
AHA	American Heart Association
Ao	aortic pressure
AP MAP	Mean Artery Pressure
Art	arterial
ATMP	barometric pressure
AUC	area under the curve
AUX	Auxiliary
aVF	left foot augmented lead
aVL	left arm augmented lead
aVR	right arm augmented lead
awRR	airway respiratory rate
BAP	brachial arterial pressure
BC	burst count
BL	baseline
BIS	bispectral index
BSA	body surface area
BT	blood temperature
BTPS	body temperature and pressure, saturated
CaO_2	arterial oxygen content
CCI	continuous cardiac index
CCO	continuous cardiac output
CCU	cardiac (coronary) care unit
Cdyn	Dynamic compliance
CE	Conformité Européenne
CFI	cardiac function index
C.I.	cardiac index
CIS	clinical information system
CISPR	International Special Committee on Radio Interference
CMOS	complementary metal oxide semiconductor
CMS	central monitoring system
C.O.	cardiac output
CO_2	carbon dioxide
COHb	carboxyhemoglobin

Abbreviation	In Full
Compl	compliance
CPI	cardiac power index
CPO	cardiac power output
CPU	central processing unit
CRT	cathode ray tube
CSA	Compressed Spectral Array
Cstat	Static compliance
CVP	central venous pressure
DBS	double burst stimulation
DC	direct current
Des	desflurane
Dia	diastolic
DO_2	oxygen delivery
DO_2I	oxygen delivery index
dpi	dot per inch
dPmx	left ventricular contractility
DSA	Density Spectral Array
DVI	digital video interface
ECG	electrocardiograph
EDV	end-diastolic volume
EEC	European Economic Community
EEG	electroencephalogram
EMC	electromagnetic compatibility
EMG	electromyograph
EMI	electromagnetic interference
Enf	enflurane
ESU	electrosurgical unit
Et	end-tidal
EtAA	end-tidal anesthetic agent
EtDes	end-tidal anesthetic agent
EtEnf	
EtHal	
EtIso	
EtSev	
$EtCO_2$	end-tidal carbon dioxide
EtN_2O	end-tidal nitrous oxide
ETO	ethylene oxide
EtO_2	end-tidal oxygen
EVLW	extravascular lung water

Abbreviation	In Full
ELWI	extravascular lung water index
FAP	femoral arterial pressure
FCC	Federal Communication Commission
FDA	Food and Drug Administration
Fi	fraction of inspired
FiAA	inspired anesthetic agent
FiDes	inspired anesthetic agent
FiEnf	
FiHal	
FiIso	
FiSev	
$FiCO_2$	fraction of inspired carbon oxygen
FiN_2O	fraction of inspired nitrous oxide
FiO_2	fraction of inspired oxygen
Flow	flow
fmand	Mandatory breathing frequency
FPGA	field programmable gate array
FRC	fractional residual capacity
FreqMIN	Minimum breath frequency
fSIMV	Frequency of SIMV
fsigh	sigh rate
fspn	Spontaneous breathing frequency
ftot	Total breath rate
F-Trigger	Inspiratory trigger level (flow trigger)
FV	flow-volume
GEDV	global end diastolic volume
GEDI	global end diastolic volume index
GEF	global ejection fraction
Hal	halothane
Hb	hemoglobin
Hct	haematocrit
HIS	hospital information system
HR	heart rate
IBP	invasive brood pressure
IBW	ideal body weight
ICG	impedance cardiography
ICP	intracranial pressure
ICT/B	intracranial catheter tip pressure transducer
ICU	intensive care unit

Abbreviation	In Full
ID	identification
I:E	inspiratory time: expiratory time ratio
IEC	International Electrotechnical Commission
IEEE	Institute of Electrical and Electronic Engineers
Ins, INS	inspired minimum
$InsCO_2$	inspired minimum carbon dioxide
Insp.Flow	inspiration flow
Insp. MAC	inspired minimum alveolar concentration
int.PEEP	Intermittent PEEP
IP	internet protocol
Iso	isoflurane
ITBI	intrathoracic blood volume index
ITBV	intrathoracic blood volume
LA	left arm
LAP	left atrial pressure
LCD	liquid crystal display
LCW	left cardiac work
LCWI	left cardiac work index
LED	light emitting diode
LL	left leg
LVET	left ventricular ejection time
LVSW	left ventricular stroke work
LVSWI	left ventricular stroke work index
MAC	minimum alveolar concentration
MAP	mean arterial pressure
MetHb	methemoglobin
MRI	magnetic resonance imaging
MV	minute volume
MVe	expiratory minute volume
MVi	inspiratory minute volume
N/A	not applied
N_2	nitrogen
N_2O	nitrous oxide
Neo	neonate
NIBP	noninvasive blood pressure
NIF	negative inspiratory force
O_2	oxygen
O_2%	oxygen concentration

Abbreviation	In Full
OR	operating room
oxyCRG	Oxygen Cardio-respirogram
PA	pulmonary artery
pArt	artery pressure
pArt-D	diastolic artery pressure
pArt-M	mean artery pressure
pArt-S	systolic artery pressure
Paw	airway pressure
PAWP	pulmonary artery wedge pressure
pCVP	central venous pressure
Ped	pediatric
PEEP	positive end expiratory pressure
PEF	peak expiratory flow
PEP	pre-ejection period
PIF	peak inspiratory flow
PIP	peak inspiratory pressure
Pleth	plethysmogram
Pmean	mean pressure
PO_2	Oxygen supply pressure
Pplat	plateau pressure
PPV	pulse pressure variation
PR	pulse rate
PVC	premature ventricular contraction
PVPI	pulmonary vascular permeability index
PVR	pulmonary vascular resistance
PVRI	pulmonary vascular resistance index
RA	right arm
RAP	right atrial pressure
Raw	airway resistance
Rec	record, recording
Resp	respiration
RL	right leg
RM	respiratory mechanics
RQ	respiratory quotient
RR	respiration rate
RSBI	rapid shallow breathing index
rSO_2	regional Oxygen Saturation
SaO_2	arterial oxygen saturation
$ScvO_2$	Central Venous Oxygen Saturation

Abbreviation	In Full
SEF	spectral edge frequency
Sev	sevoflurane
SI	stroke index
SMR	Satellite module rack
SpO_2	arterial oxygen saturation from pulse oximetry
SQI	signal quality index
SR	suppression ratio
SSI	signal strength index
STR	systolic time ratio
SV	stroke volume
SVI	Stroke volume index
SVR	systemic vascular resistance
SVRI	systemic vascular resistance index
SVV	Stroke Volume Variation
SvO_2	Venous Oxygen Saturation
Sync	synchronization
Sys	systolic pressure
TB	Blood Temperature
$tcpCO_2$	transcutaneous carbon dioxide partial pressures
$tcpO_2$	transcutaneous oxygen partial pressures
TD	temperature difference
Temp	temperature
TFC	thoracic fluid content
TFI	thoracic fluid index
TFT	thin-film technology
TI	injectate temperature
TP	total power
TRC	tube resistance compensation
TVe	expiratory tidal volume
TVi	inspiratory tidal volume
TV	tidal volume
UAP	umbilical arterial pressure
UPS	Uninterruptible power Supply
USB	Universal Serial bus
UVP	umbilical venous pressure
VAC	volts alternating current
VEPT	volume of electrically participating tissue
VI	velocity index
VAC	volts alternating current

Abbreviation	In Full
VCO_2	CO_2 production
VO_2	oxygen consumption
VO_2I	oxygen consumption index
VO_2/kg	oxygen consumption per body weight
VO_2/m^2	oxygen consumption per body surface area
VT	Tidal volume
VTapnea	apnea tidal volume
VTe/TVe	Expiratory tidal volume
VTe spn	Spontaneous breathed minute volume
VTi/TVi	inspiratory tidal volume
VTsigh	Sigh tidal volume
WOB	work of breathing
WOBimp	Imposed work of breathing

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